

What Makes Truth-Telling Obvious? An Empirical Ranking of Simplicity-Preserving Design for LLM Agents*

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Abstract

Incentive compatibility on paper does not guarantee dominant-strategy play in practice: agents must recognize that truthful play is safe. As large language model (LLM) agents begin to transact, a design question becomes operational: which levers—the extensive form, the mechanism’s description, or scaffolds for the agent’s reasoning—preserve truthful play for this new bounded reasoner? We rank these levers empirically in auction environments across four LLM families, using a testbed validated against moment-matched reconstructions of human experimental data: LLM bidders reproduce the human difficulty ranking of auction formats, winner’s-curse comparative statics, and last-minute bidding under hard-close rules. The levers form a stable tier structure. Obviously strategy-proof extensive forms rank at the top. Safety-exposing descriptions, contingent-reasoning scaffolds, and clock-framing—which helps three of the four models and harms the fourth—form a strong second tier; menu restatements have no consistent effect—small, sign-mixed across models, null on average; forward-planning and belief-formation scaffolds actively backfire. The active ingredient is a description’s invariance content: text exposing why truth-telling is safe helps, while mechanical restatement does not—and reasoning traces show a complete dissociation: the levers that move bids leave stated reasoning unchanged, and the lever that moves stated reasoning leaves bids unchanged. Human evidence agrees in sign wherever a benchmark exists, and the description finding replicates in a school-choice matching domain (appendix). At the measured frontier the ranking compresses—newer reasoning models bid truthfully without help—but presentation still has teeth: a menu restatement collapses one otherwise-perfect frontier model to severe underbidding. Yet the agreement is ordinal, not cardinal: no single tuning of the agents makes their bidding match human behavior in levels across mechanisms.

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1 Introduction

A mechanism can be incentive compatible on paper and still fail in the minds of its participants. Truthful bidding is the dominant strategy in the second-price sealed-bid (SPSB) auction (Vickrey, 1961), and the logic is short: a bid determines only *whether* the bidder wins, not *what* she pays; overbidding risks paying more than value, underbidding risks losing profitable wins. Yet human subjects have failed to bid truthfully in this auction for decades (Kagel and Levin, 1993; Kagel, 1995). The theory of simple mechanisms was built to explain such failures: obvious strategy-proofness (Li, 2017), strategic simplicity (Börger and Li, 2019), and one-step simplicity (Pycia and Troyan, 2023) characterize which *implementations* of the same allocation rule make good play cognitively feasible, and Li (2024) decomposes the underlying cognitive burden into contingent reasoning, forward planning, and belief formation. As large language model (LLM) agents begin to transact—bidding in ad exchanges, negotiating on procurement platforms, submitting preference rankings on behalf of applicants—this body of theory acquires a new and operational audience. The question is no longer only which designs are simple for humans, but which of the levers available to a mechanism designer preserve dominant-strategy play for a new kind of bounded reasoner. The stakes of the question are visible in a single cell of our data: GPT-4o’s average bid in an SPSB auction falls roughly \$3.3 below its value, against dominant-strategy bids averaging \$24.5¹—yet a one-sentence change in how the very same rule is *described* removes roughly half of that deviation.

This paper delivers an empirical ranking of those levers. We organize the design space into three families: changes to the *extensive form* (replacing a sealed bid with an ascending clock; replacing a one-shot report with an iterative query protocol), changes to the *description* of a fixed mechanism (menu restatements, safety-exposing descriptions, clock-framing), and *reasoning scaffolds* appended to the agent’s instructions (contingent reasoning, forward planning, belief formation); a fourth family of preference framings is studied in Appendix E. Our domain is the auction laboratory: an SPSB auction with three bidders and independent private values, surrounded by the first-price, third-price, common-value, and clock formats that make cross-format comparisons possible. We run the grid on four model families—GPT-4o (2024-08-06), Claude 3.5 Haiku (2024-10-22), Gemini 2.0 Flash, and Gemma 27B—and summarize each lever by a *preservation index* ρ , the fraction of the baseline deviation from dominant-strategy play that the lever removes, computed within model. We anchor the exposition on GPT-4o, which remains a widely used, well-studied general-purpose model for strategic and economic reasoning tasks in the recent literature (Horton, 2023; Manning et al., 2024). This is a design choice for interpretability rather than a claim about model superiority: a fixed anchor isolates variation induced by mechanism, information environment, and presentation, while the ranking itself is computed within each model, so it is invariant to cross-model level differences.²

A ranking computed on LLM agents is only as informative as the testbed that produces it, and a central challenge in using LLMs for mechanism design is deciding what they should be benchmarked against. Comparing bids only to theoretical equilibrium is informative but incomplete: mechanism performance in practice depends on systematic and well-documented deviations in human behavior. Auctions are a particularly useful proving ground precisely because they combine sharp theoretical predictions—equilibrium

¹Baseline provenance, here and throughout: comparisons across mechanisms (e.g., sealed bid versus clock) use the dedicated SPSB baseline cells (mean deviations $-\$0.49$, $-\$1.63$, $-\$3.28$, $-\$5.27$ for Claude, Gemini, GPT-4o, and Gemma respectively), while presentation and scaffold contrasts use the pooled axis-1/axis-3 baseline cells run in the same grid (a provenance audit reclassified the legacy axis-2 “baseline” as a treated clock-framing description; Section 6.4); both conventions are reported in the master cell dataset (Section 4).

²Appendix D brackets the capability range. A frontier reasoning model (gpt-5-mini) bids the dominant strategy almost perfectly in SPSB (SMAD 0.2%) and shades the first-price auction almost exactly to the risk-neutral equilibrium (mean deviation from the equilibrium benchmark $+\$0.002$; $-\$8.53$ from value), yet misses the third-price equilibrium badly (SMAD 60%)—frontier capability narrows, but does not close, the bounded-rationality regime our ranking speaks to. A small open-weight model (Llama-3-8B, SMAD 57.58%) marks the capability floor.

bidding and dominant-strategy truth-telling—with a deep experimental literature documenting robust regularities and mistakes, including overbidding, winner’s-curse losses, and last-minute bidding under hard-close rules (Kagel and Roth, 2020). We therefore benchmark LLM behavior against human auction data as well as against theory. Using method-of-moments reconstructions of human bidding strategies from classic experiments (Kagel and Levin, 1993, 1986; Li, 2017; Breitmoser and Schweighofer-Kodritsch, 2022; Gonczarowski et al., 2024) and a unified metric—the scaled mean absolute deviation (SMAD)—we show in Section 5 that the testbed reproduces the human difficulty ranking of auction formats: GPT-4o’s SMAD is 27.55% in first-price versus 8.69% in second-price auctions, mirroring the human 24.76% versus 5.65%. LLM bidders also reproduce the winner’s-curse comparative statics: mean winner profits in first-price common-value auctions fall from -2.54 with four bidders to -5.55 with seven, and the curse is attenuated under second-price rules. Behavior is stable across rounds, with no evidence of within-session learning, so these patterns are properties of the agents rather than artifacts of experience.

The first ranking result is that obviously strategy-proof (OSP) extensive forms sit alone at the top. Replacing the sealed bid with an ascending clock compresses the cross-model mean deviation from $-\$2.67$ to roughly $-\$0.5$ (cross-model means, here and throughout, are unweighted averages over the four models), cuts SMAD to 1.4–5.2% from baselines of 9.5–20.7% ($p < 0.001$ in every model, effect sizes up to $d = 1.00$), and eliminates the heavy left tail of severe underbids.³

The second result is that descriptions split into two sharply different sub-families, and the split is governed by what the words *contain*, not by how many of them there are. A menu restatement in the style of Gonczarowski et al. (2023)—rewording the SPSB rule as a menu of outcomes determined by others’ bids—has no consistent effect: across the four models the shift is small and sign-mixed (Gemma moves toward truthful play, $+\$2.03$, $p < 0.001$; GPT-4o away, $-\$1.16$, Welch $p = 0.013$; Claude marginally away; Gemini is null), and null on average (placebo sign test $p = 0.73$)—consistent, on average, with the modest behavioral effects of menu descriptions in human experiments (Gonczarowski et al., 2024), though with model heterogeneity humans were never tested for. Safety-exposing descriptions, by contrast, are powerful: stating the payoff-safety invariant of the auction—your bid determines *whether* you win, not *what* you pay—cuts the cross-model mean deviation from $-\$2.67$ to $-\$1.53$. Clock-framing—merely *describing* the sealed bid as an automatic exit threshold on a price clock, a change of words with no change to the game—helps three of the four models, cutting GPT-4o’s mean deviation from $-\$2.94$ to $-\$1.54$ and halving Gemma’s ($p < 0.001$ each), while pushing the fourth, Gemini, away from truthful play—a first sign of a paraphrase sensitivity that Section 6.2 measures directly. A supplementary school-choice matching experiment (Appendix H) sharpens the diagnosis by unbundling the menu description: a menu that carries the invariance statement helps, while a menu that restates only the computational mechanics makes play *worse*. The active ingredient is not rewording the rules but exposing the incentive invariant that makes truth-telling safe.

The third result is that reasoning scaffolds are double-edged, and the direction of the edge is predicted by which cognitive dimension they target. Scaffolding *contingent reasoning*—a payoff-tree prompt that walks the agent through what happens in each outcome it might face—improves play across the board, cutting auction deviations from $-\$2.67$ to $-\$1.13$. Scaffolding *belief formation* backfires: prompting first- and second-order beliefs about opponents pushes auction deviations to $-\$3.04$ and $-\$3.47$, with the damage concentrated in the weakest models while Gemini is largely immune. (A one-shot sealed bid has no forward-planning axis, so the third scaffold, forward planning, has no place to bind in the auction.) The scaffolds give models material to strategize with, but not the ability to strategize well.

Assembled, these results form the paper’s central object: an empirical simplicity-preservation hierarchy

³A dedicated clean-IPV clock cell (IPV description and draws, $\$0.5$ tick, $K = 50$) reproduces the result exactly for the two models re-runnable at the time of writing—SMAD 1.5% (GPT-4o) and 1.8% (Gemma)—so the rung does not rest on the legacy cells’ mislabeled environment (Section 6.1).

of design levers (Section 6, Figure 8, Table 5). The hierarchy is a *tier structure*, not a strict total order: OSP extensive forms at the top; safety-exposing descriptions, clock-framing, and contingent-reasoning scaffolds in a strong second tier; menu restatements statistically indistinguishable from baseline on average; belief scaffolds below baseline; and risk-averse preference framings at the bottom (Appendix E). The ordering is concordant across models—Kendall’s $W = 0.70$ in auctions ($p = 0.004$)—and sign tests separate every adjacent tier boundary except one, which we report as an honest boundary failure: restatements do not reliably beat the strategizing scaffolds below them, because the menu rung is sign-mixed rather than a clean placebo. That exception sharpens rather than muddies the lesson: the operative distinction within descriptions is *invariance content*—safety-exposing text and the invariance-stating menu help, mechanical restatement is null to harmful (the supplementary matching experiment of Appendix H unbundles the menu to make exactly this point). This localizes the binding constraint: what limits these agents is not *computing* the dominant strategy—the payoff tree shows they can execute the case analysis when guided—but *seeing that truth-telling is safe* without being shown. The models’ own reasoning traces corroborate the mechanism with a complete dissociation—no lever moves stated reasoning and bids together: the payoff-safety description and the payoff tree move bids substantially while leaving stated reasoning unchanged (the safety invariant is echoed in none of the treated traces), whereas a worst-case contingency scaffold shifts stated reasoning sharply without moving bids (Section 6).

Where human experimental evidence exists, it agrees with the ranking in sign on every rung: humans, too, improve under ascending clocks (Li, 2017)—human preservation indices $+0.62$ (open clock) and $+0.37$ (blind clock)—improve under clock-framing (Breitmoser and Schweighofer-Kodritsch, 2022)—as do three of our four families—and are essentially unmoved by menu restatements (Gonczarowski et al., 2024), whose human index of -0.07 matches the LLM null on average even as individual models diverge in sign. The agreement is more informative than it first appears, because the two populations make *opposite* baseline mistakes: humans overbid in SPSB auctions—67.2% of bids strictly exceed value, or 92.2% among bids that deviate at all—while GPT-4o overbids in only 0.4% of baseline bids and errs instead by shading down (Section 5).⁴ That the same levers repair opposite errors in both populations indicates that the levers operate on the strategic problem itself—the recognizability of the dominant strategy—rather than on any psychology particular to humans. It also disciplines interpretation: our LLM testbed is a model of a boundedly rational player, not a simulation of a human subject. The rungs with only partial human anchors are then delivered as predictions (Section 7): most cheaply, that a safety-exposing description—an honest, one-sentence statement of the mechanism’s incentive invariant—will shift human bidding toward value in auctions, where, unlike matching (Gonczarowski et al., 2024; Katuščák and Kittsteiner, 2024), no invariant-statement experiment yet exists.

Two boundaries discipline the scope of these findings, and the paper closes by measuring both. The first is capability. Frontier reasoning does not uniformly dissolve the constraints: gpt-5-mini plays the dominant strategy almost perfectly in SPSB and applies the textbook equilibrium-shading formula in first price, yet misses the third-price equilibrium badly—it dissolves the constraints exactly where optimal play is textbook, and nowhere else (Appendix D). Nor does frontier capability immunize against presentation: in a four-model frontier intervention grid, a content-free menu restatement collapses one otherwise exactly-truthful model to severe underbidding (Section 6.4). The second is the depth of the human agreement. Everything above establishes *ordinal* agreement: LLM populations rank formats and levers the way humans do. Section 7 then asks what it would take to agree *cardinally*—whether any tuning of the agents (temperature, risk personas, preference framings, rule explanations) makes their bidding match human behavior in levels and direction. The answer is that no single tuning does: the best level-matching knob is mechanism-specific, and the only

⁴Direction shares use the repository-canonical convention: the share of all bids strictly above value, with the share among deviating bids reported alongside. Earlier drafts reported a 96.0% human overbid share under a $\pm 2\%$ truth-telling-band convention.

knob that reproduces human overbidding in second- and third-price auctions—a risk-seeking persona—simultaneously introduces dominated overbidding in the first-price auction, where humans almost never bid above value. Ordinal fidelity is a property of the agent population; cardinal fidelity is, at best, a mechanism-by-mechanism calibration.

The paper makes four contributions. First, the ranking itself: a tiered, cross-model ordering of the mechanism designer’s levers by how much dominant-strategy play they preserve, with a common preservation index making rungs comparable, established in the auction laboratory. Second, a sharpening of the descriptions result: a supplementary school-choice matching experiment unbundles the menu description and shows that its invariance-carrying content, not its mechanics-restating form, is what helps—anchored to the human descriptive-mechanism evidence (Appendix H). Third, a methodology: moment-matched reconstruction of human experimental benchmarks and the SMAD metric family, which turn decades of laboratory results into a validation layer for agentic testbeds—together with a characterization of where that validation stops, at the ordinal–cardinal boundary; we release the framework as an open-source benchmark. Fourth, design guidance. For a designer who must keep a direct mechanism, an honest safety-exposing description recovers most of the benefit of full OSP redesign at the cost of a sentence; conversely, prompting agents to plan ahead or model others’ beliefs—precisely the “reasoning enhancements” an agent vendor might advertise—systematically backfires. If the cognitive constraints that the theory of simplicity was built to address also bind on artificial agents, then the case for engineering simplicity no longer rests on human psychology alone: it becomes a design principle for markets that must work for any kind of reasoning agent.

The paper proceeds as follows. Section 2 reviews the related literature. Section 3 derives the lever taxonomy from simplicity theory and states our hypotheses. Section 4 describes the mechanisms, environments, treatments, models, metrics, and the reconstruction of human benchmarks. Section 5 establishes the testbed’s behavioral fidelity against human experimental and field evidence. Section 6 presents the ranking in the auction domain: extensive forms, then descriptions, then scaffolds, then the assembled hierarchy, together with the reasoning-trace evidence on how the levers act. Section 7 moves from ordinal to cardinal agreement with human behavior: it anchors the ranking rung by rung to human evidence, issues out-of-sample predictions for unanchored rungs, and asks whether any tuning of the agents achieves cardinal agreement across mechanisms. Section 8 draws out design implications, scope conditions, and limitations, and Section 9 concludes. Appendices collect the human-data reconstructions, simulation procedures, prompt library, additional ablations, and a supplementary school-choice matching experiment demonstrating the menu-content split (Appendix H).

2 Related Work

Auction theory and experiments. There is a vast quantity of theoretical and experimental literature on auctions. While Krishna (2009)’s textbook and Kagel and Roth (2020)’s handbook provide invaluable general resources, we focus on citations relevant to the results in this paper. Starting with the IPV case, the benchmark of revenue equivalence in the risk-neutral case is developed in Myerson (1981). The experimental evidence for departures due to risk-aversion in the FPSB auction is documented in Coppinger et al. (1980) and Cox et al. (1988)’s survey. The experimental evidence for the common error of bidding *above* one’s value in the SPSB auction is documented in Kagel and Levin (1993); as we show in Section 5, LLM bidders instead tend to bid *below* value in the SPSB auction—a qualitative divergence from human behavior whose implications for the design ranking we take up in Section 7. For common value settings, experimental evidence for the winner’s curse was first given by Bazerman and Samuelson (1983); we primarily follow the experimental design and analysis framework in Kagel and Levin (1986) and Kagel et al. (1987). Levin et al.

(1996) extend this line to English (ascending) common-value auctions, where the information released by observed dropouts attenuates, but does not eliminate, the curse. For eBay-style auctions, the phenomenon of bid sniping has been studied by Roth and Ockenfels (2002) and Ockenfels and Roth (2006), while Einav et al. (2011) provide an empirical study of hidden reserves on eBay.

Simple mechanism design. Li (2017) formalized obvious strategy-proofness (OSP) and showed that ascending clock auctions satisfy it while SPSB auctions do not. Related notions include *strategic simplicity* (Börger and Li, 2019), which requires that optimal play not depend on higher-order beliefs, and *one-step simplicity* (Pycia and Troyan, 2023), which requires that optimal play emerge from considering only the immediate next step rather than full backward induction. The SPSB auction satisfies strategic simplicity—truthful bidding is dominant regardless of beliefs about others—but is not OSP. Ascending clock auctions satisfy both. Li (2024) decomposes mechanism complexity into three cognitive dimensions: contingent reasoning, forward planning, and belief formation. We use this decomposition to structure the reasoning-scaffold family of our lever taxonomy (Section 3).

Recent work emphasizes the connection between strategic complexity and apparent irrationality. Oprea (2024) argues that many anomalies attributed to non-standard preferences are better explained by computational complexity: subjects make mistakes not because their preferences are irrational but because the problem is hard. This framing aligns with our approach. We do not assume LLMs have “biases” in the behavioral economics sense (though we probe prospect-theoretic preference framings in Appendix E); rather, our primary question is whether the computational structure of strategic problems creates failure modes for LLM agents—and which design levers remove them.

Mechanism descriptions and framing. A complementary literature holds the mechanism fixed and varies only how it is described. Gonczarowski et al. (2023) study descriptions of strategy-proof mechanisms that aim to make incentive properties transparent, presenting short menu-based descriptions in which a participant’s report is used only to select from a menu determined by others’ reports; they develop such descriptions both for the SPSB auction and for deferred acceptance (DA), though their menu-description treatment of an SPSB auction did not perform well empirically. In an incentivized lab experiment, Gonczarowski et al. (2024) find that a menu-based explanation significantly improved participants’ measured understanding of DA’s strategyproofness relative to standard descriptions, though average behavioral effects were modest. Breitmoser and Schweighofer-Kodritsch (2022) shows that re-framing a static sealed-bid auction as an ascending clock auction—a change of description, not of extensive form—can improve the rate of truth-telling behavior. Three further results discipline how description treatments should be read. Guillén and Hakimov (2018) find in a field experiment on top trading cycles that describing the strategy-proofness *property* increases truth-telling while describing the mechanism’s *mechanics* significantly reduces it; Katuščák and Kittsteiner (2024) show that a description isolating what does and does not depend on a participant’s own report increases truthful play in TTC, more so for higher-numeracy subjects; and Masuda et al. (2022) show that direct *advice* to bid truthfully in a Vickrey auction raises truthful bidding by roughly 24 percentage points net of experimenter-demand effects—a treatment type we deliberately exclude from our lever space, since advice reveals optimal play rather than exposing why it is safe. These strands—menu restatements with weak behavioral effects, framings that import the salience of a dynamic format, and content that exposes the incentive-relevant invariant as opposed to the mechanics—correspond to distinct description sub-families in our taxonomy (Section 3) and to distinct rungs of our ranking (Section 6.2).

Matching and deferred acceptance. There is a deep literature in market design for two-sided matching. Gale and Shapley (1962) introduced the DA algorithm, which produces stable matchings, and Roth (1982)

showed that DA is strategy-proof for the proposing side. The question of simplicity in DA was addressed first by Ashlagi and Gonczarowski (2018), who show that DA is generally not OSP-implementable. However, under *acyclic* priority structures—a condition introduced by Ergin (2002)—the proposer-optimal stable matching rule can be implemented in an OSP manner (for the proposer). Acyclicity rules out configurations where student a has priority over b at school s , while b has priority over c at school s' , and c has priority over a at school s'' . Under acyclic priorities, rejection chains remain local, enabling an iterative query protocol that is OSP. Our iterative DA treatment implements exactly this protocol; to our knowledge, it constitutes the first experimental test of an OSP implementation of DA with any kind of agent. In this paper the matching domain serves as a robustness and generalization check on the auction results: the full lever grid is replicated in DA, with results reported in Appendix H.

LLMs as economic agents. While LLMs are able to generate human-like text and reasoning (Achiam et al., 2023; Bubeck et al., 2023), results also show that they can have limited planning abilities and exhibit various cognitive biases (Wan et al., 2023). A growing literature explores using these human-like AI models as simulated agents in economics and social science. Horton (2023) introduces the “homo silicus” framework, showing that LLMs can replicate qualitative patterns from classic experiments in labor economics, bargaining, and social preferences; Aher et al. (2023) demonstrate that LLMs can reproduce human-subject results across a range of paradigms; Brand et al. (2023) use LLMs for market research, finding that they approximate human consumer behavior; and Park et al. (2023) and Manning et al. (2024) develop simulation and automated-experimentation infrastructure around LLM agents. Beyond simulation, there is growing interest in connecting LLMs with game-theoretic modeling of strategic environments more broadly—for example, deriving extensive-form structure, incentives, or interaction graphs from unstructured narratives (Daskalakis et al., 2024), or integrating LLMs into mechanism-design pipelines and general-sum strategic settings (Huang et al., 2025; Soumalias et al., 2025; Jiang et al., 2025; Wang et al., 2026).

Within auction settings specifically, there has been a proliferation of work since early versions of this paper, which can be broadly divided into three categories.

LLMs as first-class participants. The first category treats LLMs as first-class entities—bidders or agents who actively participate in a marketplace—and studies their strategic behavior in its own right. Fish et al. (2024) study collusive behavior in first-price sealed-bid auctions. Chen et al. (2023) study how to make LLMs better at playing auctions. More recent work includes studies of information disclosure with LLM agents (Zhang et al., 2025), comparisons across frontier models in various auction formats (Lu, 2025), and early feasibility checks on LLM bidding behavior (Tsuchihashi, 2023). Chen et al. (2024) also study a budget-constrained repeated ad auction (GSP-style), where bidders optimize campaign objectives over time. Much of this work uses auctions primarily as agent benchmarks for planning or collusion, without validating distributional similarity to human lab bidding, or focuses on narrow mechanism families without cross-format replication.

LLMs as proxies for human behavior in economic design. The second category asks whether LLM agents can serve as useful proxies for human participants, with the goal of informing mechanism design. The closest work to ours here is Corrigan et al. (2025), who take distributional similarity seriously as the criterion of usefulness and show that a light-touch calibration step using a small sample of human bids can bring LLM bid distributions closer to human benchmarks. Their calibration agenda is complementary to ours and directly anchors the question we take up in Section 7: our validation layer (Section 5) benchmarks out-of-the-box behavior across multiple canonical formats against moment-matched human data—the ordinal fidelity evidence that licenses using LLM populations to rank design levers—and Section 7 then asks the Corrigan et al.-style question of what it takes to agree *cardinally*, finding that the best tuning knobs are mechanism- and model-specific rather than portable.

Auction design for LLM-powered platforms. A third, distinct line of work studies the design of auction mechanisms *within* LLM-powered platforms—for example, how to allocate and price advertising slots or sponsored content when an LLM generates the display surface (Dütting et al., 2024b; Hajiaghayi et al., 2024). Here the research question is about mechanism design for a new marketplace, rather than about how design choices shape the behavior of LLM participants, and is thus orthogonal to our agenda.

Our contribution differs in focus from all three categories: we use the conceptual vocabulary of simple mechanism design—contingent reasoning, forward planning, belief formation—to probe which implementation, description, and scaffolding levers preserve dominant-strategy play for LLM agents, and to ask whether the same theory that was built to characterize bounded rationality in humans also organizes theirs.

Automated and computational mechanism design. Our work also connects to computational and automated mechanism design, which uses algorithms and optimization to search over (or learn) allocation and payment rules in complex environments. A prominent learning-based line trains neural networks to maximize an objective such as revenue while controlling incentive-compatibility violations—as in RegretNet and its successors (Dütting et al., 2024a; Curry et al., 2025) and architectures such as GemNet that build strategyproofness into the mechanism class itself (Wang et al., 2024)—with extensions to auctions with budgets, combinatorial auctions, and data markets (Feng et al., 2018; Wang et al., 2025; Ravindranath et al., 2023). Our contribution is orthogonal: rather than learning a new mechanism, we measure how implementation, description, and scaffolding choices affect the behavior of a fixed population of agents, and the resulting simplicity-preservation ranking can be used as an input to (and behavioral diagnostic for) automated design pipelines.

Relative to these literatures, our contribution is a single empirical object: a ranking of the simplicity-preserving levers available to a mechanism designer—extensive-form changes, mechanism descriptions, and reasoning scaffolds—measured on a common metric in canonical auction environments across four LLM families (Section 6) and replicated in a school-choice matching domain (Appendix H), licensed by benchmarking LLM behavior against moment-matched reconstructions of decades of human experimental evidence (Section 5), and anchored rung-by-rung to human data wherever a human experiment exists (Section 7).

3 The Lever Space: A Conceptual Framework

Fix the outcome rule: a strategy-proof direct mechanism in which truthful reporting is a dominant strategy. Everything a designer can still change—the extensive form through which reports are elicited, the language in which the rules are explained, and the reasoning the interface invites the participant to perform—constitutes the *lever space* of this paper. The theory of simple mechanisms tells us these residual choices are not innocuous. A second-price sealed-bid (SPSB) auction and an ascending clock auction implement the same outcome rule, yet only the clock is obviously strategy-proof (Li, 2017); a mechanism and its description are formally distinct objects, and descriptions can be engineered to expose strategyproofness (Gonczarowski et al., 2023); and the cognitive demands of a mechanism decompose into distinct axes—contingent reasoning, forward planning, and belief formation—that different designs tax differently (Li, 2024). The SPSB auction makes this lever space fully operable. It is strategy-proof, and *strategically simple* in the sense of Börgers and Li (2019)—the dominant strategy is independent of beliefs entirely—yet its direct form is not obviously strategy-proof, and it admits an OSP implementation: the ascending clock (Li, 2017). That wedge between strategy-proofness on paper and obviousness in play is the identifying variation the levers act on.

Auctions are the paper’s domain, and every lever family is evaluated there in the main text (Section 6). One lever—the description of a fixed mechanism—is additionally probed in a supplementary school-choice matching experiment (student-proposing deferred acceptance under Ergin-acyclic priorities (Ergin, 2002; Ashlagi and Gonczarowski, 2018)), reported in Appendix H, where the menu description can be unbundled into its invariance content and its mechanics; we draw on it in the main text only where it sharpens the descriptions result.

We organize the levers into three main families—**A** (extensive form), **B** (descriptions), and **C** (cognitive scaffolds)—plus an auxiliary family **D** (preference framings) reported in Appendix E. The taxonomy, and the predicted sign of every lever in it, is derived *ex ante* from published theory rather than from exploratory prompt search; roughly half the taxonomy is predicted to be inert or to *backfire*, and those predictions are part of what the experiments test. Table 1 summarizes the space; Section 4 gives the operational treatments and Appendix F the verbatim prompt texts.

3.1 Three Families of Levers

Family A: Extensive-form implementation. The strongest lever changes the game itself while holding the outcome rule fixed. A mechanism is obviously strategy-proof (OSP) if, at every information set where a deviation first diverges from the dominant strategy, the *worst* outcome from following the dominant strategy is at least as good as the *best* outcome from the deviation (Li, 2017). Verifying dominance in an OSP mechanism therefore requires no hypothetical, case-by-case reasoning about unobserved opponent moves: the comparison can be made on-path, one information set at a time. The ascending clock auction is the canonical example—it is OSP where SPSB is not, and it is additionally *one-step simple* in the sense of Pycia and Troyan (2023): comparing “stay one more tick” against “exit now” suffices at each price, with no backward induction over the price trajectory. Our Family A lever is the ascending clock, in open AC and closed AC-B variants (Section 4).

Family B: Mechanism descriptions. The second family holds the extensive form fixed and changes only the words. Gonczarowski et al. (2023) formalize *strategyproofness-exposing* descriptions: honest accounts of a mechanism from which its incentive properties are transparent. We distinguish three sub-families, because they turn out to behave very differently. **(B1) Menu restatement** re-describes the outcome function in menu form: the opponent bids fix a menu (a price at which you may win), and your report merely selects from it. This is the description Gonczarowski et al. (2023) propose for second-price auctions; in incentivized human experiments, a menu-based explanation of DA improved measured *understanding* of strategyproofness while average *behavioral* effects were modest (Gonczarowski et al., 2024). In the matching domain we deploy the menu description in two variants—one that includes the invariance statement (your report cannot change the menu you face, only which item you receive from it), and one that restates only the mechanics by which the menu is computed—so that the menu *format* can be separated from its invariance *content*. The distinction is drawn *ex ante* by Gonczarowski et al. (2023), who separate descriptions engineered so that incentive properties are transparent from the text from alternative, equally correct accounts of the same outcome function. **(B2) Safety-exposing descriptions** state, in one or two sentences, the structural invariant that makes truthful play safe. In the auction, this is the pivotal logic at the heart of Vickrey (1961)’s original argument: your bid determines *whether* you win, never *what* you pay (*Payoff Safety*). In DA, it is the property that historically distinguished DA from the Boston mechanism (Abdulkadiroğlu and Sönmez, 2003): rejections only redirect, never eliminate—your ranking determines the order in which you are considered, not whether you are considered (*Rejection Safety*). In the matching domain we additionally test a plain textbook statement of strategy-proofness itself, the bluntest member of this sub-family (Appendix H). **(B3)**

Clock-framing describes the sealed bid as an automatic exit threshold in a hypothetical ascending clock—“your bid is the price at which you would stop bidding”—following Breitmoser and Schweighofer-Kodritsch (2022), who show that such OSP-style *descriptions* of SPSB increase truthful bidding in humans without any change to the mechanism. We emphasize the accounting: clock-framing is a description of a sealed-bid game and belongs to Family B; the actual ascending clock is a different extensive form and belongs to Family A. The two are never pooled in our analysis.

Family C: Cognitive scaffolds. The third family holds both the mechanism and its rule description fixed and instead prompts the agent to *reason* in a particular way—scaffolding without revealing the dominant strategy. We index the scaffolds by the three cognitive axes of Li (2024). **(C1) Contingent reasoning:** the core demand of non-OSP strategy-proof mechanisms is case-by-case reasoning over unobserved opponent moves—“calculating payoffs for each profile of opponent bids” (Li, 2024). Our *Payoff Tree* scaffold performs this construction for the agent, laying out the explicit tree of payoff contingencies (win and pay the second-highest bid; lose and pay nothing). It says what *happens* under each action, never what to *do*. **(C2) Forward planning:** Pycia and Troyan (2023) formalize limited foresight over an agent’s own future decision points. A one-shot sealed bid has no such axis—there are no future own moves to plan over—so forward planning has no place to bind in our auction domain and does not enter the ranking. **(C3) Belief formation:** since the mechanism is strategy-proof, optimal play is independent of beliefs about opponents—this is precisely strategic simplicity (Börger and Li, 2019). First-order (“what do you expect the others to do?”) and second-order (“what do the others think *you* will do?”) belief scaffolds are therefore irrelevant to a participant who has understood dominance. Their inclusion is deliberately diagnostic, as we formalize in H3 below.

Family D: Preference framings (appendix). Finally, we test framings that manipulate induced preferences rather than reasoning: risk-attitude personas and prospect-theoretic loss/endowment frames (Kahneman and Tversky, 1979; Dreyfuss et al., 2022). Under a dominant strategy, these are payoff-irrelevant to optimal play—bidding one’s value is dominant for any monotone utility—so they should be inert for a rational agent. They are not levers a designer would pull to *improve* play, but they complete the bottom rungs of the ranking and calibrate how much damage an adversarial or careless framing can do; results are in Appendix E.

3.2 Hypotheses

The theory above yields three ordered hypotheses, which we state before presenting any results. They pre-commit the paper’s predictions and, because several predict null or negative effects, they discipline the exercise against the objection that prompt variations were searched until something worked.

H1 (Extensive form dominates). Family A levers produce the largest and most robust improvements in truthful play. They are the only levers that change the game the agent actually faces: under OSP, verifying dominance no longer requires contingent reasoning at all (Li, 2017), and the clock is additionally one-step simple (Pycia and Troyan, 2023). Every other family asks the agent to do the same hard verification with better hints; Family A removes the verification problem.

H2 (Descriptions substitute when—and only when—they expose the invariant). Within Family B, descriptions that state the invariant making truthful play safe—the safety descriptions (B2), clock-framing

Lever (family)	Domain	Theoretical anchor	Ex-ante prediction
Ascending clock, AC/AC-B (A)	Auction	Li (2017); Pycia and Troyan (2023)	Largest improvement
Menu restatement (B1)	Auction, DA ^a	Gonczarowski et al. (2023)	Weak/none (human behavioral null)
Payoff / Rejection Safety (B2)	Auction	Gonczarowski et al. (2023); Vickrey (1961)	Improvement, approaching OSP
Clock-framing (B3)	Auction	Breitmoser and Schweighofer-Kodritsch (2022)	Improvement, approaching OSP
Payoff Tree (C1)	Auction	Li (2024)	Improvement (axis binds)
Beliefs, 1st/2nd order (C3)	Auction	Börgers and Li (2019)	Inert if dominance understood; any effect diagnoses mis-strategizing
Risk / loss framings (D)	Auction	Kahneman and Tversky (1979)	Payoff-irrelevant; inert for a rational agent (Appendix E)

Table 1: The lever space. Every lever holds the outcome rule fixed; families differ in what else they hold fixed (Family A changes the extensive form; B changes only the description; C changes only the invited reasoning; D changes only the induced preferences). Predictions are stated ex ante from the cited theory. All lever cells are reported in the auction main text (Section 6); the menu description is additionally probed in a supplementary matching experiment (Appendix H). Verbatim prompt texts and the old-to-new label mapping appear in Appendix F. ^a In the matching experiment the menu description is deployed in two variants—invariance-property and mechanics-only—to separate the menu format from its invariance content (Appendix H).

(B3), and the invariance-property variant of the menu—approach the OSP benchmark *without* any change to the extensive form, while descriptions that merely restate the outcome function in different words—the plain menu restatement and its mechanics-only variant—do not, and may even backfire. The operative distinction is thus *invariance-exposing* versus *mechanically-restating* text, a line drawn ex ante by Gonczarowski et al. (2023) between descriptions engineered so that incentive properties are transparent and alternative, equally correct accounts of the same outcome function. The logic: if the binding constraint is not *computing* the dominant strategy but *seeing that truthful play is safe*, then supplying the safety certificate verbally should substitute for supplying it structurally—while supplying more words without the certificate should not. The mechanical restatements are the built-in placebo for this claim: they change wording, length, and salience while adding no invariant. The human evidence already points this way: menu descriptions improve measured comprehension with modest behavioral effects (Gonczarowski et al., 2024), whereas OSP-style clock descriptions shift human bids toward truthfulness (Breitmoser and Schweighofer-Kodritsch, 2022). A confirmation of H2 would be the paper’s most useful design result, since descriptions are the cheapest lever a practitioner can pull.

H3 (Scaffolds are axis-specific; belief scaffolds are diagnostic). A Family C scaffold helps only on the axis where the mechanism’s cognitive demand actually binds. Concretely: (i) C1 (contingent reasoning) should help, because case-by-case reasoning over opponent profiles is the core demand of the non-OSP sealed-bid format (Li, 2024); (ii) C2 (forward planning) has no axis to relax in a one-shot sealed bid, so it does not enter the auction ranking; (iii) C3 (beliefs) should be exactly inert, because dominant strategies are belief-free (Börgers and Li, 2019). The diagnostic content of H3 is in its failure modes. Any measured effect of a belief scaffold—in either direction—is evidence that the agent is not playing the dominant strategy but treating the interaction as a Bayesian game, forming beliefs about competitors and best-responding to

them. Belief scaffolds that activate strategic simulation without resolving it hand the agent material to strategize with; if the agent has not internalized strategy-proofness, we predict such scaffolds *backfire*, and most severely for the weakest reasoners. Backfiring is thus not an embarrassment for the taxonomy but its theoretically expected signature: a designer’s lever space contains anti-levers, and knowing which is which is part of the ranking this paper delivers (Section 6).

A note on taxonomy provenance. The operational labels in our experiment repository predate the final taxonomy, and three assignments were corrected during its construction (the full old-to-new mapping is in Appendix F, Table 15). First, the *Payoff Tree* scaffold was originally filed under forward planning (repo label `axis2_forward_tree`); it is classified here as contingent reasoning (C1), because the tree enumerates payoff *contingencies*—win versus lose, and what each implies—not future own moves, and a one-shot sealed bid has no forward-planning axis for it to occupy. Second, *Rejection Safety* was drafted within a DA “monotonicity” scaffold family (`axis2_da_monotonic_safety`); it scaffolds no reasoning at all—it states a property of the mechanism—and is therefore a safety-exposing description (B2). Third, one repository template name (`intervention_proxy_breitmoser`) was historically shared between the sealed-bid clock-framing *description* and a true iterative clock; in this paper the two are strictly separated as B3 and Family A respectively, with distinct prompt identifiers (`desc_clock_framing` vs. `osp_clock_iterative`) throughout. In the harmonized cell dataset (Section 4) the two conditions appear as separate cells for all four models—the sealed-bid clock-framing description and the closed ascending clock—so rungs A and B3 are now distinguished empirically as well as by design.

With the lever space fixed and its predictions on record, the next section turns the taxonomy into an experimental design: environments, treatment implementations, models, metrics, and the human benchmarks against which the testbed is validated.

4 Experimental Design

Our design crosses the lever families of Section 3 with the canonical strategy-proof auction domain—second-price sealed bidding and its clock implementations—and four LLM families, plus one field-inspired environment (a stylized eBay marketplace) that serves validation only. A supplementary school-choice matching experiment (student-proposing deferred acceptance), used to unbundle the menu description, is described and reported in Appendix H. The unit of observation throughout is a fully specified, seed-reproducible *market instance*: a draw of values (or preferences), a mechanism, a description, and a set of LLM agents whose actions mechanically determine allocations and payments. This section describes the mechanisms and environments (Section 4.1), the treatment grid (Section 4.2), the models and simulation protocol (Section 4.3), the unified deviation metrics and the preservation index that aggregates them into the ranking (Section 4.4), and the reconstruction of human benchmarks (Section 4.5). Table 2 pins the canonical calibration that all cells share; where legacy runs deviate from this calibration, the deviation is flagged in the table notes and resolved by the harmonized re-runs.

4.1 Mechanisms and Environments

4.1.1 Auctions

We generate synthetic auction data by simulating auctions in which n LLM agents bid against one another under fully specified information structures and auction rules. Each experimental *setting* is defined by (i) a *value model* (how bidder values or signals are drawn), (ii) a *mechanism environment* (the action space,

allocation and payment rules, and the information revealed during play), and (iii) a *prompt framing* (how the mechanism is described to the bidder). For each setting, we repeat the simulation K times with independent random seeds, producing a dataset of realized values (or signals), bidder actions, allocations, prices, profits, and free-text plans. “Actions” refers to: a numeric bid in sealed-bid formats, a *stay/exit* decision at each clock tick in ascending auctions, and an *increase/hold* decision over a bidder’s maximum bid in the eBay proxy-bidding environment.

Value models and numerical calibration. We implement three canonical information structures. All parameters (support ranges, bidder count n , and bid/price increments) are specified in configuration files and held fixed within a setting. In sealed-bid auctions, bids are restricted to a discrete grid with increment \$0.1; in ascending clock auctions, the clock price increases by \$0.5 per tick.⁵

- **Independent private values (IPV).** Each bidder’s value is an independent private draw from the integer grid

$$v_i \sim \text{Unif}\{0, 1, \dots, 49\},$$

so that the expected truthful bid in the second-price auction is $\mathbb{E}[v] = 24.5$. This is the baseline environment for the lever grid.

- **Affiliated private values (APV).** Values share a common component plus an idiosyncratic private shock:

$$c \sim \text{Unif}[0, 29], \quad \epsilon_i \sim \text{Unif}[0, 20], \quad v_i = c + \epsilon_i.$$

This induces affiliation: higher realized values for one bidder make higher values for others more likely. Each bidder observes her own realized v_i , so the dominant strategy in strategy-proof formats is unchanged.

- **Common value (CV).** The item has an underlying common value V shared by all bidders, while each bidder observes a noisy private signal:

$$V \sim \text{Unif}[20, 29], \quad \eta_i \sim \text{Unif}[-20, 20], \quad s_i = V + \eta_i.$$

Bidder i conditions on s_i when bidding, while realized payoffs depend on V , generating the winner’s-curse selection effect.

Sealed-bid auctions. In sealed-bid auctions, each bidder submits a single bid b_i simultaneously. Let $b_{(1)} \geq b_{(2)} \geq b_{(3)}$ denote the order statistics and let i^* be the highest bidder. The winner pays a price p determined by the mechanism:

- **First-price sealed-bid (FPSB).** i^* wins and pays $p = b_{(1)}$.
- **Second-price sealed-bid (SPSB).** i^* wins and pays $p = b_{(2)}$.
- **Third-price sealed-bid (TPSB).** i^* wins and pays $p = b_{(3)}$.

Profits are computed as $\pi_i = v_i - p$ for winners (or $\pi_i = V - p$ in CV) and 0 for losers. SPSB is the strategy-proof baseline on which the lever grid operates; FPSB and TPSB appear only in the behavioral-fidelity battery of Section 5, where their Bayes–Nash benchmarks (Section 4.4) discipline the comparison with human data.⁶

⁵The predecessor draft of the ranking study stated a \$0.01 sealed-bid increment; the configuration files specify \$0.1, which we adopt as canonical here (Table 2).

⁶The repository also contains FPSB and TPSB *intervention* batteries; these are outside the scope of the ranking, which concerns strategy-proof mechanisms whose benchmark is a dominant strategy rather than a Bayes–Nash equilibrium. They enter the paper

Ascending clock auctions. In an ascending clock auction, the price starts at 0 and increases by a fixed increment $\Delta_{\text{clock}} = \0.5 at each tick. At each posted price, each active bidder chooses *stay* or *exit*; exit is irreversible. The last remaining bidder wins and pays the exit price of the second-to-last bidder, implementing the standard second-price logic in a dynamic interface. We distinguish two information regimes: in the *open* clock (AC), bidders are informed as rivals drop out; in the *closed* or *blind* clock (AC-B), exit decisions are private and bidders are never told when others leave. Both regimes preserve the dominant strategy of remaining active exactly while the price is below one’s value. The clock format imported from the ranking pipeline is AC-B.

Why the sealed-bid-versus-clock comparison runs in both value environments. The extensive-form lever (Family A) appears twice in the auction design—once under APV and once under IPV—and the two cells do different work. The *APV cells exist for human-benchmark comparability*: the laboratory evidence that clock formats improve truthful play (Li, 2017; Breitmoser and Schweighofer-Kodritsch, 2022) is the evidence base for our moment-matched human reconstructions (Section 4.5), and it is in this environment that humans and LLMs can be placed on the same SMAD scale across SP, AC, and AC-B. The *IPV cells exist for identification*: under independent private values, a bidder who knows her own value learns nothing payoff-relevant from the clock—no information about her value exists in others’ dropout decisions—so any improvement from the clock format must operate through a purely cognitive channel. In a common- or affiliated-value setting, by contrast, an open clock could in principle improve play by revealing information about others’ values, confounding the cognitive interpretation. Keeping both cells, purpose-labeled, lets the APV cell carry the human anchor and the IPV cell carry the causal claim. The imported-pipeline clock cells are APV (common component $[0, 29]$) with open and blind variants and mixed increments (\$1 vs. \$0.5)—not the IPV clock the predecessor draft describes—so the clean IPV-clock identification cell is run separately (IPV template, IPV draws, tick \$0.5, $K = 50$). Two of the four canonical cells are complete—GPT-4o (SMAD 1.5%) and Gemma (SMAD 1.8%), folded into Section 6. **[TODO: IPV clock for Claude 3.5 Haiku and Gemini 2.0 Flash pending native keys (neither is servable on OpenRouter, E2).]**

Stylized eBay proxy-bidding environment (validation only). Separately from the laboratory mechanisms, we implement a stylized eBay environment with *proxy bidding*: each bidder maintains a maximum bid (a willingness-to-pay cap), and the platform automatically raises the standing bid only as needed to keep the current leader ahead. Each eBay simulation has $n = 3$ bidders, IPV values $v_i \sim \text{Unif}[0, 99]$, and $T_{\text{eBay}} = 10$ discrete bidding periods; treatments cross a hard versus soft (auction extension) closing rule with the presence of a hidden reserve price. Because this environment studies a richer, marketplace-inspired microstructure rather than a canonical laboratory mechanism, it enters the paper only as a field validation exhibit (Section 5); the eBay prompts are deliberately minimal and descriptive, so that behavior is not steered by didactic instruction. Full rules, treatments, and procedures appear in Appendix I.

4.1.2 Supplementary matching experiment

A supplementary school-choice matching experiment—student-proposing deferred acceptance in a 4×4 market with Ergin-acyclic priorities (Ergin, 2002; Ashlagi and Gonczarowski, 2018), in which LLMs play the role of students submitting rank-order lists—probes the descriptions lever outside the auction laboratory, unbundling the menu description into its invariance content and its mechanics. Because it is not a co-equal

instead through the cardinal-calibration analysis of Section 7, where the non-strategy-proof formats discipline the search for prompt-level tunings that match human levels and directions.

Table 2: Master calibration. Canonical parameterization of every environment in the design; all cells use temperature $T = 0.5$ and a one-shot protocol (no feedback across instances). K = repetitions per cell (pinned targets; deviations of legacy runs flagged in the notes).

Environment	Mechanisms	Value model	Action grid	N	K per cell	Model coverage
Sealed bid, IPV	SPSB (anchor); FPSB, TPSB	$v_i \sim \text{Unif}\{0, \dots, 49\}$	bid grid \$0.1	3	$100 / \geq 50^a$	canonical four + tiers ^b
Sealed bid, APV	SP	$v_i = c + \epsilon_i$; $c \sim U[0, 29]$, $\epsilon_i \sim U[0, 20]$	bid grid \$0.1	3	≥ 50	canonical four + tiers ^b
Clock, APV	AC, AC-B	as APV above	tick \$0.5	3	50^c	canonical four + tiers ^b
Clock, IPV	AC-B	as IPV above	tick \$0.5	3	50^d	canonical four ^d
Sealed bid, CV	FPSB, SPSB	$V \sim U[20, 29]$; $s_i = V + \eta_i$, $\eta_i \sim U[-20, 20]$	bid grid \$0.1	4–7	— ^e	GPT-4o
eBay (validation)	proxy bidding; closing rule $\times$ reserve	$v_i \sim U[0, 99]$	raise/hold max bid	3	30^e	GPT-4o
Matching (DA) ^f	direct (menu descriptions)	$v_{i,s} = c_s + \epsilon_{i,s}$; $c_s \sim U[40, 70]$, $\epsilon_{i,s} \sim U[0, 20]$	rank-order list	4×4	50	canonical four

Notes. ^a $K = 100$ for the anchor cell (SPSB-IPV baseline, per model); $K \geq 50$ for lever-grid and robustness cells. **[TODO: E8: harmonized baseline re-run pending; some legacy robustness cells ran $K = 30$ –40.]** ^b Canonical four = GPT-4o (2024-08-06, anchor), Claude 3.5 Haiku (2024-10-22), Gemini 2.0 Flash, Gemma 27B; appendix tiers = Llama-3-8B (capability floor) and the frontier tier gpt-5-mini, gpt-5, claude-sonnet-5, and gemini-2.5-flash (full lever grid), Appendix D. Gemma ran the six-format sealed-bid fidelity battery (Table 11); its two ascending-clock cells are pending native-key runs. ^c Clock K : the newer clock cells (the harmonized IPV-clock cells and the frontier-battery AC-B cells) run the pinned $K = 50$; some legacy incumbent APV clock cells still stand at $K = 40$ (configs repetitions: 40) and will be topped up in the harmonized re-runs. ^d IPV clock (identification cell): the clean $K = 50$ cells now exist for GPT-4o (SMAD 1.5%) and Gemma (SMAD 1.8%, $K = 44$ after parse drops) via OpenRouter, folded into Section 6; Claude 3.5 Haiku and Gemini 2.0 Flash are not servable on OpenRouter and their two IPV-clock cells await native keys. (The legacy imported clock cells are APV with mixed increments, \$1 vs. \$0.5, not the IPV clock described in the predecessor draft.) ^e eBay: $K = 30$ runs per treatment cell, verified from the recovered logs (29 for the soft-close $r = 60$ cell; Appendix I). **[TODO: Pin K for the CV cells from the recovered run logs (E5/E10).]** ^f Supplementary matching experiment (menu-description split): design summary in Section 4.1.2; implementation details and results in Appendix H.

domain, we defer its full specification—environment, priority orders, prompt skeletons, elicitation protocol, and results—to Appendix H, which reports it end to end.

4.2 Treatments

The treatment grid crosses the lever families of Section 3 with the auction domain. Family A (extensive form) is implemented as a change of *mechanism*: the ascending clock replaces the sealed-bid interface, with value draws, allocation and payment rules, model, and temperature all held fixed. Families B and C are implemented as *prompt treatments* on the sealed-bid baseline: each treatment modifies only the rule-explanation block of the prompt while holding the value distribution, bidder identities, payment rules, and the structured response format fixed, so that any behavioral change is attributable to the description or scaffold alone, under the same underlying game.

Family B: descriptions. Three sub-families restate or expose properties of the mechanism without changing its extensive form.

- **B1, menu restatement.** Bidders are instructed to choose a “maximum acceptable price”: they win if the highest rival bid (the market price) is below their chosen maximum, and if they win they pay that market price. This is the menu-style description of the second-price auction studied by Gonczarowski et al. (2023). The DA analogue restates DA through the menu of schools a student’s report can obtain given others’ reports, and is implemented in two variants—a *property* variant that includes the menu’s invariance statement (the student’s report cannot change which schools are obtainable) and a *mechanics* variant that restates the computation only—so that the content of a restatement can be separated from its form (Appendix H, Appendix F). This split turns out to be load-bearing for the ranking’s interpretation (Section 6.4).

- **B2, safety-exposing descriptions.** One-sentence statements of the incentive invariant that makes truthful play safe: *Payoff Safety* in auctions (“your bid determines *if* you win, not *what* you pay”) and *Rejection Safety* in DA (“rejections only redirect, never eliminate”). Exact texts appear in Section 6.2 and Appendix F.
- **B3, clock-framing.** The sealed bid is *described* as an automatic exit threshold in a simulated ascending clock: conceptually, the clock rises until bidders “drop out” at their thresholds, and the winner pays the second-highest threshold. Both B1 and B3 implement exactly the same allocation and payment rule as the SPSB auction; only the prompt language differs. B3 is a description, *not* a mechanism change, and must never be conflated with the true clock formats AC/AC-B of Family A.⁷

Family C: reasoning scaffolds. Two sub-families scaffold a mode of reasoning without revealing the dominant strategy. **C1, contingent reasoning** (the *Payoff Tree*): the prompt lays out the explicit decision tree of outcomes—win versus lose paths in the auction—telling the agent *what happens* under each action, not what to do. **C3, belief formation** (first- and second-order beliefs): the prompt directs attention to what opponents will do, or to what opponents think the agent will do. (Forward planning, C2 in the taxonomy of Section 3, has no axis in a one-shot sealed bid and therefore does not enter the auction grid.)

Family D: preference framings. Prospect-theory-inspired personas (risk aversion, loss aversion, endowment framing) alter the agent’s stated objective rather than the mechanism’s presentation. They are not designer levers in the same sense—no designer would instruct participants to be risk-averse—but they bound the ranking from below and are reported in Appendix E.

Inventory and coverage. Every treatment template, its exact prompt text, its mapping from legacy template names to the taxonomy above, and its run status appear in Appendix F (Table 15). The menu-restatement (B1) and clock-framing (B3) auction cells, GPT-4o-only in the predecessor draft, now cover all four canonical models at $K \approx 50$ (Section 6.2), as do the DA menu variants (Appendix H).

4.3 Models and Protocol

Models. The canonical model set comprises four widely deployed, non-reasoning model families from three commercial API providers and one open-weight family: OpenAI’s GPT-4o (2024-08-06) (OpenAI, 2024), which serves as the anchor model present in every cell; Anthropic’s Claude 3.5 Haiku (2024-10-22) (Anthropic, 2024); Google’s Gemini 2.0 Flash (2025-02-05) (DeepMind, 2024); and Google’s open-weight Gemma 27B (Riviere et al., 2024). Two further models delimit the capability range in Appendix D: gpt-5-mini (2025-08-07), a reasoning model that serves as the frontier tier, and Meta-Llama-3-8B-Instruct, a small open-source model that serves as the capability floor. Our goal is to study the interaction between cognitive constraints and mechanism design, which requires models that are capable enough to parse the rules of a mechanism yet still susceptible to the reasoning failures that mechanism design aims to remedy. We therefore deliberately select well-established, widely deployed model families rather than the most powerful frontier reasoning models: at the frontier, raw reasoning ability may mask the very cognitive bottlenecks our levers target, collapsing our ability to identify which design features matter and why. The frontier is a measured scope condition, not an omission—and it is two-sided. In the fidelity battery, gpt-5-mini plays the

⁷[TODO: The repository template name `intervention_proxy_breitmoser` historically referred to both the clock-framing description and the true iterative clock; prompt IDs are split into `desc_clock_framing` (B3) and `osp_clock_iterative` (A) per plan §5.6, and the provenance of the imported clock cells is being audited.]

textbook formats essentially perfectly (SPSB-IPV SMAD 0.2%; FPSB bids within pennies of the risk-neutral equilibrium shading rule), yet it misses the third-price format’s equilibrium by a wide margin (Appendix D): frontier reasoning narrows, but does not uniformly dissolve, the bounded-rationality regime in which the ranking applies.

Temperature. All non-reasoning models are run at temperature $T = 0.5$, the most common setting in LLM-based simulation (Horton, 2023; Horton et al., 2024); for reasoning models (gpt-5-mini), temperature is ignored by the API. For the anchor model we ablate $T \in \{0.1, 1.0\}$: effects are modest and non-monotonic across formats, so $T = 0.5$ is retained as a comparability convention rather than a performance choice (Appendix D, Table 10).

One-shot protocol and the learning ablation. Our main experiments use *single-instance* simulations: each repetition k is an independent market instance with freshly drawn values or preferences, and agents do not receive feedback across instances. We adopt this design to minimize reliance on cross-instance learning and long-term adaptation, which remains unreliable for language models (Dong et al., 2024). This choice is distinct from whether a mechanism is *iterative within* an instance: clock auctions still unfold over multiple within-instance steps, but we do not run repeated markets with feedback by default. In Appendix C, we run a repeated-auction ablation in which bidders observe outcomes from prior instances; consistent with the motivation above, we find no meaningful learning effects across repetitions, which also justifies pooling one-shot observations across the two source studies.

Simulation protocol. Each experiment proceeds as follows, repeated for K independent repetitions:

1. **Market realization.** For repetition k , we draw values (IPV/APV), signals (CV), or school-choice valuations (DA) for all agents under a fixed random seed.
2. **Prompted action.** We construct an individualized prompt for each agent containing (i) the mechanism rules (and treatment text, if applicable), (ii) the agent’s private value, signal, or preference information, and (iii) in the learning ablation only, a feedback transcript summarizing prior instances. We request a structured response with explicit tags: `<PLAN> ... </PLAN> / <ACTION> ... </ACTION>` in auctions and `<REASON> ... </REASON> / <DECISION> ... </DECISION>` in DA, where the action tag contains a numeric bid (sealed bid), a stay/exit decision (clock), an increase/hold decision (eBay), or a rank-order list (direct DA).
3. **Parallel elicitation and parsing.** We query the LLM for each agent using separate API calls in parallel, and parse the action from the action tag using a deterministic regex-based extractor.
4. **Outcome computation.** Given the submitted actions, we determine allocations and payments mechanically—winner and price under the auction rules, or the final matching by executing student-proposing DA with the true school priorities.

Unless otherwise noted, auction settings use $n = 3$ bidders, and each configuration is repeated K times with independent seeds (Table 2). Step-by-step procedures for the sealed-bid, clock, and eBay environments appear in Appendix B.

4.4 Metrics

The ranking requires a single deviation scale that is comparable across mechanisms, environments, and models. We build it in three steps: a scaled deviation metric for bids, its analogue for rank-order reports, and a preservation index that expresses every lever’s effect relative to its own baseline.

Unified deviation metric for auctions (applied to all bids). Consider an auction *environment* and *format* indexed by m . The index m collects all primitives needed to define the theoretical benchmark, including the number of bidders N , the value (or signal) distribution, the information structure (independent private value/common value), and the mechanism rules (e.g., first-price vs. second-price, sealed-bid vs. clock). Let $\mathcal{I}_i$ denote bidder i ’s information at the moment of bidding under environment m (e.g., $\mathcal{I}_i = v_i$ in private values, or $\mathcal{I}_i = s_i$ in common values). Let b_i denote an *observed submitted bid* by bidder i (we apply the definition to *all bids*, not only winning bids), and let $b_m^*(\mathcal{I}_i)$ denote the *theoretical benchmark bid* under m given information $\mathcal{I}_i$: the risk-neutral Bayes–Nash equilibrium in non-strategy-proof formats,

$$b_{\text{FPSB}}^*(v) = \frac{n-1}{n} v, \quad b_{\text{TPSB}}^*(v) = \frac{n-1}{n-2} v,$$

and truthful bidding, $b^*(v) = v$, in strategy-proof formats (second price, ascending clock).

We define the absolute bid error for a bid with information $\mathcal{I}_i$ as

$$d_m(\mathcal{I}_i) \equiv |b_i - b_m^*(\mathcal{I}_i)|.$$

To compare deviations across formats on a common scale, we normalize by the expected benchmark bid level,

$$\mu_m^* \equiv \mathbb{E}_m[b_m^*(\mathcal{I})],$$

where $\mathbb{E}_m[\cdot]$ denotes expectation with respect to the distribution of the information object $\mathcal{I}$ induced by environment m (i.e., the value/signal distribution and any randomization in the experimental design). We then define the *scaled mean absolute deviation* (SMAD) for auction format m as

$$\Delta_m \equiv 100 \cdot \frac{\mathbb{E}_m[d_m(\mathcal{I})]}{\mu_m^*} = 100 \cdot \frac{\mathbb{E}_m[|b - b_m^*(\mathcal{I})|]}{\mathbb{E}_m[b_m^*(\mathcal{I})]}.$$

This metric is interpretable as a percentage of the expected benchmark bid: $\Delta_m = 10$ means that, on average, bids deviate from the benchmark by an amount equal to 10% of the expected benchmark bid.

Given a sample of bids $\{(b_k, \mathcal{I}_k)\}_{k=1}^n$ from format m , the plug-in estimator is

$$\hat{\Delta}_m = 100 \cdot \frac{\frac{1}{n} \sum_{k=1}^n |b_k - b_m^*(\mathcal{I}_k)|}{\hat{\mu}_m^*}, \quad \hat{\mu}_m^* \equiv \frac{1}{n} \sum_{k=1}^n b_m^*(\mathcal{I}_k).$$

When we report a standard error for the (unscaled) mean absolute deviation $\frac{1}{n} \sum_k |b_k - b_m^*(\mathcal{I}_k)|$, we propagate it to $\hat{\Delta}_m$ by scaling with $\hat{\mu}_m^*$. We construct 95% confidence intervals using a normal approximation when an analytic standard error is available. Alongside SMAD, we report the *signed* mean deviation (bid – benchmark, in dollars) where the direction of error is itself of interest; the two statistics can disagree about model orderings, and we treat that disagreement as informative rather than choosing between them.

Deterministic benchmarks. SMAD requires a single-valued benchmark mapping $\mathcal{I} \mapsto b_m^*(\mathcal{I})$. Accordingly, we restrict attention to auction formats where the theoretical prediction is deterministic given the bidder’s information.

Matching analogue: normalized Kendall- τ error. For matching we construct the same kind of object: a unit-free percentage deviation from truthful play. For student i with true preference ranking π_i^* and submitted ranking π_i over the school set S ($|S| = 4$), the *normalized Kendall- τ error* is the fraction of school pairs ranked in opposite order,

$$\kappa_i \equiv 100 \cdot \frac{\#\{\{s, s'\} : \pi_i \text{ and } \pi_i^* \text{ disagree on } s \text{ vs. } s'\}}{\binom{|S|}{2}},$$

so $\kappa_i = 0$ if and only if the report is truthful and $\kappa_i = 100$ for a complete reversal. The parallel to SMAD is deliberate but not literal: SMAD normalizes the deviation by the expected benchmark bid, while κ normalizes by the maximum number of pairwise inversions. Both are unit-free percentages that equal zero exactly at truthful play and increase in the size of the deviation, which is what the ranking requires. Because 17% of simulated students draw at least one pair of exactly tied school values, we also compute a tie-robust variant of κ that does not count inversions among tied schools; headline DA numbers use the tie-robust variant, since apparent “misreports” that merely permute tied schools are artifacts of the deterministic tie-break (Appendix H). Even small κ can be consequential: under deferred acceptance, a single misreport can cascade through rejection chains and alter the final matching for $\Theta(n)$ participants (Dubins and Freedman, 1981).

Baseline declaration. Two baseline families appear in the auction analysis, and we fix their use once. *Cross-format* comparisons—sealed bid versus clock, the OSP rung of the ranking—are made against the dedicated SPSB baseline cells, whose mean deviations (−0.49, −1.63, −3.28, −5.27 for Claude 3.5 Haiku, Gemini 2.0 Flash, GPT-4o, and Gemma 27B) match the predecessor draft’s published values to rounding. *Within-format* description and scaffold contrasts (Families B–D) are evaluated against the same model’s axis-specific baseline cells run in the same batch (pooled across axes where a treatment spans them), which differ materially from the dedicated cells for some models (e.g., +0.48 versus −0.49 for Claude 3.5 Haiku). Both baseline families are reported in the released cell-level data, and no comparison in this paper mixes them.

Preservation index. For lever L , domain $d \in \{\text{auction, DA}\}$, and model m , let $D(L, d, m)$ denote the mean scaled deviation from dominant-strategy play—SMAD for auctions, mean normalized Kendall- τ error for DA—and let $D(\text{base}, d, m)$ denote the same object under the domain’s baseline mechanism and description (SPSB sealed bid; direct DA). The *preservation index* is

$$\rho(L, d, m) \equiv 1 - \frac{D(L, d, m)}{D(\text{base}, d, m)}.$$

By construction $\rho = 1$ means the lever restores perfect dominant-strategy play, $\rho = 0$ means it has no effect, and $\rho < 0$ means it *backfires*, pushing behavior further from the dominant strategy than the untreated baseline. Because ρ is computed within each model $\times$ domain cell, the ranking it induces is invariant to cross-model differences in baseline levels (and to the choice between absolute and signed deviation measures for ordering models)—a lever is credited only for what it does relative to the behavior it inherits.

Tier-claim discipline. We pre-commit to reporting the ranking as a *tier structure* (a partial order over levers), not a strict total order. Adjacent tiers are separated only where pairwise sign tests across the model cells support the separation; concordance of the full ordering across cells is summarized by Kendall’s W with bootstrap confidence intervals; and within-tier reversals (e.g., contingent scaffolds versus safety descriptions) are reported as findings, not smoothed away. Section 6.4 implements this discipline.

Statistical standard. Every reported cell receives a bootstrap standard error and 95% confidence interval (resampling market instances, 2,000 replications, seed 1299), and every treatment effect is evaluated against the baseline declared above with both a Mann–Whitney test and a Welch t -test, along with Cohen’s d ; we report both tests because description levers can move dispersion without moving the mean (and vice versa), and the two tests can disagree in exactly those cases. Where an error rate of exactly zero is observed, we report one-sided rule-of-three upper confidence bounds rather than declaring perfection. Concordance of the lever ordering across model cells (Kendall’s W with bootstrap confidence intervals), adjacent-tier sign tests, and the double-bootstrap intervals for the pooled preservation indices are reported in Section 6.4. Knife-edge cells (e.g., near-zero error rates under the safety descriptions) are topped up to $K = 100$ before adjacent rungs are declared separated.

4.5 Human Benchmark Reconstruction

Reconstructing human bid distributions via moment matching. Because micro-level bid data are rarely available from published experiments, we reconstruct synthetic human bid distributions by calibrating a mixture model to aggregate statistics reported in each source. We model bids as:

$$b = b_m^*(v) \cdot r$$

where $b_m^*(v)$ is the equilibrium bid and r is a bid-to-equilibrium ratio drawn from a three-component mixture:

$$\begin{aligned} \text{With probability } p_{\text{eq}} : \quad & r \sim \mathcal{N}(1, \sigma^2) \\ \text{With probability } p_{\text{over}} : \quad & r \sim 1 + \text{Exp}(\lambda) \\ \text{With probability } p_{\text{under}} : \quad & r \sim \text{Uniform}(1 - \delta, 1) \end{aligned}$$

This structure maps directly to how experimental papers categorize behavior: equilibrium bidders ($r \approx 1$), overbidders ($r > 1$), and underbidders ($r < 1$).

Source papers and equilibrium benchmarks. We draw on four sources: Li (2017) and Breitmoser and Schweighofer-Kodritsch (2022) for second-price and ascending clock auctions; Kagel and Levin (1993) for first-price, second-price, and third-price sealed-bid auctions with $n \in \{5, 10\}$ bidders; and Gonczarowski et al. (2023) for second-price auctions comparing mechanism descriptions. For each format, we use the risk-neutral Bayes–Nash equilibrium benchmarks of Section 4.4.

Parameter identification. Mixture weights ($p_{\text{eq}}, p_{\text{over}}, p_{\text{under}}$) are set directly from reported rates. For example, Li (2017) reports $\sim 50\%$ dominant-strategy play and $\sim 40\%$ overbidding, yielding $p_{\text{eq}} = 0.50$, $p_{\text{over}} = 0.40$, $p_{\text{under}} = 0.10$. The overbid scale λ is calibrated to match a secondary moment:

- For Li (2017) and Breitmoser and Schweighofer-Kodritsch (2022): the mean bid-to-value ratio (e.g., $\mathbb{E}[b/v] \approx 1.08$).
- For Kagel and Levin (1993): the regression coefficient β from bid regressions.
- For Gonczarowski et al. (2023): mean absolute deviation ($\text{MAD} \approx \$0.51\text{--}0.55$).

The equilibrium noise σ is derived from regression R^2 where reported, via $\sigma^2 = \beta^2 \text{Var}(v)(1 - R^2)/R^2$. The underbid bound δ is set consistently across papers; this is a parsimony choice as papers do not report underbid tail shapes separately. Full parameter values appear in Appendix A.1 (Table 9).

Why a mixture model rather than QRE. Quantal response equilibrium predicts symmetric deviations around equilibrium, contradicting the well-documented asymmetric overbidding in second-price auctions (Kagel and Levin, 1993). The mixture model directly maps to how experimental papers categorize and report behavior.

Limitations and epistemic status. Some reported statistics are difficult to reconcile within any parametric model—for instance, the regression coefficients and bidding frequencies in Kagel and Levin (1993) cannot both be matched under Gaussian noise, suggesting heteroskedasticity in the original data. Our approach matches key aggregate patterns while acknowledging this limitation. More fundamentally, the reconstructions are model-based simulations of published moments, not samples of human behavior, and we hold every use of them in this paper to a corresponding standard: *comparisons to reconstructed humans are made at the level of rankings, directions, and comparative statics only; levels and significance tests are reserved for theory benchmarks.* We run no hypothesis tests against reconstructed distributions—such tests would be statistically circular, reflecting our calibration choices rather than sampling variation in human behavior—and where reconstructed anchors appear in figures, their uncertainty is represented by Monte-Carlo bands obtained by re-drawing the mixture parameters within the ranges the source statistics admit. The ranking in Section 6 never depends on the reconstructions; they enter only in Section 5 (fidelity) and Section 7 (anchoring). Full reconstruction details, per-paper calibrations, and the reconstruction-uncertainty procedure appear in Appendix A.

5 Testbed Validation Against Human Data

The results in this section establish that the testbed’s behavioral responses to mechanism variation track those of human experimental subjects, licensing its use to rank design levers in Section 6. We proceed through three validation exercises of increasing institutional richness—the cross-mechanism difficulty ordering, the winner’s-curse comparative statics, and behavioral stability—and close with a validity map that states explicitly which phenomena transfer from human to LLM populations and which do not. Throughout, comparisons to the moment-matched human reconstructions of Section 4.5 are made at the level of rankings, directions, and comparative statics; levels and statistical tests are reserved for theory benchmarks.

5.1 Difficulty Ordering and the Direction of Mistakes

The experimental literature documents that human bidders deviate systematically from theoretical predictions across canonical auction environments, but that the *severity* and *direction* of mistakes depend strongly on the mechanism and the information environment. We therefore ask whether LLM agents (i) reproduce the same cross-mechanism ranking of “difficulty” that appears in human data, and (ii) make similar kinds of mistakes once we look beyond aggregate deviation magnitudes. Performance is measured by SMAD, the scaled mean absolute deviation of bids from the relevant theoretical benchmark (Section 4.4): the Bayes–Nash equilibrium bid function in first-price formats and truthful bidding in strategy-proof formats. We complement SMAD with a directional decomposition of deviations into underbidding and overbidding.

Figure 1 summarizes the benchmark battery, with formats ordered by human SMAD so that mechanisms appear in decreasing order of measured difficulty for human bidders. The central pattern is that GPT-4o broadly reproduces the human ordering: mechanisms that are empirically harder for humans also generate larger deviations for the LLM. The sharpest comparison is between the two sealed-bid IPV payment rules, which hold the allocation rule fixed while changing only the mapping from bids to payments: human bidders

exhibit FPSB SMAD of 24.76% versus SPSB SMAD of 5.65% (a $4.4\times$ difficulty gap), while GPT-4o shows 27.55% versus 8.69% (a $3.2\times$ gap). In the affiliated private-values (APV) environment, GPT-4o’s sealed-bid second-price deviation (10.73%) again sits close to the human level (9.31%); the clock formats in the same figure, where the format effect is the object of interest rather than a validation moment, are analyzed as the top rung of the ranking in Section 6.1. That GPT-4o reproduces the qualitative ranking suggests it is sensitive to the additional strategic complexity introduced by paying one’s own bid, and that the benchmark set captures genuine variation in strategic and cognitive difficulty. The fidelity battery now also exists for Gemma 27B (Appendix D, Table 11); its difficulty ordering is much flatter than GPT-4o’s—Gemma is nearly as far from equilibrium in the strategy-proof formats as in first price—so the clean human-like ordering documented here for the anchor model is itself partly model-specific, a point the validity map records.

Aggregate SMAD, however, masks a sharp divergence in *how* the two populations deviate. Figure 2 decomposes errors directionally after removing near-optimal bids (within a $\pm 2\%$ band of the benchmark). In the first-price IPV setting, both populations primarily *overbid* relative to the risk-neutral equilibrium, but GPT-4o’s overbidding is far more extreme: humans overbid 56.5% of the time (with 10.4% of bids near equilibrium removed), whereas GPT-4o overbids 85.9% of the time and almost never lands near the benchmark (0.7% removed)—consistent with systematically more aggressive competition by LLM bidders in FPSB (Cox et al., 1988).

The second-price IPV setting *reverses direction*, and this reversal is load-bearing for the rest of the paper. Human bidders are overwhelmingly on the overbidding side: 67.2% of reconstructed human bids lie strictly above value, and among bids that deviate from value at all, 92.2% are overbids,⁸ consistent with the classic right-skewed bid distributions documented by Kagel and Levin (1993), where most bidding mass lies above value.⁹ GPT-4o, by contrast, produces a predominantly *left-skewed* distribution with most mass on underbidding—74.1% of bids lie strictly below value and only 16.7% above (81.6% of deviating bids are underbids)—despite truthful bidding being a dominant strategy; it also places substantial probability mass at or near value (29.2% of bids within the $\pm 2\%$ band of Figure 2), reflecting a sizable fraction of exactly-truthful bids in our runs. The same reversal appears in the APV sealed-bid cell (humans 64.7% overbid; GPT-4o 92.4% underbid, under the figure’s band convention). Humans err toward competitive overbidding under an incentive-compatible rule; GPT-4o errs toward conservatism. We record this as a declared divergence in the validity map below (Table 4), and Section 7 shows why it strengthens, rather than undermines, the transferability of the lever ranking: the same design levers repair opposite-signed errors in both populations.

5.2 The Winner’s Curse in Common-Value Auctions

Common-value auctions provide a sharp test of whether bidders correctly condition on the adverse selection implicit in winning. In the canonical design of Kagel and Levin (1986), each bidder observes a noisy private signal x about a common value x_0 ; winning is “bad news” because, conditional on holding the highest of n signals, the expected common value lies strictly below x , with the gap widening in n .¹⁰ Bidders who neglect

⁸Direction shares are reported under the repository-canonical convention: the share of bids strictly above (below) value among *all* bids, with the share among *deviating* bids (the complement of exactly-truthful bids) in parentheses. An earlier draft reported 96.0% human overbidding under a band-removal convention (deviations outside a $\pm 2\%$ band of the benchmark), which Figure 2 retains; the qualitative reversal is invariant to the choice of convention.

⁹In an SPSB auction with 5 players, Kagel and Levin (1993) observed that the distribution of bids was right-skewed, with the majority of the bidding mass (67%) representing over-bidding relative to private values. However, this figure appears inconsistent with their reported regression of bids on values; specifically, both the intercept and the coefficient exceed 1, which would typically imply a more pervasive pattern of over-bidding than the percentage suggests.

¹⁰In the standard symmetric design, $\mathbb{E}[x_0 \mid X = x_{1:n}] = x - \frac{n-1}{n+1}\varepsilon$ away from boundary regions, where ε controls signal noise—competition exacerbates adverse selection.

this conditioning overbid relative to $\mathbb{E}[x_0 \mid \text{win}]$ and earn negative profits—the classic *winner’s curse*. The laboratory evidence delivers two comparative statics: winner losses worsen as n rises (e.g., approximately $-\$1.68$ with $n = 4$ versus $-\$3.68$ with $n = 7$ in Kagel and Levin’s benchmark conditions), and formats that reveal more information during price discovery attenuate the curse relative to first-price sealed bidding (Levin et al., 1996). Table 3 shows that GPT-4o bidders reproduce both. In common-value FPSB auctions, winners lose money on average in every treatment, losses grow monotonically with competition (mean winner profit -2.54 with 4 bidders down to -5.55 with 7), and the share of winners earning positive profits collapses from 24% to 0%; one-sided tests reject nonnegative winner profit in all FPSB treatments (Kagel and Levin, 1986).

Under a common-value *second-price* rule with the same information structure, the same comparative-static forces appear but the curse is attenuated: winners earn positive average profits with 4 bidders (mean 1.30), roughly break even with 5, and suffer statistically significant losses only once competition is sufficiently intense ($n \geq 6$). LLM behavior thus displays a substantially stronger winner’s curse in FPSB than in SPSB, mirroring the qualitative format ranking emphasized in the experimental literature (Levin et al., 1996).

Auction	Bidders	Mean (Std.)	Median	% Positive	p (Profit < 0)
FPSB	4	−2.54 (3.87)	−3.00	24.0%	< 0.001***
FPSB	5	−4.40 (3.63)	−4.00	6.0%	< 0.001***
FPSB	6	−5.46 (3.82)	−5.05	4.0%	< 0.001***
FPSB	7	−5.55 (2.93)	−5.30	0.0%	< 0.001***
SPSB	4	1.30 (4.87)	1.00	58.0%	0.967
SPSB	5	0.07 (4.35)	0.00	46.0%	0.542
SPSB	6	−2.82 (3.50)	−1.75	18.0%	< 0.001***
SPSB	7	−2.83 (2.78)	−2.00	12.0%	< 0.001***

Table 3: Winner profit in common-value auctions with GPT-4o bidders. The last column reports one-sided p -values for the hypothesis that winner profit is negative. *** denotes $p < 0.001$.

5.3 Stability

Two ablations establish that the behaviors above are stable properties of the agents rather than artifacts of run structure or decoding noise. First, in a multi-round ablation with full history feedback, GPT-4o shows no cross-round learning over 15 rounds: mean FPSB SMAD is 0.2523 in the first five rounds versus 0.2524 in the last five (t -test $p = 0.997$; Mann–Whitney $p = 0.913$), with the SPSB comparison likewise null (0.1282 vs. 0.1447, n.s.), and trend tests detect no drift (Appendix C). This justifies the one-shot design used throughout and implies that the lever effects in Section 6 cannot be attributed to within-session adaptation. Second, sampling-temperature effects on deviations are modest and non-monotonic across formats; $T = 0.5$ is retained as a convention for comparability rather than for performance, and model-tier ablations (a frontier reasoning model and a capability floor) are reported alongside (Appendix D).

5.4 A Validity Map

Table 4 collects the section’s evidence into an explicit accounting of what transfers from human populations to LLM agents and what does not. Four phenomena transfer: the cross-mechanism difficulty ordering, the clock/OSP format effect, the winner’s-curse comparative statics, and—on average across models—the menu-description null, though the menu case transfers with a caveat: the average effect is null, but individual models shift in both directions (Section 6.2). Two phenomena do not transfer, and we declare them as the map’s red cells: the *direction* of deviations under strategy-proof sealed-bid rules (humans overbid; GPT-4o underbids) and the *levels and distributional shape* of deviations (LLMs place a point mass at exactly truthful play that humans lack, and levels are not calibrated across populations).

The two red cells discipline everything that follows. Because deviation levels and directions do not transfer, the ranking in Section 6 is never built on cross-population level comparisons: every lever is scored by the preservation index ρ , a within-model ratio to that model’s own baseline, which is invariant to both the sign and the scale of a population’s characteristic error. And because transfer cannot be assumed, Section 7 treats it explicitly, anchoring each rung of the ranking to human evidence where it exists—where, as the map’s transferring rows already suggest, the levers move both populations in the same direction even when their baseline errors point opposite ways.

6 The Simplicity-Preservation Ranking

This section delivers the paper’s central product: an empirical ranking of the design levers of Section 3 by how well each preserves dominant-strategy play. Auctions are the domain throughout; one lever—the menu description—is additionally probed in a supplementary school-choice matching experiment (deferred acceptance) reported in Appendix H, drawn on below where it sharpens the descriptions result. We proceed in three passes over the lever taxonomy. Section 6.1 evaluates the extensive-form lever (family A): replacing the sealed-bid auction with an ascending clock. This is the lever for which simplicity theory makes its sharpest predictions, and it sits at the top of the ranking—but it bundles several simplifications at once. Sections 6.2 and 6.3 therefore decompose the bundle with presentation-only levers that hold the extensive form fixed: descriptions of the mechanism (family B) and reasoning scaffolds appended to it (family C). Section 6.4 assembles the cells into the ranking itself (Table 5 and Figure 8), and Section 6.5 closes the section by reading the agents’ own stated reasoning against their realized bids.

Throughout, $D(L, m)$ denotes the deviation of model m from dominant-strategy play under lever L : the mean bid deviation from value (bid – value, in dollars), with the scaled mean absolute deviation (SMAD; normalizer $E[b^*] = \$24.5$, Section 4.4) reported in parallel. Every cell-level statistic below—bootstrap confidence intervals (2,000 draws, seed 1299), Mann–Whitney and Welch tests, Cohen’s d —is computed from the merged run-level dataset in the replication package (`results/merged_ranking/auction_cells.csv`). Two auction baseline conventions appear in what follows, and we fix them here once. *Format-level* comparisons (sealed bid vs. clock) use the dedicated SPSB cells, with per-model mean deviations of -0.49 (Claude), -1.63 (Gemini), -3.28 (GPT-4o), and -5.27 (Gemma)—cross-model mean $-\$2.67$. *Presentation and scaffold* contrasts are tested against the pooled axis baselines run in the same grid as each treatment—with one correction made in this draft: a provenance audit showed that the legacy “axis-2 baseline” cell was itself a treated clock-framing description (Section 6.4), so the pool comprises the axis-1 and axis-3 baseline cells only, with per-model means of $+0.54$ (Claude), -1.25 (Gemini), -2.94 (GPT-4o), and -6.53 (Gemma)—cross-model $-\$2.55$. The two conventions differ materially for Claude, Gemini, and Gemma, and both are recorded cell by cell in the replication file. The comparative statistic is the preservation index ρ of Section 4.4, computed within model on the absolute-deviation scale and pooled

Phenomenon	Human finding	LLM finding	Verdict
Cross-mechanism difficulty ranking	FPSB $\gg$ SPSB in SMAD (24.76% vs. 5.65%); harder formats harder everywhere	Same ordering (27.55% vs. 8.69%); ordering preserved across the battery (Fig. 1)	Transfers (✓)
Clock/OSP format effect	Clock formats reduce deviations vs. sealed SPSB-APV: 3.54% (AC), 5.83% (AC-B) vs. 9.31% (Li, 2017; Breitmoser and Schweighofer-Kodritsch, 2022)	Same direction, larger effect: 0.81% (AC), 0.47% (AC-B) vs. 10.73% (Section 6.1)	Transfers (✓)
Winner’s-curse comparative statics	CV-FPSB winner losses grow with n ; attenuated under formats revealing more information (Kagel and Levin, 1986; Levin et al., 1996)	Mean winner profit $-2.54 \rightarrow -5.55$ as $n: 4 \rightarrow 7$; % positive 24% $\rightarrow$ 0%; SPSB attenuated (Table 3)	Transfers (✓)
Menu-description null	Menu restatement of SPSB leaves human bidding essentially unchanged (Gonczarowski et al., 2023)	Small and sign-mixed across the four models: null on average (placebo sign test $p = 0.73$); slightly worse for Claude ($p = 0.042$) and GPT-4o ($p = 0.003$), null for Gemma ($p = 0.94$), better for Gemini ($p < 0.001$) (Section 6.2)	Transfers on average (✓), heterogeneous
Deviation <i>direction</i> (strategy-proof sealed bid)	Overbidding: 67.2% of bids above value; 92.2% of deviating bids (SPSB-IPV); 64.7% of out-of-band deviations (APV)	Underbidding: 74.1% of bids below value; 81.6% of deviating bids (SPSB-IPV); 92.4% (APV, band convention)	Diverges (×)
Deviation <i>levels</i> and shape	Continuous, right-skewed bid distributions; little mass exactly at value	Point mass at truthful play (29.2% within $\pm 2\%$ in SPSB-IPV; 71.7% in AC vs. 26.8% for humans); levels not calibrated across populations	Diverges (×)

Table 4: **Validity map: which behavioral phenomena transfer from human experimental populations to LLM agents.** Human findings are from the cited experimental and field literatures via the moment-matched reconstructions of Section 4.5; LLM findings are from this section and Section 6. Verdicts are qualitative: transfers = same sign and comparative statics; diverges = opposite sign or uncalibrated levels. [TODO: Add a rank-concordance statistic with a bootstrap CI to the difficulty-ordering row (residual of E6; the other transferring rows now carry cell-level tests).]

as the median across cells: $\rho = 1$ is perfect preservation of dominant-strategy play, $\rho = 0$ no effect, and $\rho < 0$ a lever that *backfires*. Because ρ is computed within model, the ranking is invariant to cross-model differences in baseline levels.

6.1 Extensive-Form Levers: The Ascending Clock

A strategy-proof mechanism and its OSP counterpart implement the same outcome rule—the same mapping from reports to allocations and payments—but differ in extensive form. The non-OSP version requires *contingent reasoning*: to see that truth-telling is dominant, an agent must trace through hypothetical scenarios about others’ behavior and verify that no deviation improves payoffs in any case. The OSP version restructures the game so that the dominant strategy is *obviously* optimal: at every information set, the best outcome from following the strategy weakly dominates the worst outcome from any deviation, without requiring any reasoning about what others might do (Li, 2017). Section 5 established that LLM agents deviate from truthful play in sealed-bid strategy-proof formats. Here we ask how much of that deviation the extensive form itself removes, across all four model families.

The human-anchored APV cell. Clock auctions are dynamic formats in which the price rises incrementally and bidders decide whether to stay in or drop out at each price point. Although the sealed-bid second-price format and the ascending clock share the same dominant strategy, human participants behave quite differently across them: a robust pattern in the experimental literature is that bidders do much better in clock-style auctions than in sealed-bid auctions. Li (2017) provides an argument for why clock auctions are easier to recognize as strategy-proof, formalized as obvious strategy-proofness, and Breitmoser and Schweighofer-Kodritsch (2022) provide additional evidence that clock auctions improve truthful play, with behavior under the open clock (AC, where dropouts are observed) slightly better than under the closed clock (AC-B).

We evaluate the clock lever in two purpose-labeled cells (Table 2). The first is an affiliated private values (APV) environment chosen for comparability with the human evidence of Li (2017) and Breitmoser and Schweighofer-Kodritsch (2022). While affiliation could in principle change incentives through correlation in values, the three formats we consider are strategically equivalent: in the second-price sealed-bid (SP), the open ascending clock (AC), and the closed clock (AC-B), it remains a dominant strategy to bid—or drop out at—one’s value. Reconstructed human bidders deviate substantially less from truthful bidding in the clock formats than in sealed-bid SP: SMAD is 3.54% in AC and 5.83% in AC-B, compared to 9.31% in SP (Figure 1). For GPT-4o the clock advantage is even more pronounced. In the APV clock auctions GPT-4o achieves near-truthful play, with SMAD falling to 0.81% in AC and 0.47% in AC-B (a 77%–92% reduction relative to the human clock baselines), whereas in sealed-bid SP it remains at 10.73%—close to the human level of 9.31%. Breaking the decision into a sequence of simple threshold choices (“stay iff the price is below value”) yields an order-of-magnitude improvement over the one-shot sealed-bid interface, even though the dominant-strategy benchmark is identical.

The direction of the residual mistakes matters as much as their size. Figure 2 decomposes bidding errors by direction using a $\pm 2\%$ tolerance band around value (dropping near-equal bids and classifying the remainder as under- vs. over-bids). Despite comparable aggregate deviations in sealed-bid SP-APV (9.31% vs. 10.73% SMAD), the nature of the deviations differs strikingly: human bids are predominantly *above* value (64.7% overbidding vs. 35.3% underbidding), while GPT-4o bids are predominantly *below* value (92.4% underbidding vs. 7.6% overbidding)—humans exhibit competitive overbidding even under an incentive-compatible rule, while GPT-4o is overly conservative in the sealed-bid interface. The clock aligns not only the magnitude but the *shape* of behavior: in AC, humans and GPT-4o display an almost perfectly

symmetric under/over split (49.2/50.8 vs. 50/50), with GPT-4o concentrating far more mass near truthful bidding (71.7% within $\pm 2\%$ vs. 26.8% for humans). We return to what this direction reversal does—and does not—imply for the ranking in Section 7.

The four-model IPV cell. The second cell runs the same lever in the independent private values (IPV) environment of Table 2, across all four model families; its purpose is identification rather than human comparability. In the SPSB baseline (Figure 3, left column, gray), all four models deviate from truthful bidding, with mean deviations of $\mu = -0.49$ (Claude), -1.63 (Gemini), -3.28 (GPT-4o), and -5.27 (Gemma), for a cross-model mean of $-\$2.67$; these are the dedicated SPSB cells of the baseline declaration above. The negative sign indicates systematic *underbidding*—the mirror image of the human overbidding documented in Section 5. The distributions also exhibit long tails: some bids deviate by $\$10$ – 20 from value, cases in which the model substantially misplays the auction.

The closed ascending clock corrects deviations toward truthful play in every model. Mean deviations compress to $\mu = -0.05$ (Claude), -1.24 (Gemini), -0.45 (GPT-4o), and -0.33 (Gemma)—a cross-model mean of roughly $-\$0.5$ —and SMAD falls to 1.4%, 5.2%, 1.9%, and 1.9% against pooled-axis baselines of 9.5%, 11.7%, 11.4%, and 20.7%. The shift is significant for every model (Welch $p < 0.001$ in all four cells), with effect sizes up to Cohen’s $d = 1.00$ (Gemma). The improvement is most dramatic for the models with the worst SPSB performance: Gemma’s mean deviation falls from -5.27 to -0.33 , a near-complete correction. Beyond the mean shifts, the clock *eliminates the long tails* visible in the SPSB distributions: the catastrophic deviations effectively disappear. This variance reduction may be as practically important as the mean correction—OSP does not merely improve average play, it also removes the worst-case failures. Because this cell is run under IPV, the improvement cannot be informational.¹¹ Two provenance notes complete this cell’s accounting. The environment audit (plan E2) found that the legacy `ascending_clock_closed` runs pair an affiliated-values *description* with independent private-value *draws* (the configuration’s value model sets the common component to zero), so the informational channel was already absent in the data—but the prompt and the environment disagreed. A dedicated clean IPV cell (`ascending_clock_ipv_closed`: IPV description, IPV draws, $\$0.5$ tick, seed 1299, $K = 50$) now exists for GPT-4o and Gemma and reproduces the result exactly—SMAD 1.5% (GPT-4o) and 1.8% (Gemma) against sealed baselines of 12–27%—so the identification claim does not rest on the mislabeled cells. The remaining IPV-clock cells for Claude 3.5 Haiku and Gemini 2.0 Flash await native-API access (both model snapshots have been retired from the aggregator used for this draft’s re-runs); given the exact replication in the two available families and the direction of every legacy cell, we do not expect the remaining two cells to change the rung.

The bridge: which margin does the work? In summary, LLMs deviate from dominant-strategy play in the sealed-bid strategy-proof auction, and the OSP ascending clock substantially corrects these deviations. As a design lesson, however, the extensive-form result is coarse, because OSP changes several things at once: it changes the game tree, it coarsens the action space to simple binary or menu choices, and it makes the safety property of truth-telling salient at every decision point. The clock eliminates the need for contingent reasoning, reduces each decision to a stay/exit choice, and makes the connection between action and outcome immediate. Which of these simplifications actually drives the improvement? The theory does not need to answer this question—any OSP implementation bundles them—but a designer does, because the margins have very different costs: re-engineering the extensive form of a deployed mechanism is expensive, while

¹¹In a common- or affiliated-values setting, the clock format could improve play by revealing information about others’ values through their dropout decisions. Under IPV, no such information exists, so the improvement must be purely cognitive.

changing its description, or the reasoning it invites, is nearly free. The next two subsections decompose the bundle with presentation-only levers that hold the sealed-bid extensive form fixed: descriptions of the mechanism (family B, Section 6.2) and reasoning scaffolds appended to it (family C, Section 6.3).

6.2 Descriptions: Restating the Rules versus Exposing the Safety Invariant

Family B holds the mechanism fixed and varies only how it is described. Recent studies ask whether behavior in strategy-proof mechanisms depends on how the mechanism is *presented*, holding the underlying incentives fixed. Gonczarowski et al. (2023) propose strategyproofness-exposing descriptions, including an extensive-form *menu* description that compresses the dominant-strategy logic of a second-price auction into a short menu statement intended to make strategy-proofness salient; in their laboratory experiments, however, this menu framing does not meaningfully change human bidding relative to the traditional SPSB description, and Gonczarowski et al. (2024) likewise find that menu-based explanations of DA improve measured *understanding* while average behavioral effects remain modest. In contrast, Breitmoser and Schweighofer-Kodritsch (2022) show that *reframing* SPSB as a dynamic ascending clock substantially increases truthful play by human subjects. The taxonomy of Section 3 accordingly splits the family into three sub-levers—menu restatement (B1), safety-exposing descriptions (B2), and clock-framing (B3)—and they turn out to behave very differently.

Menu restatement (B1): null on average, sign-mixed underneath. The menu restatement has no consistent effect on bidding. Across the four models, its effects are small and sign-mixed relative to each model’s pooled axis baseline (Figure 5, top row): Gemma is a clean null (mean deviation -4.50 vs. baseline -4.54 ; Welch $p = 0.94$), Claude moves slightly *away* from truthful play ($+0.48$ to $+1.37$, $\Delta = +0.88$ above value, $p = 0.042$, $d = 0.24$), GPT-4o moves *away* as well (-2.78 to -4.10 , $\Delta = -1.32$, $p = 0.003$, $d = -0.37$), and Gemini alone moves *toward* truthful bidding (-2.88 to -1.13 , $\Delta = +1.75$, $p < 0.001$, $d = 0.50$). Pooling across cells, the restatement is a null on average: a two-sided sign test of the B1 cells against their baselines cannot reject no effect ($p = 0.73$; Section 6.4). The pattern also survives payment-rule variation: in GPT-4o’s first- and third-price variants of the same grid, the menu cells are indistinguishable from their baselines ($p = 0.63$ and 0.51). We therefore read the menu cell as consistent with the human null *on average* (Gonczarowski et al., 2023, 2024), with the caveat that the average conceals genuine model-level heterogeneity—a point the human experiments, run on a single population, cannot speak to. The menu family turns out to be more interesting in the supplementary matching experiment, where it splits into two sub-variants with opposite effects; we return to this in Section 6.4.

Clock-framing (B3): describing the bid as a clock exit. The clock-framing description keeps the one-shot sealed-bid elicitation—the agent still submits a single number—but describes that number as the threshold at which an automatic proxy would exit an ascending clock on the bidder’s behalf. This is a description, not a mechanism change: no prices are posted, no dropouts occur, and no information is revealed; it should not be conflated with the AC/AC-B formats of Section 6.1. The two rungs are empirically distinct cells in the merged dataset: the clock-framing cells record exactly one sealed bid per bidder per auction, while the true clock cells record per-tick stay/exit decisions (Appendix F). Against the corrected pooled baselines (Section 6.4), clock-framing moves play significantly in every model family but not always in the same direction (Figure 5, bottom row): SMAD falls from 12.1% to 6.3% for GPT-4o and from 26.8% to 13.8% for Gemma—each roughly halving its deviation—and from 8.3% to 7.2% for Claude (all Welch $p < 0.001$), while Gemini moves the *wrong* way, from 5.1% to 9.5% ($p < 0.001$). The sign agrees with the human evidence that clock reframing increases truthful play (Breitmoser and Schweighofer-Kodritsch, 2022) in three

of the four families, and the exception is coherent rather than noisy: Gemini is also the one model that the two-stage clock-exit variant below sharply harms. Pooled, the rung is modest and dispersed—median $\rho = +0.30$ (95% CI $[-0.77, +0.52]$, the width driven by the Gemini reversal)—recovering, where it works, a third to a half of the gap to truthful play, where the true extensive-form clock recovers nearly all of it (SMAD 1.4–5.2%; Section 6.1). Words that borrow the clock’s framing can help, and for some models help substantially; the clock itself helps every model, and helps far more.

The rung comes with a measured warning about paraphrase sensitivity. A provenance audit of the legacy run grid (documented in the replication package) uncovered a *second* clock-framing text that had been mislabeled as an axis-2 baseline cell: a two-stage description that presents the sealed bid as an “exit price” at which an ascending price clock would remove the bidder. Its manipulation check passes—clock-and-exit language appears in 14.3% of treated traces against 0% in the corrected baselines (wild-cluster $p = 0.002$)—but its behavioral effects, unlike the canonical clock-framing text above, are violently heterogeneous: it nearly fixes Gemma (mean deviation -6.53 to -0.59), sharply *backfires* for Gemini (-1.25 to -6.12), and leaves Claude and GPT-4o unmoved, with GPT-4o inert in language as well as bids (zero echo of the framing), so that the pooled effect is a null that conceals offsetting model-level responses ($p = 0.88$ pooled). The two texts agree on one regularity—both harm Gemini, the family most reactive to clock vocabulary—but otherwise produce different model-level profiles from the same clock analogy: the canonical framing helps three families, the two-stage variant helps only Gemma. This is direct evidence that description effects are sensitive to wording in a way that extensive-form changes are not (the true clock helps every model; Section 6.1), and a caution against reading any single description cell, including ours, as *the* effect of its family.

Safety-exposing descriptions (B2): stating the invariant. The safety-exposing sub-family does not restate the rules at all; it states the structural property of the mechanism that makes truth-telling safe. The key step in Vickrey (1961)’s original argument for the strategy-proofness of the second-price auction concerns being *pivotal*: your bid determines only *that* you win, not *what* you pay. This inspires our *Payoff Safety* intervention for the auction setting: “Think of your bid as setting your maximum price. Your bid determines IF you win, not WHAT you pay. If you win, you pay what others bid, not what you bid.” Its analogue in the supplementary matching experiment, *Rejection Safety*, states the property that made deferred acceptance preferable to the Boston mechanism in practice (Abdulkadiroğlu and Sönmez, 2003): “Rejections only redirect, never eliminate. Your ranking determines the *order* in which you are considered, not *whether* you are considered.” Neither intervention scaffolds a particular mode of reasoning: they do not ask the agent to trace through contingencies, simulate the algorithm, or reason about opponents. They simply state the key property of the mechanism that makes it strategy-proof.

In auctions, *Payoff Safety* reduces the mean deviation from $\mu = -2.67$ to -1.53 (Figure 4), with the deviation shrinking in magnitude for every model: Claude from -0.49 to $+0.30$, Gemini from -1.63 to -0.38 , GPT-4o from -3.28 to -1.88 , and Gemma from -5.27 to -4.17 .¹² Against the axis-specific baselines the improvement is significant for Gemini ($p < 10^{-30}$, $d = 1.85$) and GPT-4o (Mann–Whitney $p < 10^{-5}$); for Claude the mean effect is null while dispersion halves (SMAD 12.0% to 4.8%); Gemma is the one negative mark, moving away from its axis-2 baseline—though that baseline is anomalously low relative to Gemma’s other baselines, a caveat we quantify in Section 6.4. For comparison, the ascending clock achieves $\mu \approx -0.5$ (Section 6.1); the description closes roughly half the gap.

That a brief description of a mechanism’s incentive properties can nearly replicate OSP’s effect—without

¹²The Claude endpoint is $+0.30$ (overbidding) in the raw logs; the ES manuscript reported it as -0.30 . The magnitude is identical and the qualitative claim—Claude ends closest to truthful—is unchanged, but the sign correction means Claude’s residual error flips from under- to over-bidding under this description.

changing the extensive form—is a surprising result. It suggests that the cognitive barrier for LLMs in non-OSP mechanisms is not in *computing* the dominant strategy given the game tree, but in understanding the mechanism well enough to *see that truth-telling is safe*. OSP mechanisms resolve both problems simultaneously: they simplify the game tree and make the safety guarantee transparent. Our results suggest it is the latter that does most of the work, at least for LLMs.

Descriptions, then, are two levers, not one—and the split is not between “more text” and “less text” but between what the text *asserts*. Restating the rules, even in the theoretically grounded menu form, is null on average and sign-mixed underneath; exposing the safety invariant recovers most of what full OSP implementation delivers, in every model whose baseline leaves room to improve. The menu cell still functions as a placebo in the aggregate—not every additional sentence about the mechanism moves behavior, so the large effects of B2 cannot be attributed to generic prompt sensitivity—but it is a noisier placebo than a single-population experiment would suggest. What the effective descriptions share, and the restatement lacks, is that they *assert* the invariant that makes truth-telling safe rather than leaving the agent to derive it. The supplementary matching experiment sharpens this reading to a within-family contrast, discussed in Section 6.4: a menu description that adds the invariance statement behaves like a safety description, while a menu description that merely restates the computation backfires.

6.3 Reasoning Scaffolds: Contingencies Help, Strategizing Hurts

Family C holds both the mechanism and its rule description fixed and appends a reasoning scaffold. Following Li (2024), who decomposes the cognitive demands of mechanisms into three dimensions, we scaffold thinking *contingently* about unobserved moves by other players (C1), *planning* for one’s own future moves (C2), and reasoning about other players’ *beliefs* (C3). Each scaffold directs reasoning without revealing the dominant strategy; each modifies only the rule-explanation block of the prompt while the mechanism, value distributions, and response format are unchanged (full prompt texts in Appendix F). Prospect-theory-inspired preference framings (family D) are reported in Appendix E; they form the ranking’s bottom rungs.

6.3.1 Contingent reasoning (C1): the payoff tree

Li (2024) identifies contingent reasoning as the core cognitive demand of non-OSP mechanisms. To see that truth-telling is dominant in SPSB, a bidder must consider each possible profile of opponent bids and check that no deviation improves payoffs in any case—a “case-by-case comparison, calculating payoffs for each profile of opponent bids” (Li, 2024). The ascending clock eliminates this demand: at each price, the worst case from following the dominant strategy weakly dominates the best case from any deviation, without reference to opponents. The question is whether scaffolding contingent reasoning—without changing the extensive form—can partially recover these gains. Our *Payoff Tree* intervention presents the agent with the explicit decision tree of outcomes: “PATH A: your bid exceeds others $\rightarrow$ you WIN, and pay the second-highest bid. PATH B: your bid falls below $\rightarrow$ you LOSE, and pay nothing. Key insight: your bid determines which path you are on, not the payment in Path A.”

In auctions, *Payoff Tree* reduces the mean deviation from $\mu = -2.67$ to $\mu = -1.13$, roughly halving the distance from truthful play (Figure 6). The improvement is consistent across models: GPT-4o moves from -3.28 to -0.86 , Gemma from -5.27 to -2.34 , and Gemini from -1.63 to -0.53 ; Claude, already close to truthful (-0.49), is the one model showing null improvement. The scaffold closes roughly half the gap to the OSP benchmark (clock $\mu \approx -0.5$) without changing the extensive form.

Laying out the payoff tree does not tell models *what* to do; it tells them *what happens* under each action. This is precisely the cognitive gap between OSP and non-OSP mechanisms: in OSP mechanisms

the extensive form makes the payoff-relevant contingencies visible at each decision point, whereas in non-OSP mechanisms the agent must construct them herself. The *Payoff Tree* scaffold does this construction for the agent, and the consistent improvement across models suggests that the difficulty of constructing payoff-relevant contingencies is a real constraint that binds on LLM agents.

6.3.2 Belief formation (C3): making opponents salient

The third axis is belief formation. Börgers and Li (2019) introduce *strategic simplicity*: a mechanism is strategically simple if optimal play does not depend on higher-order beliefs about what opponents believe. Strategy-proof mechanisms are trivially strategically simple—the dominant strategy is independent of beliefs entirely. Scaffolding belief formation should therefore be irrelevant at best. But if models are already treating these mechanisms as Bayesian games, attempting to best-respond to perceived competition rather than playing the dominant strategy, then making beliefs more salient could amplify mistakes. We test two levels of belief scaffolding: *First-Order Beliefs* (“What do you expect the other players to do? ... Does this affect what you should do?”) and *Second-Order Beliefs* (“What do the other players think YOU will do? ... And does that, in turn, affect what you should do?”).

Belief scaffolding almost uniformly worsens play (Figure 7). In auctions, *First-Order Beliefs* moves the cross-model mean deviation from $\mu = -2.67$ to -3.04 and *Second-Order Beliefs* to -3.47 : models prompted to think about opponents shade further below value, undercutting perceived competition. Interestingly, Gemini is almost completely immune to belief scaffolding, while Gemma is the most susceptible to this failure mode. These results suggest that LLM reasoning styles are not monolithic; the heterogeneity across model families in susceptibility to belief-based reasoning may itself be relevant to mechanism design when markets serve diverse artificial agents.

6.4 The Ranking Assembled

Table 5 assembles the preceding cells into the paper’s central object. For each lever we report the raw auction deviations and the preservation index ρ (Section 4.4), computed within model and pooled as the median across the available cells, with 95% confidence intervals from a double bootstrap that resamples cells and propagates each cell’s own bootstrap uncertainty (2,000 draws, seed 1299). Because ρ is computed against each model’s own baseline, the ranking is invariant to disputes about cross-model levels; it asks only how much of each model’s own deviation a lever removes.

The tier structure, tested. Our pre-committed reading of the data is a *tier structure*—a partial order, not a strict total order:

$$\underbrace{A}_{\substack{\text{extensive form;} \\ \text{ascending clock}}} \succ \underbrace{\{B2, B3, C1\}}_{\substack{\text{safety descriptions,} \\ \text{clock-framing, payoff tree}}} \succ \underbrace{\text{baseline} \approx B1}_{\text{menu restatement}} \succ \underbrace{C3}_{\substack{\text{belief} \\ \text{scaffolds}}} \succ \underbrace{D\text{-averse.}}_{\substack{\text{risk-averse} \\ \text{persona}}}$$

The concordance statistics now exist, and they support most—not all—of this structure. Treating each model as a judge ranking the levers, Kendall’s W is 0.70 in auctions ($p = 0.004$): the lever ordering is far from independent across models. (This statistic is computed on the corrected baselines described below; correcting the contaminated axis-2 “baseline” cell *raised* auction concordance from 0.43 to 0.70—the mislabeled treatment had been injecting noise into every axis-2 contrast.) Exact binomial sign tests on adjacent tiers confirm three of the boundaries: the extensive form beats the strong presentation tier in 7/8 cells ($p = 0.070$), the strong presentation tier beats the restatement tier in 8/8 ($p = 0.008$), and the

strategizing scaffolds beat the risk-averse persona in 8/8 ($p = 0.008$). The boundary that *fails* is baseline-vs-menu against the strategizing scaffolds: restatements beat lookahead and beliefs in only 4/8 cells ($p = 1.0$). The failure is informative rather than noise: the menu restatement is not a clean placebo tier but a sign-mixed family, null on average (two-sided placebo sign test against baseline: 3/8 cells above zero, $p = 0.73$) yet with genuine model-level backfires (GPT-4o’s auction menu cell, Welch $p = 0.003$) that land it below some strategizing cells.

The sharpened story: invariance content is the active ingredient. The matching-domain menu cells (new in this draft; Appendix H) resolve what the sign-mixed auction menu leaves ambiguous, by splitting the restatement family in two. A menu description that *includes the invariance statement*—“your ranking cannot change your obtainable set, only which school you get within it”—behaves like a safety-exposing description: pooled error falls from 4.2% to 1.8%, with significant improvements for the two weak-baseline models (Claude 5.8% to 2.1%, Mann–Whitney $p = 10^{-4}$; Gemini 7.6% to 0.3%, $p < 10^{-4}$) and nulls for the strong ones. A menu description that restates the *computation* without the invariance statement *worsens* play (pooled 4.2% to 6.9%; GPT-4o 1.0% to 4.2%, Gemma 2.6% to 13.7%). Together with the auction results, the pattern across all of family B is consistent: safety-exposing text (B2, pooled $\rho = +0.90$) and the invariance-carrying menu variant (+0.65) help; mechanical restatement (auction menu -0.22 , mechanics-only DA menu -1.67) is null-to-harmful. The active ingredient of an effective description is not restating the mechanism but asserting the invariant that makes truth-telling safe. The human experimental record, read through the same split, independently supports it: invariance-carrying descriptions improve human understanding and (modestly) behavior in matching mechanisms (Gonczarowski et al., 2024; Katuščák and Kittsteiner, 2024), while mechanics-only descriptions of a strategy-proof mechanism have been found to *reduce* truth-telling in the field (Guillén and Hakimov, 2018)—the same backfire our mechanics-only cell produces.

Within-tier order and an honest caveat. We deliberately do not order the safety description and the payoff tree within the strong tier: their pooled preservation indices ($\rho = +0.78$ and $+0.59$) have overlapping confidence intervals, and the payoff tree beats the safety description in only 3/4 of the auction cells, so the data do not support a strict order between them. Clock-framing, now estimated on all four models, is a reliable but modest member of the tier ($\rho = +0.29$ [$+0.17, +0.45$]). One caveat from the concordance analysis qualifies individual cells without moving any rung: what previously appeared as an anomalously good Gemma axis-2 baseline (SMAD 8.6% vs. 20.7% pooled across its axes) turns out to have a mechanical explanation: a provenance audit of the run templates shows the legacy “axis-2 baseline” cell was not a plain SPSB description at all but a two-stage sealed-bid-as-clock-exit description—a clock-framing variant that Gemma responds to strongly (Section 6.2). The pooled axis baseline used throughout therefore excludes that cell (axis-1 and axis-3 cells only), the axis-2 treatment contrasts are computed against the corrected pool, and the cell itself enters the description family as a treatment.

The frontier: the ranking compresses, and the failures change kind. The full lever grid now also exists for four frontier models (gpt-5-mini, gpt-5, Claude Sonnet 5, Gemini 2.5 Flash; Appendix D), and it turns the frontier scope condition from an extrapolation into a measurement. Three facts summarize it. First, *the ranking’s positive rungs have nothing left to do*: all four models bid essentially exactly truthfully at every sealed-bid baseline (mean deviations 0.00–0.30, SMAD 0–2.3%), so the safety descriptions, payoff tree, and belief scaffolds leave play unchanged—the preservation index is undefined as $D(\text{baseline}) \rightarrow 0$, exactly the compression anticipated in Section 8. Second, *the backfire rungs retain their teeth, heterogeneously*: the menu restatement—inert to slightly harmful in the incumbent grid—collapses Claude Sonnet 5 from exact

truthfulness to a mean deviation of $-\$8.71$ (SMAD 35.5%, Cohen’s $d = -3.8$, $p < 0.001$) and degrades Gemini 2.5 Flash (SMAD 2.3% $\rightarrow$ 9.1%, $p < 0.001$), while leaving gpt-5 and gpt-5-mini exactly truthful. The traces say exactly what broke: under the menu wording, Claude Sonnet 5 announces that “the standard first-price auction equilibrium bid is $(n - 1)/n$ of my value” and executes that formula flawlessly—its $-\$8.71$ is almost exactly the $-\frac{1}{3} \mathbb{E}[v] \approx -\8.17 the first-price rule implies—despite the prompt stating that the winner pays the highest *rival* bid. The unfamiliar wording did not degrade its play into noise; it re-classified the game and retrieved the wrong textbook solution, executed perfectly. A description with no incentive content can still break an otherwise perfect strategic reasoner, and which reasoner it breaks is not predictable from baseline capability. Third, what first appears as an *inversion of the top rung*—the ascending clock, which improves every incumbent model, makes Claude Sonnet 5 *worse* than its own sealed baseline (SMAD 0.27% sealed vs. 8.6% clock, $p < 0.001$, with 75% of its exits more than a dollar early)—turns out on inspection to be the same retrieval failure as the menu cell, not a property of the extensive form. The clock cells inherit the legacy affiliated-values *description* (Section 6.1), and Claude Sonnet 5’s exit rationales explicitly invoke the winner’s curse (“others’ persistence signals possibly similar high values, risking overpayment”), a script that is correct in common-value clocks and irrelevant here, where the bidder knows its own value. Rerunning the same model on the clean IPV-described clock restores near-perfect play: SMAD 2.0%, early exits 3%, and marginal-reasoning traces (“price equals my value; drop”). One sentence of affiliation language costs an otherwise exactly-truthful model eight SMAD points by activating the wrong playbook—while the same sentence leaves the incumbent models’ clock gains intact, because they have no winner’s-curse script to misfire. (gpt-5 and gpt-5-mini shift by $+0.3$ – 0.6 on the clock, mostly tick granularity, with clean marginal reasoning in their traces.) The design lesson for mixed fleets is asymmetric: simplifying levers are (at worst) wasted on frontier agents, but presentation choices can still harm them, so the practitioner guidance of Section 8—expose invariants, avoid gratuitous restatements—remains binding even where the bounded-rationality regime has closed. (The gpt-5 column carries reduced cell sizes, $K = 5$ – 50 : the model frequently omits the required response tags, and unparseable repetitions are dropped rather than imputed; Appendix D reports per-cell n .)

What the ranking localizes. Read from the bottom up, the assembled ranking localizes the constraint that simple mechanism design relieves. Scaffolding strategizing—beliefs, lookahead—makes play strictly worse, so the problem is not too little strategic reasoning. Mechanically restating the rules is null on average and harmful where it displaces the safety-relevant content, so the problem is not raw comprehension bandwidth either. What works, wherever it appears and in whatever syntactic form—the safety one-liners, the invariance-carrying menu variant, the textbook strategy-proofness statement—is text that asserts the invariant that makes truth-telling safe; and full OSP implementation, which makes that invariant visible at every decision point, delivers the remainder. As Section 6.2 put it: the binding cognitive barrier in non-OSP mechanisms is not *computing* the dominant strategy given the game tree, but understanding the mechanism well enough to *see that truth-telling is safe*. The ordering, moreover, parallels the theoretical hierarchy from which the taxonomy was derived *ex ante* (Section 3): the levers that theory says relax binding cognitive constraints sit at the top, and the levers that theory says should be irrelevant for strategy-proof mechanisms are, for agents who do not fully trust that irrelevance, actively harmful. Section 6.5 adds a final, mechanism-level piece of evidence: the effective descriptions move bids *without* changing what the agents say they are doing. Section 7 then closes the loop, anchoring each rung to the human experimental record where one exists and issuing the unanchored rungs as predictions.

Tier	Lever	Family	Theory tag	Deviation D			Human anchor
				Auct. (\$)	DA (%)	Pooled ρ [95% CI]	
1	Extensive form (AC-B clock / iterative DA)	A	OSP (Li, 2017; Ashlagi and Gonczarowski, 2018)	-0.5	0.0	+0.97 [+0.83, +1.00]	AC +0.62 [+0.39, +0.79]; AC-B +0.37 [-0.12, +0.72]
2	Safety description	B2	SP-exposing (Gonczarowski et al., 2023)	-1.53	0.2	+0.78 [+0.39, +1.00]	— (lab prediction)
3*	Menu, invariance property (DA)	B1	menu + invariant	—	1.8	+0.65 [-0.52, +0.95]	✓ agrees (sign) (Gonczarowski et al., 2024; Katusiřák and Kittsteiner, 2024)
2	Payoff tree	C1	contingent reasoning (Li, 2024)	-1.13	1.7	+0.59 [+0.46, +0.68]	— (lab prediction)
2	Clock-framing	B3	OSP description (Breitmoser and Schweighofer-Kodritsch, 2022)	-1.93 [†]	—	+0.30 [-0.77, +0.52]	✓ agrees (sign, 3/4 models)
3	Baseline (SPSB / direct DA)	—	—	-2.67	4.2	0	✓ agrees
4	Forward planning (lookahead)	C2	one-step simplicity (Pycia and Troyan, 2023)	-1.65 [‡]	7.8 / 5.1	-0.00 [-2.12, +0.26]	— (lab prediction)
3	Menu restatement (auction)	B1	menu description (Gonczarowski et al., 2023)	-2.09 [†]	—	-0.14 [-1.00, +0.27]	✓ agrees on avg. (-0.07 [-0.12, -0.02])
4	Second-order beliefs	C3	strategic simplicity (Břrgers and Li, 2019)	-3.47	9.1	-0.35 [-1.36, +0.03]	— (lab prediction)
4	First-order beliefs	C3	strategic simplicity (Břrgers and Li, 2019)	-3.04	5.1	-0.41 [-0.88, -0.04]	— (lab prediction)
3*	Menu, mechanics only (DA)	B1	menu, computation restated	—	6.9	-1.67 [-5.01, +0.57]	✓ agrees (backfire) (Guillén and Hakimov, 2018)
5	Risk-averse persona	D	preference framing (Kahneman and Tversky, 1979)	App. E	App. E	-1.82 [-5.73, -0.72]	— (lab prediction)

Table 5: The simplicity-preservation ranking, rows sorted by pooled ρ . D = mean deviation from dominant-strategy play: bid — value in dollars for auctions (cross-model mean over GPT-4o, Claude 3.5 Haiku, Gemini 2.0 Flash, Gemma 27B), normalized Kendall- τ error in percent for the supplementary matching experiment (Appendix H; lookahead entries are one-/two-step). $\rho = 1 - D(\text{lever}) / D(\text{baseline})$ on the absolute-deviation scale, computed within model $\times$ domain against the baseline conventions of Section 6; pooled value = median over available cells, with double-bootstrap 95% CIs (2,000 draws, seed 1299). Tier = pre-committed partial order; within-tier order is not asserted. *New matching-domain menu cells assigned ex ante to the restatement family (tier 3); their measured positions straddle the ranking—the invariance-property variant performs like a tier-2 safety description, the mechanics-only variant like a tier-4 backfire (see text). [†]Intervention-grid cells tested against pooled-axis baselines (corrected pool: axis-1 and axis-3 cells only, cross-model mean -2.55; see the axis-2 provenance discussion above) rather than the dedicated SPSB cells (-2.67). [‡]The auction C2 cell is a backward-induction *framing* of the static sealed bid (SPSB has no true planning axis); it enters the pooled ρ alongside the DA lookahead cells. Human anchor: ✓ = a human experiment exists for this rung and its effect agrees in sign with the LLM effect (reconstructed human ρ [CI] shown where computable); — = no human experiment exists and the rung is issued as a lab prediction (Section 7).

6.5 What the Traces Say: Stated Reasoning and Realized Bids

Every bid in our experiments is accompanied by a short free-text plan in which the agent explains its choice (median 68 words). We analyze all 21,990 plans from the sealed-bid and clock cells with a transparent dictionary of eighteen keyword features—stated intent to shade below or bid above value, rehearsal of the second-price payment rule, dominance language, worst-case and profit-margin rhetoric, and explicit mechanism confusion (calling the auction “first-price”), among others (full regular expressions and per-cell prevalence in the replication package, `analysis/build_trace_features.py`).¹³

Two facts establish what the traces are. First, the modal plan applies *first-price* logic to the second-price auction: 53% of sealed-bid second-price traces state an intent to bid below value, 55% worry about “overpaying,” and 31% invoke a “profit margin” or “buffer,” while only 2% use dominance language and only 4% correctly rehearse that the winner pays the *second*-highest bid; Claude explicitly mislabels the mechanism “first-price” in 10% of its traces, a mistake no other model makes more than 0.1% of the time. Second, the traces are not cheap talk: among traces stating an intent to shade, 97.1% of bids fall below value (mean deviation $-\$1.84$); among traces stating an intent to bid above, 87.6% fall above ($+\$1.88$); and the largest errors come from the 4,541 traces that state no value-anchored intent at all (mean deviation $-\$4.84$), whose plans anchor on uninformative quantities such as “slightly above half my value.” Stated direction almost always matches the realized bid; the failure is in which plan is chosen. A pooled regression of $|b - v|$ on the dictionary features nonetheless explains only a modest share of the error variance (incremental $R^2 \approx 0.075$ over model and lever-family fixed effects); the regression table and robustness checks are in Appendix G.

Most important for interpreting the ranking, the levers of Sections 6.2–6.3 produce a complete *dissociation* between stated reasoning and behavior (Table 6): across every lever we test, language and bids never move together. *Payoff Safety* (B2) moves bids substantially while leaving stated reasoning untouched: the invariant it asserts (“your bid determines *if* you win, not *what* you pay”) is echoed in zero of its 600 treated traces, and payment-rule rehearsal is flat relative to baseline (2.8% treated vs. 3.9%, $p = 0.60$); the plans keep the same shading rhetoric and simply shrink the shade to pennies (“I will bid \$31.9 to avoid over-bidding,” value \$32). The *worst-case* scaffold (C1) does the opposite: it massively shifts stated reasoning (worst-case language rises from 2% to 43%) while bids barely move ($3.24 \rightarrow 3.07$, $p = 0.81$). The *Payoff Tree* is a second bids-only lever: it cuts mean $|b - v|$ from 3.20 to 1.29 while leaving payment-rule rehearsal exactly at its baseline rate (33.0% vs. 32.7%, $p = 0.90$)—an earlier reading on which the tree also moved rule-rehearsal language was an artifact of the contaminated axis-2 baseline cell documented in Section 6.4, whose clock narrative crowded out rule vocabulary. The *menu* restatement moves neither, serving as the placebo row. Because the first stage is null for every verbalized-understanding measure, a classical mediation decomposition bounds the share of Payoff Safety’s behavioral effect that could run through stated comprehension at roughly one third (95% upper bound 35%; point estimate ≈ 0), under assumptions the within-treatment association (rule-rehearsing traces err \$1.5 less, CI $[-2.15, -0.56]$, non-causal) cannot tighten. Stated comprehension and behavioral improvement are separable outcomes, and our effective descriptions improve the latter without the former—which is why we describe them as changing behavior, not stated understanding, and why the traces should not be read as faithful mediators of the lever effects. [TODO: LLM-judge strategy taxonomy (trace option 1): validate the keyword dictionary against a judged taxonomy on a 100-trace human-labeled set before the “2% dominance recognition” figure is made load-bearing.]

Finally, the traces speak to *why* comprehension fails. In the GPT-4o first-price and third-price variants of the same grid, the model deploys a nearly identical shading script—shading language in 70%/79%/72%

¹³The plans are stated strategies, not full chains of thought; all claims in this subsection concern what models *say*, with Table 6 showing that what they say tracks what they do.

of second-/first-/third-price traces and mean bids of 0.88/0.87/0.84 of value—even though the optimum differs sharply across formats (truthful in second-price; roughly $\frac{2}{3}v$ in our three-bidder first-price cell; above value in third-price). The same words justify underbidding where shading is optimal and where it is an error: the reasoning script is a property of the model, not of the mechanism it faces. [TODO: The cross-mechanism script-invariance result is currently GPT-4o-only (recovered V12 logs); run FPSB/TPSB cells for Claude, Gemini, and Gemma with the frozen dictionary before claiming generality.]

Lever (family)	Stated reasoning		Behavior	
	target language	Δ vs. baseline	mean $ b - v $	baseline
Payoff Safety (B2)	payment-rule rehearsal	3.9% $\rightarrow$ 2.8% ($p = 0.60$)	1.95 ($p = 0.015$)	3.20
Worst-case scaffold (C1)	worst-case rhetoric	2% $\rightarrow$ 43% ($p = 0.007$)	3.07 ($p = 0.81$)	3.24
Payoff Tree (C1)	rule rehearsal	33.0% $\rightarrow$ 32.7% ($p = 0.90$)	1.29 ($p = 0.017$)	3.20
Menu restatement (B1)	—	—	3.55 ($p = 0.66$)	3.20

Table 6: Dissociation between stated reasoning and bidding behavior (pooled over four models, sealed-bid second-price cells; wild-cluster bootstrap p -values clustered at the run level, with model-cluster permutation cross-checks and duplicate-trace robustness in `results/traces/mediation/`). No lever moves both columns: Payoff Safety and the Payoff Tree improve bids without changing what agents say; the worst-case scaffold changes what agents say without improving bids; the menu restatement moves neither. The baseline is the corrected pooled axis baseline (axis-1 and axis-3 cells only; the legacy axis-2 “baseline” cell was itself a clock-framed description and is analyzed as a treatment in Section 6.2); under the dedicated-SPSB convention the Payoff Safety contrast is the $-2.67 \rightarrow -1.53$ of Section 6.2. No language entry is reported for the menu row because the menu prompt replaces the “second-highest” vocabulary wholesale, so rule-rehearsal prevalence is zero by construction (prompt echo, not comprehension).

7 Ordinal Agreement and the Limits of Cardinal Calibration

The ranking of Section 6 is estimated on populations of LLM agents. This section asks, in two steps, what that ranking is worth to a designer whose participants are human—and, symmetrically, what five decades of human experimental evidence contribute to the credibility of the ranking. The first step is ordinal: on every rung of the ranking where a human benchmark exists, the sign of the treatment effect agrees, even though the two populations make opposite baseline mistakes—a fact we argue converts the validity map’s most awkward cell into the strongest available evidence that the ranking measures properties of *mechanisms* rather than of any particular reasoner. The second step is cardinal, and its answer is negative in an instructive way: searching a grid of the tuning knobs available to a practitioner—sampling temperature, risk personas, framing manipulations, rule explanations—we find that no single setting produces cardinal agreement with human play across mechanisms. Level agreement is achievable mechanism by mechanism; direction agreement is achievable with exactly one knob that breaks the level agreement elsewhere; and the best tuning for one model family overshoots badly on another. Together the two steps justify the discipline imposed since Section 4.5: the testbed is an instrument for signs, orderings, and comparative statics, and the boundary of that instrument is now measured rather than assumed.

7.1 Rung-by-Rung Ordinal Agreement

We walk down the ladder of Table 5, rung by rung, comparing each lever’s effect on LLM play with its effect on human subjects where a human experiment exists. The human-anchored rungs are marked as diamonds in Figure 8; human quantities below are computed from the moment-matched reconstructions of Section 4.5 and anchored to the original published findings.

Extensive form (Family A). Clock implementations help humans. Li (2017) showed experimentally that ascending clock auctions, which are obviously strategy-proof, produce markedly more dominant-strategy play than the strategically equivalent sealed-bid format, and Breitmoser and Schweighofer-Kodritsch (2022) replicate and extend this pattern; in our reconstruction, human SMAD falls from 9.31% under the sealed-bid second-price format to 3.54% under AC and 5.83% under AC-B, for a preservation index of +0.62 [+0.39, +0.79] (AC) and +0.37 [−0.12, +0.72] (AC-B). The same lever helps every model we test, and helps it more: in the harmonized four-model grid, the closed clock cuts SMAD to 1.4–5.2% from sealed-bid baselines of 9.5–20.7% ($p < 0.001$ for all four models; Cohen’s d up to 1.00; Section 6.1).

Clock-framing (B3). Breitmoser and Schweighofer-Kodritsch (2022) show that *describing* the sealed-bid second-price auction as an automatic exit threshold in an ascending clock—changing words, not the mechanism—substantially increases truthful bidding among human subjects. The same one-paragraph description now has four-model evidence, and it agrees with the human finding in three of the four families: SMAD falls from 12.1% to 6.3% for GPT-4o, 26.8% to 13.8% for Gemma, and 8.3% to 7.2% for Claude (each Welch $p < 0.001$), while Gemini moves away from truthful play (5.1% to 9.5%, $p < 0.001$)—the one family that also reacts badly to a second clock-framing paraphrase (Section 6.2). The pooled preservation index is +0.30 [−0.77, +0.52], the wide interval reflecting the Gemini reversal. The anchor therefore holds at the majority-of-families level rather than universally, and the magnitude, where positive, remains well below the true clock’s.

Menu restatement (B1). In the laboratory experiments of Gonczarowski et al. (2023), a menu description that compresses the dominant-strategy logic of the second-price auction does not meaningfully change human bidding (reconstructed $\rho = -0.07$ [−0.12, −0.02]). Across our four model families the restatement has no consistent effect either, but the agreement is an agreement *on average* rather than cell by cell: against the corrected baselines, the restatement moves Gemma *toward* truthful play (+\$2.03, $p < 0.001$), moves GPT-4o away (−\$1.16, Welch $p = 0.013$, Mann–Whitney $p = 0.54$), moves Claude marginally away (+\$0.82, Welch $p = 0.064$, Mann–Whitney $p < 0.001$), and leaves Gemini essentially unchanged ($p = 0.56$). Pooling, the effects are small and sign-mixed, and a two-sided sign test cannot reject a zero median ($p = 0.73$; Section 6.2)—in sharp contrast to the uniformly signed effects of safety-exposing text. We therefore state this rung’s anchor precisely: LLM play is consistent with the human null *on average*, with model-level heterogeneity that the single human point estimate cannot adjudicate. The rung retains its placebo value in one direction—the testbed does not manufacture a reliable gain from restating the mechanics—but it is not the clean, uniformly inert tier the human evidence might suggest.

Baseline. Both populations deviate under the direct mechanisms. Human subjects systematically overbid in sealed-bid second-price auctions (Kagel and Levin, 1993; Kagel, 1995) and submit dominated rank-order lists in deployed matching mechanisms (Rees-Jones, 2018); all four LLM families underbid in SPSB (Section 5, Section 6), and all four misreport under direct DA in the supplementary matching experiment (Appendix H).

In sum: the clock helps both populations, the clock-framing description helps both in three of four model families, the menu restatement moves neither on average, and the baseline direct mechanisms fail both; in the matching domain, invariance-carrying descriptions help and mechanics-only descriptions backfire in both populations as well (Appendix H and the anchors discussed below). Every rung with a human benchmark agrees in sign at the pooled level; on two description rungs (clock-framing, menu) the agreement is over model-heterogeneous cells, and we state the heterogeneity rather than smooth it.

7.2 Opposite Failures, Common Levers: Substrate Independence

The validity map’s red cell is the direction reversal: humans overbid in SPSB-IPV—67.2% of all bids exceed value, and 92.2% of the bids that deviate from value deviate upward—while GPT-4o’s deviations are overwhelmingly (81.6%) underbids; the same reversal appears in the APV sealed-bid cell (64.7% human overbidding against 92.4% GPT-4o underbidding).¹⁴ Read as a test of the proxy hypothesis—“LLMs reproduce human behavior”—this cell is a failure, and we declared it as such in Section 5. Read against the ranking, it becomes the paper’s sharpest identification result. Two populations fail in *opposite* directions, yet they are corrected by the *same* levers, in the same order: the clock repairs human overbidding and LLM underbidding alike, the clock-framing description does the same, and the menu restatement repairs neither. If the deviations were driven by population-specific psychology—a joy of winning or spite pushing humans above value, a trained conservatism pulling LLMs below it—there would be no reason for a shared correction structure. The parsimonious reading is that the binding constraint lives in the strategic problem itself: in the difficulty of verifying, from a direct mechanism’s description, that truth-telling is safe (Li, 2017; Oprea, 2024). What fills the gap left by that missing verification is population-specific—competitive overbidding in humans, conservative shading in LLMs—but the gap, and the levers that close it, are shared. This is the sense in which the lever ordering is substrate-independent, echoing the hypothesis structure of Section 3.

The reversal also rules out a deflationary explanation of the rung-by-rung agreement. If LLMs merely reproduced human experimental regularities memorized from training corpora, agreement on the levers would be uninformative—but a population imitating the published human data would *overbid* in second-price auctions, and ours does not. The agreement on which levers work is therefore not inherited from the human literature; it is re-derived by a different reasoner failing in a different way.

Partial anchors and an out-of-sample prediction. The safety-description tier—the cheapest lever in the designer’s toolkit and, on LLM populations, among the most powerful—is where the human record is thinnest, but it is no longer empty, and the recent evidence splits along exactly the line our decomposition draws. On the invariant-exposing side, the *Menu-SP* treatment of Gonczarowski et al. (2024)—a stripped-down statement that a participant’s menu of obtainable outcomes is computed from everyone else’s reports—produced the largest gain in measured understanding of strategy-proofness in their five-treatment design (71% versus 54–62% elsewhere) and a directionally consistent but modest gain in truthful play (59% versus 48–53%); Katuščák and Kittsteiner (2024) find that an analogous description of top trading cycles—isolating what does and does not depend on one’s own report—increases truthful play outright. On the mechanics side, Guillén and Hakimov (2018) find that describing a strategy-proof mechanism’s *mechanics* significantly reduces truth-telling in the field—the same backfire our mechanics-only menu cell produces in DA (Appendix H), so that rung enters Table 5 as anchored rather than predicted. What remains genuinely

¹⁴Direction shares depend on the convention. Throughout this section we use the repo-canonical convention—the share of bids strictly above value among *all* bids—under which the human SPSB overbid share is 67.2%; conditioning on bids that deviate from value gives 92.2%. The directional decomposition of Section 5, which removes a $\pm 2\%$ band around the benchmark before classifying, yields slightly different deviator-conditional shares; no conclusion in this paper turns on the choice.

untested in human subjects is the auction half of the tier and the magnitude question. No human experiment has tested a bare invariant statement in an auction—“your bid determines whether you win, not what you pay”—as distinct from direct advice to bid truthfully, which Masuda et al. (2022) show is powerful (a ~ 24 -percentage-point net gain in the Vickrey auction) but which reveals optimal play rather than exposing why it is safe. And where the DA evidence exists, it moves understanding much more than behavior, whereas our LLM populations show large *behavioral* gains from the same content. We therefore predict, concretely: in matched human experiments, (i) a Payoff Safety description will shift human SPSB bid distributions toward value where the menu framing of Gonczarowski et al. (2023) did not, and (ii) the behavioral gain from description treatments will concentrate in invariance-stating text and be absent-to-negative for mechanics-restating text, in both domains. The prediction is cheap to test and falsifiable in both directions: the synthetic version of each treatment cell cost roughly \$10 to run, and an incentivized human counterpart is a single-session design at the per-subject payment scale of existing description experiments—on the order of \$15–26 per subject (Gonczarowski et al., 2024)—rather than the substantially larger scale of a dedicated multi-hundred-subject laboratory study (Table 8).

7.3 From Ordinal to Cardinal: Is There a Consistent Tuning?

Everything so far is ordinal, and deliberately so: where levels can be compared, they diverge. LLMs *over-comply* in exactly the formats where the levers work best—in the clock implementations, model SMAD of 0.5–0.8% sits an order of magnitude below the human 3.5–5.8%—while in third-price auctions they sit far above any human benchmark. A natural response, and an increasingly common proposal in the literature on LLMs as experimental proxies, is to *tune*: adjust temperature, personas, or framings until the synthetic population matches the human one cardinally, then read levels off the tuned simulator. This subsection asks whether any such tuning exists in our environment, using every knob for which we have data: sampling temperature ($T \in \{0.1, 0.5, 1.0\}$; GPT-4o, eight mechanisms), risk personas (averse/neutral/seeking; second price for all four models, first and third price for GPT-4o), prospect-theoretic framings (loss/gain/mixed/endowment/WTB–WTP), and rule explanation on/off—a grid of 95 mechanism $\times$ knob cells. Distance to the human benchmarks is measured on three axes: *level* (the gap in SMAD), *direction* (the gap in the share of bids above value), and *shape* (Wasserstein-1 distance between the scaled deviation distributions and the moment-matched human reconstructions). Table 7 reports the core sealed-bid grid.

Level: the best knob is mechanism-specific. On the level axis alone, cardinal agreement is often achievable—but with a different knob in every format. In FPSB, temperature 1.0 closes the gap almost exactly (SMAD 25.50 [23.49, 27.69] against a human 24.76; untuned anchor 28.11). In SPSB, the WTB–WTP frame is best (7.24 [6.02, 8.56] against 5.65; -3.57 SMAD relative to its own baseline, Welch $p = 0.005$), while both temperature deviations from the anchor make play *worse*. In the affiliated-values sealed-bid format the untuned anchor is already the best available cell (10.95 against 9.31). The clock formats invert the problem: LLMs are *too good*—SMAD 0.83 (AC) and 0.48 (AC-B) against human targets of 3.54 and 5.83—so the only knob that moves them toward humans is raising temperature to 1.0, which works purely by injecting noise into near-perfect play.¹⁵ And third-price auctions are not tunable at all: the best third-price cell anywhere in the grid has SMAD 44.1 (temperature 0.1, $N=5$) against a human 7.66, and at $N=3$ —where the equilibrium bid is $2v$ —every persona and frame cell sits at SMAD 106–131, because the models bid approximately their value under every setting.

¹⁵The high-temperature closed-clock cell is small (8 non-winner exits); we read the clock rows as suggestive. Two grid cells are unavailable (both clock formats at $T=0.1$), and the third-price persona and frame cells exist only at $N=3$.

Table 7: Cardinal calibration under tuning knobs (GPT-4o, sealed-bid IPV). Each cell reports SMAD % and the share of bids above value; human targets in the top row. $\text{SMAD} = 100 \cdot \mathbb{E}|b - b^*(v)|/24.5$ with each environment’s own equilibrium b^* (FPSB $N=3$: $2v/3$; SPSB: v ; TPSB $N=3$: $2v$). $\dagger$: differs from its own baseline at $p < 0.05$ (Welch tests for SMAD, two-proportion tests for shares); baselines are the pooled axis baselines (personas), the loss-aversion baseline (frames), and the $T=0.5$ anchor (temperature, explanation). The human third-price target is the Kagel and Levin (1993) $n=5$ environment, the closest available benchmark for the $N=3$ knob cells.

Setting	First price		Second price		Third price ($N=3$)	
	SMAD	over %	SMAD	over %	SMAD	over %
<i>Human target</i>	24.8	0.4	5.6	67.2	7.7 ^{$n=5$}	89.8
Untuned anchor ($T=0.5$)	28.1	0.0	8.9	16.7	108.3	19.4
Axis-pooled baseline	23.4	0.0	11.4	0.4	120.2	0.2
Temperature 0.1	28.6	0.0	15.1 ^{$\dagger$}	10.2 ^{$\dagger$}	112.5	1.3 ^{$\dagger$}
Temperature 1.0	25.5	0.7	13.2 ^{$\dagger$}	9.8 ^{$\dagger$}	123.7 ^{$\dagger$}	2.7 ^{$\dagger$}
Risk-averse persona	15.0 ^{$\dagger$}	0.0	18.3 ^{$\dagger$}	0.0	129.2	0.0
Risk-neutral persona	20.9	0.0	11.0	2.7 ^{$\dagger$}	121.3	6.7 ^{$\dagger$}
Risk-seeking persona	28.0 ^{$\dagger$}	18.0 ^{$\dagger$}	10.5	45.3 ^{$\dagger$}	105.8 ^{$\dagger$}	56.5 ^{$\dagger$}
WTA-WTP frame	23.0	0.0	7.2 ^{$\dagger$}	0.0	114.0	0.0
Rule explanation off	28.1	0.0	8.7	16.0	111.3	12.6 ^{$\dagger$}

Direction: one knob moves it, at a price. The direction axis is where the human-LLM reversal of Section 7.2 lives, and exactly one knob addresses it. Under the risk-seeking persona, GPT-4o’s share of above-value bids rises from 0.4% to 45.3% in SPSB and from 0.2% to 56.5% in third price (both $p < 10^{-4}$), against human targets of 67.2% and 89.8%—direction gaps fall from 66.8 to 21.9 percentage points and from 89.6 to 33.3. The same persona also improves the third-price *level* (-14.4 SMAD, $p = 0.011$) and the second-price *shape* (Wasserstein-1 from 23.8 to 14.8 points). No other knob moves direction: under every temperature, frame, and explanation setting, the SPSB overbid share never exceeds 17%, and the risk-averse persona drives all overbid shares to zero. But the price is paid in first price, the one format where humans almost never bid above value: the risk-seeking persona induces dominated bids in 18.0% of FPSB decisions (human: 0.4%; $p < 10^{-4}$) and raises FPSB SMAD by 4.6 points ($p = 0.002$). In the direction rankings the same setting is first of fifteen in both SPSB and third price and *last* of fifteen in FPSB—the sharpest possible statement of the trade-off. Matching human overbidding where humans overbid requires a disposition that violates the sharpest regularity humans display where they do not.

Tuning is model-specific. The trade-off does not even transfer across model families. The same risk-seeking persona applied to SPSB moves Gemma from 20.7 to 5.87 SMAD—against the human 5.65, the single best cardinal match in the grid—and Gemini from 11.7 to 4.9, but leaves GPT-4o essentially unchanged (11.4 to 10.5) and explodes Claude from 9.5 to 35.0, with 93% of bids above value. A persona calibrated on one model family, then deployed on another, can convert mild underbidding into catastrophic overbidding.

The remaining knobs are not dials. Temperature is non-monotone where it matters: SPSB SMAD is 15.1, 8.9, and 13.2 at $T = 0.1, 0.5$, and 1.0 respectively (both deviations from the anchor significant at $p \leq 10^{-4}$), so it cannot be used to trade off fidelity smoothly; its only clean cardinal use is adding human-

like noise to the too-perfect clock formats. Turning the rule explanation on or off has no detectable effect on any axis (level differences of at most 0.16 SMAD in the first-, second-, and affiliated-value formats, all $p \geq 0.53$).¹⁶ The remaining prospect-theoretic frames mostly *hurt* the second-price level relative to their own baseline.

Verdict. There is no single consistent tuning. Formally: no setting in the grid is in the top half of both the level ranking and the direction ranking on all three core mechanisms; the level winner by worst-case rank (the WTA–WTP frame) and the direction winner (the risk-seeking persona) are disjoint, and the direction winner breaks the level axis in first price. Cardinal agreement is therefore achievable only mechanism-by-mechanism, and sometimes only model-by-model—a boundary that is structural rather than an accident of our grid, since the direction axis demands a disposition that the first-price format punishes. Two implications follow. First, the boundary bounds the proxy use-case: a designer who needs calibrated *levels*—revenue predictions, welfare magnitudes—cannot obtain them from a single tuned synthetic population that spans mechanisms, which is why every cross-population claim in this paper is ordinal. Second, it clarifies what *would* be needed: per-mechanism (and, our results add, per-model) calibration against a small sample of human data from the target environment, in the spirit of Corrigan et al. (2025)—a local correction that our grid suggests cannot be amortized across formats, because the knob that fixes one format’s distribution is the knob that breaks another’s.

8 Discussion

8.1 Design Implications

The ranking’s most actionable feature is that a nearly costless lever sits close to its top. A one- or two-sentence honest description of a mechanism’s safety invariant recovers a large share of what a full obviously strategy-proof redesign buys, without touching the allocation rule, the message space, or the extensive form: Payoff Safety cut the cross-model auction deviation from $-\$2.67$ to $-\$1.53$, and the clock-framing description substantially reduced SMAD in three of the four model families while backfiring in the fourth (pooled $\rho = +0.30$, wide interval; Section 6.2). The supplementary matching experiment pushes the same point further: Rejection Safety cut direct-DA error from 4.2% to 0.2%, and a plain textbook statement of strategy-proofness drove it to 0.0% in all four models (Appendix H). For a platform whose participants increasingly include LLM agents, these are deployable as interface text or API documentation at approximately zero engineering cost. They are also honest in the sense of Gonczarowski et al. (2023): they state true properties of the mechanism rather than steering behavior through misdirection, so a designer can adopt them without incurring the credibility costs of manipulative framing. The comparison with restatements disciplines the recommendation, and more sharply than a simple placebo: it is not “more explanation” that helps. Menu restatements of the auction are null on average and sign-mixed across models (Section 6.2), and in the matching domain the same menu format splits by content—a menu that includes the invariance statement improves play like a safety description, while a menu that merely restates the outcome computation makes play significantly *worse* (Appendix H). The active ingredient is the invariance content, not the explanation.

As LLMs increasingly participate in markets, the design of the mechanisms they interact with will matter. Our results suggest that simple mechanisms, and clear (and honest) descriptions of mechanism properties, can serve as a form of alignment: channeling artificial agents toward the outcomes the mechanism was

¹⁶We phrase this as “no detectable effect” rather than as an estimated effect of removing explanation: the manipulation cannot be verified from the archived prompts, and the contrast rests on the run directories’ provenance.

designed to produce, without requiring that they fully understand why. And because every lever with a human anchor moves human participants in the same direction it moves LLM agents (Section 7), a designer serving a mixed human–AI market need not choose between populations: simplicity, on this evidence, is a common good across participant types.

The reasoning traces add a caution about *how* these text levers work—and about how their efficacy should be audited. The dissociation of Section 6.5 shows that the effective levers change behavior without changing stated reasoning: Payoff Safety cuts the mean absolute deviation from \$3.20 to \$1.95 while its invariant is echoed in none of the six hundred treated traces and the rate of correct payment-rule statements is flat, so a classical mediation analysis fails at the first stage and bounds any comprehension-mediated share of the effect at roughly one third; the payoff tree behaves the same way (bids move, rule-rehearsal language does not). Conversely, the worst-case scaffold visibly changes what models say—worst-case language jumps from 2% to 43% of traces—without moving bids, and the menu restatement moves neither: across every lever tested, language and bids never move together. The channel through which descriptions operate is therefore behavioral rather than deliberative: the text changes what agents *do* without passing through what they *articulate*. For practitioners this cuts in both directions. A designer cannot certify a description by checking that agents verbalize the right principle (the best lever produces no such verbalization), and cannot dismiss one because comprehension checks fail; symmetrically, a passed manipulation check on stated reasoning is no evidence that behavior has moved. Interventions on LLM participants should be evaluated on realized play, not on elicited understanding.

The backfiring tier carries an equally practical warning. Forward-planning and belief-formation scaffolds are not exotic: they are precisely the kind of “think ahead, model the other side” instructions that agent frameworks and prompt libraries routinely inject into deployed LLM pipelines (Fish et al., 2024). In strategy-proof environments, our evidence says such instructions are at best inert and at worst substantial liabilities, with the damage concentrated in the weakest models (Section 6.3). We summarize the operational content of the ranking as follows.

Practitioner guidance: fielding strategy-proof mechanisms to LLM participants.

- *Prefer OSP implementations where available.* The ascending-clock format produced the best play in every model we measured, consistent with its record in human markets (Li, 2017; Milgrom and Segal, 2020).
- *If you must keep a direct mechanism, state its safety invariant—do not merely restate its mechanics.* Add a safety-exposing sentence (“your bid determines *whether* you win, not *what* you pay”; “rejections only redirect, never eliminate”) and, where the interface permits, a payoff-tree explanation of what happens under each outcome. Each is a one-paragraph text change; each independently closed roughly half the gap to the OSP benchmark in auctions, and the safety description nearly closed it in DA. Mechanical restatements of the outcome mapping are useless-to-harmful: menu-style restatements were null on average and sign-mixed in auctions, and a menu that restated the computation *without* the invariance statement significantly worsened matching play. The invariance content is the active ingredient.
- *Never prompt agents to plan ahead or to model opponents.* Lookahead and belief-formation scaffolds worsened play in nearly every configuration tested—most severely for the weakest models—by activating strategizing that the mechanism makes unnecessary.
- *Audit vendor “reasoning enhancements” as potential backfire levers.* Agent frameworks that silently inject planning steps or opponent modeling into prompts are, in our taxonomy, Family-C interventions of the backfiring kind; benchmark them against truthful play before deployment.

8.2 Scope Conditions

The ranking is a statement about a bounded-rationality regime, and both of its boundaries are measured rather than assumed. The upper boundary is now measured with the full lever grid rather than inferred from baseline play alone: we ran the complete intervention battery on four frontier reasoning models (gpt-5-mini, gpt-5, Claude Sonnet 5, Gemini 2.5 Flash; Section 6.4, Table 14), and the picture is three-sided. First, the ranking’s positive rungs *compress to nothing*: all four models bid essentially exactly truthfully at every sealed-bid baseline (mean deviations 0.00–0.30, SMAD 0–2.3%), so safety descriptions, the payoff tree, and belief scaffolds leave play unchanged—the preservation index ρ of Section 4.4, a ratio to the baseline deviation, becomes uninformative as $D(\text{baseline}) \rightarrow 0$. Second, the constraints *migrate* rather than vanish across formats: gpt-5-mini plays second price and (correctly shaded) first price essentially perfectly, yet still fails the harder third-price format outright, bidding roughly its value where the equilibrium calls for bidding above it (three-bidder mean deviation +\$14.76; SMAD 60%; Appendix D), so wherever equilibrium reasoning is genuinely required a bounded-rationality regime persists even at the frontier. Third, and the reason compression is not the whole story, *the backfire rungs keep their teeth*: the menu restatement collapses Claude Sonnet 5 from exact truthfulness to a 35.5% deviation ($d = -3.8$, $p < 0.001$) and degrades Gemini 2.5 Flash, and one sentence of affiliation language in a clock description triggers a misapplied winner’s-curse script in the same model (SMAD 0.3% $\rightarrow$ 8.6%), removed entirely by an IPV-worded rerun (Section 6.4). The frontier failure mode is thus *retrieval selection*: surface wording chooses which memorized playbook runs, and a wrong match is executed flawlessly. A presentation choice with no incentive content can still break an otherwise perfect strategic reasoner, so the practitioner guidance above remains binding even where the simplifying levers have nothing left to preserve. Our main model grid deliberately comprises well-established, widely deployed model families rather than frontier reasoning models: the design question requires agents capable enough to parse the rules of a mechanism yet still susceptible to the reasoning

Instrument	Population	Scale	Approx. cost
OSP laboratory experiment (Li, 2017)	Human	548 subjects (144 main + 404 follow-up)	$\approx$ \$5,400 [†]
Description experiments (Gonczarowski et al., 2024)	Human	542 subjects, incentivized lab	\$15–26/subject [‡]
This paper: full lever battery	LLM (4 families)	>1,000 auctions; >5,000 decisions	< \$400
This paper: one treatment cell	LLM	50 repetitions	$\sim$ \$10

Table 8: Approximate cost of instruments for testing simplicity-preserving design levers. [†]Li (2017)’s main experiment paid 144 subjects an average of \$37.54 (including a \$20 show-up fee), $\approx$ \$5,400 total; documented for the main experiment only. The paper’s one-shot follow-up added 404 subjects (\$5 show-up plus incentive payments), for 548 subjects overall, but does not report an average payment for the follow-up, so a total cost across both experiments is not recoverable from the published figures. [‡]Gonczarowski et al. (2024) paid Prolific participants an average of £15.0 ($\approx$ \$19) and Cornell participants an average of \$25.5 across 542 subjects (Appendix A.1 of that paper).

failures that simple design aims to remedy. Deployed fleets will occupy this regime for the foreseeable future: high-volume agentic transactions run on small, cheap models for cost and latency reasons, so the population of economic agents that mechanisms will actually face looks more like our grid than like the frontier. At the lower boundary, Llama-3-8B posts a mean SMAD of 57.58%—worse than the reconstructed human benchmark in six of seven formats (Appendix D)—indicating a capability floor below which agents cannot reliably parse the mechanism at all, and beneath which no descriptive lever can be expected to help. The ranking applies to the wide band between these boundaries.

8.3 What the Testbed Costs

The traditional instruments for questions like ours are incentivized human experiments and observational field data, and both are expensive ways to search a large design space: the laboratory test of obvious strategy-proofness engaged 548 subjects across a main experiment and a one-shot follow-up, with documented payments on the order of \$5,400 for the main experiment alone (Li, 2017); incentivized description experiments recruited 542 subjects with average payments of \$15–26 per subject (Gonczarowski et al., 2024); and observational platform data require controls for seller and bidder heterogeneity that complicate the causal inference a lever ranking needs (Manning et al., 2024). The synthetic testbed sits at the exploratory end of this pipeline, extending the cost reduction that online laboratories brought to behavioral economics (Horton et al., 2011): the full battery reported in this paper—more than 1,000 auctions and 5,000 individual agent decisions across two domains, four model families, and the complete lever grid—cost under \$400 in API fees, and a single treatment cell of 50 repetitions costs roughly \$10 (Table 8). These economics are most useful read jointly with Section 7: cheap synthetic screening ranks the candidate levers; expensive human experiments are reserved for the rungs where the ranking makes novel, falsifiable predictions.

8.4 Limitations

The human reconstruction is model-based. Our human benchmarks are moment-matched mixture reconstructions of published summary statistics, not raw experimental microdata (Section 4.5, Appendix A). We therefore confine human–LLM comparisons to rankings, directions, and comparative statics, attach no significance tests to reconstructed levels, and note that the ranking itself (Section 6) never conditions on the reconstructions—they anchor transferability only. [TODO: Reconstruction-uncertainty bootstrap bands on every human anchor (Appendix A) pending.]

Prompt-template specificity. Each lever is instantiated as one template per domain, so the estimated effects are, strictly, effects of particular wordings. Three features argue against a prompt-engineering artifact: the taxonomy was fixed *ex ante* from published theory (Section 3), the menu restatement functions as a built-in restatement control (null on average, though sign-mixed across models; Section 6.2), and half the taxonomy backfires—a pattern opportunistic prompt search would not produce and would not publish. But invariance to paraphrase remains an assumption rather than a measurement. [TODO: Paraphrase robustness: three paraphrases $\times$ two headline levers (plan §6).]

Direction shares are convention-dependent. Statements about the *direction* of deviations (overbidding versus underbidding shares) depend on the classification convention—whether shares are taken over all bids or only over bids that deviate from value, and how a tolerance band around the benchmark is drawn. We use the repo-canonical convention throughout Section 7 (share of bids strictly above value among all bids; the human SPSB overbid share is 67.2%, or 92.2% among deviating bids), while the directional decomposition of Section 5 removes a $\pm 2\%$ band before classifying. The qualitative reversal—humans overbid, LLMs underbid—is robust to the choice, but individual shares should not be compared across conventions.

Model-version dependence. All results are estimated on pinned model snapshots (Section 4); provider updates, changes to default system prompts, and fleet turnover can shift behavior in ways we cannot forecast. The ranking should be read as a measurement protocol to be re-estimated as agent populations evolve, not as a constant of nature—and the frontier migration documented above, with dominant-strategy formats solved first and equilibrium-reasoning formats persisting, already indicates the likely direction of drift.

9 Conclusion

Which levers available to a mechanism designer keep dominant-strategy play intact for boundedly rational agents? Across sealed-bid auctions—the paper’s domain—and four LLM families, the answer is a stable hierarchy of tiers. Obviously strategy-proof extensive forms sit at the top. A strong second tier contains levers that change no rule of the game, and its content is sharper than “more explanation helps”: the active ingredient is the *invariance*. Honest text that exposes why truth-telling is safe, and scaffolds that lay out the mechanism’s payoff contingencies, deliver large gains—while restatements of the mechanics carry none of it, coming in null on average and sign-mixed in auctions and actively harmful in matching when the invariance statement is omitted. Below the baseline, scaffolds that prompt agents to plan ahead or to model opponents actively backfire, most severely for the weakest models. The ordering parallels the theoretical simplicity hierarchy built for human reasoners, and it localizes the binding constraint: not *computing* the dominant strategy, but *seeing* that truth-telling is safe.

The human experimental record agrees in sign on every rung where it speaks, at the pooled level—clock formats, clock-framing (in three of four families), the menu restatement’s null, invariance-carrying versus mechanics-only descriptions, the failing baselines—even though the two populations fail in opposite directions, which is precisely why the agreement is informative: the constraints, and the levers that relieve them, belong to the strategic problem rather than to any one kind of mind. The agreement is ordinal, and we measure that boundary rather than assume it: no single tuning of the agents—temperature, risk personas, framings, explanations—produces cardinal agreement with human play across mechanisms, because level calibration is mechanism- and model-specific and the one knob that restores the human direction of error in second- and third-price auctions breaks first-price play. The testbed ranks levers; it does not simulate levels. Where the human record is silent, the ranking speaks first: its sharpest untested claim is that safety-exposing

descriptions will deliver for human subjects the behavioral gains that mechanical restatements have not—a prediction testable in a single incentivized session at a small fraction of the cost of the synthetic program that generated it.

Three directions strike us as most valuable. First, frontier tracking: the constraints migrate rather than vanish as capability grows—the four frontier models we test master the dominant-strategy sealed-bid auction, yet a content-free menu restatement still collapses one of them, and the third-price format still defeats the frontier where we can measure it—so the ranking should be re-estimated on the formats and presentation margins where reasoning still binds as deployed fleets evolve. Second, combining complementary levers—pairing a safety-exposing description with a payoff-tree scaffold—to trace the minimal intervention that closes the gap to the rational benchmark. Third, the method itself: cognitive scaffolds as diagnostic probes, together with the heterogeneity we document across model families, offer a general way to characterize the structure of an artificial agent’s reasoning failures—in mechanism design and beyond.

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A Human Data Reconstruction

This appendix records, source by source, how we map the aggregate statistics reported in each human experiment into the unified SMAD metric of Section 4.4. The mixture model used to reconstruct bid-level distributions is specified in Section 4.5; the calibrated parameters for every source–treatment pair appear in Appendix A.1 below, and the closing paragraph of this appendix states exactly which forms of uncertainty our reconstruction bands do and do not capture.

Li (2017) and Breitmoser–Schweighofer-Kodritsch (2022): SPSB, AC, and AC-B under APV. These papers (Li, 2017; Breitmoser and Schweighofer-Kodritsch, 2022) report mean absolute deviation (MAD)¹⁷ from truthful bidding. In Li (2017), the reported deviation is based on a group-level statistic involving the second-highest bid and second-highest value; in Breitmoser and Schweighofer-Kodritsch (2022), deviations are computed from submitted bids (often focusing on non-winning bids). In each case we adhere to the source paper’s unit of observation, but the SMAD definition conceptually applies to the underlying submitted bids. Because the equilibrium optimum under AC and SPSB is truthful bidding $b^*(v) = v$, we have $d_m = |b - v|$ and thus

$$\Delta_m = 100 \cdot \frac{\text{MAD}_m}{\mu_m^*}, \quad \mu_m^* = \mathbb{E}[v].$$

Here $\mathbb{E}[v]$ is implied by the experimental value distribution reported by the source paper: $\mathbb{E}[v] = 70$ in Li (2017), and $\mathbb{E}[v] = 65$ in Breitmoser and Schweighofer-Kodritsch (2022) given their stated common-value and idiosyncratic components.¹⁸ If the paper reports $\text{SE}(\text{MAD}_m)$ (often using group means as the unit of observation), we propagate it via

$$\text{SE}(\hat{\Delta}_m) = 100 \cdot \frac{\text{SE}(\widehat{\text{MAD}}_m)}{\hat{\mu}_m^*}, \quad \text{CI}_{95\%}(\Delta_m) = \hat{\Delta}_m \pm 1.96 \text{SE}(\hat{\Delta}_m).$$

Kagel and Levin (1993): FPSB, SPSB, and TPSB under IPV. Kagel and Levin (1993) report bid regressions and over-/under-bidding frequencies for fixed bidder counts (e.g., $N \in \{5, 10\}$). We treat each N separately: the regression form $\hat{b}(x) = \hat{\alpha} + \hat{\beta}x$ is fit for a *fixed* number of bidders, and is not pooled across varying N . Theory gives linear equilibrium bid functions under IPV with $x \sim U[x_{\min}, x_{\max}]$:

$$\begin{aligned} b_{\text{FPSB-IPV}}^*(x) &= x - \frac{x - x_{\min}}{N} = \frac{N-1}{N}x + \frac{1}{N}x_{\min} \\ b_{\text{SPSB-IPV}}^*(x) &= x \\ b_{\text{TPSB-IPV}}^*(x) &= x + \frac{x - x_{\min}}{N-2} = \frac{N-1}{N-2}x - \frac{1}{N-2}x_{\min} \end{aligned}$$

(When $x_{\min} = 0$, these reduce to the familiar slope-only formulas.) Because only summary moments are reported, we reconstruct the bid distribution via a parametric data-generating process (DGP) consistent with the bid regressions and over-/under-bidding frequencies reported in Table 2 of Kagel and Levin (1993). For each auction series and fixed N , we assume $x \sim U[x_{\min}, x_{\max}]$ (bounds from the source paper), and generate bids as

$$b = \hat{\alpha} + \hat{\beta}x + \varepsilon, \quad \varepsilon \sim \mathcal{N}(\mu, \sigma^2),$$

¹⁷For clock auctions, since the winner did not submit a final sealed bid, the reported MAD is computed only from bidders who revealed a bid by dropping out (i.e., those who exited the clock), excluding the winner’s unobserved terminal willingness-to-pay.

¹⁸When auctions are repeated, we follow the source paper’s convention and report post-learning blocks (e.g., dropping the first three rounds and using the next reported block).

where (μ, σ) are calibrated by moment matching to the reported frequencies (e.g., $\Pr(b > x)$ and $\Pr(b > b^*(x))$). Given the calibrated DGP, we estimate

$$\widehat{\text{MAD}}_m = \mathbb{E}[|b - b^*(x)|], \quad \hat{\mu}_m^* = \mathbb{E}[b^*(x)], \quad \hat{\Delta}_m = 100 \cdot \frac{\widehat{\text{MAD}}_m}{\hat{\mu}_m^*}.$$

In practice, we run the Monte Carlo procedure 100 times and report the across-run mean $\hat{\Delta}_m$ as the point estimate; uncertainty is obtained from across-run variability (percentile or normal approximation).

Levin, Kagel, and Richard (1996) and related CV papers: first-price CV and English (second-price proxy). In common-value auctions (Levin et al., 1996), bidders observe signals s_i about a common value V (e.g., $s_i = V + \varepsilon_i$ as specified in the original paper).

Theory gives the following equilibrium bid functions under common values:

$$\begin{aligned} b_{\text{FPSB-CV}}^*(x) &= x - \epsilon + Y \\ Y &= \left(\frac{2\epsilon}{N+1} \right) \exp \left[- \left(\frac{N}{2\epsilon} \right) (x - (\underline{x} + \epsilon)) \right] \\ b_{\text{SPSB-CV}}^*(x) &= x \end{aligned}$$

The theoretical benchmark is the equilibrium bidding strategy $b_m^*(s)$ implied by the source paper's signal structure and mechanism. We therefore compute SMAD *on bids* exactly as in private values:

$$d_m = |b - b_m^*(s)|, \quad \Delta_m = 100 \cdot \frac{\mathbb{E}[|b - b_m^*(s)|]}{\hat{\mu}_m^*}.$$

Because there is a very large common part in each player's valuation, we take the following normalization factor:

$$\hat{\mu}_m^* = \mathbb{E}[b_{\text{max}}^*(x) - b_{\text{min}}^*(x)]$$

When micro-data are unavailable, we reconstruct the bid distribution by simulating the source paper's DGP (signals and value draws) together with the theoretical bidding rule $b_m^*(\cdot)$, and calibrate any auxiliary noise parameters by moment matching to the summary statistics reported (e.g., frequencies of overbidding/high bids, or mean prices/profits as available). We run Monte Carlo 100 times and report the across-run mean $\hat{\Delta}_m$ as the point estimate; uncertainty is obtained from across-run variability (percentile or normal approximation).

A.1 Mixture-Model Calibration Details

Table 9 reports the calibrated parameters of the three-component mixture model of Section 4.5 for every source–treatment pair. The mixture weights $(p_{\text{eq}}, p_{\text{over}}, p_{\text{under}})$ are set directly from the behavioral rates reported in each source; the overbid scale λ is calibrated to match the secondary statistic listed in the final column (a mean bid-to-value ratio, a regression fit, or a reported MAD, depending on what the source publishes).

Source	Treatment	p_{eq}	p_{over}	λ	Calibration Target
Li (2017)	SPSB	0.50	0.40	0.219	Mean ratio = 1.08
Li (2017)	AC	0.67	0.18	0.198	Mean ratio = 1.02
Breitmoser (2022)	SPSB	0.44	0.40	0.283	Mean ratio = 1.10
Breitmoser (2022)	AC	0.83	0.12	0.154	Mean ratio = 1.01
Kagel-Levin (1993)	FPSB, $n=5$	0.75	0.12	0.100	$R^2=0.88$
Kagel-Levin (1993)	SPSB, $n=5$	0.27	0.67	0.153	$R^2=0.97$
Kagel-Levin (1993)	TPSB, $n=5$	0.58	0.27	0.924	$R^2=0.94$
Gonczarowski et al. (2023)	Traditional	0.37	0.41	0.450	MAD = \$0.51
Gonczarowski et al. (2023)	Menu	0.34	0.43	0.450	MAD = \$0.55

Table 9: Calibrated mixture model parameters by source and treatment. Mixture weights from reported rates; λ calibrated to match secondary statistic listed; $p_{under} = 1 - p_{eq} - p_{over}$.

Reconstruction uncertainty. All reconstructed human benchmarks are Monte Carlo objects: for each calibrated DGP we simulate 100 independent replications and report the across-run mean of $\hat{\Delta}_m$ as the point estimate, with uncertainty summarized by percentile bands (or a normal approximation) across replications. These bands capture simulation noise *conditional on* the calibrated parameters; they do not capture uncertainty in the calibration itself, because the mixture weights and the overbid scale λ are point-identified from reported aggregates that typically come without standard errors. For this reason, we never run hypothesis tests against reconstructed human distributions anywhere in the paper (see also Appendix D): comparisons to reconstructed humans are made at the level of rankings, directions, and comparative statics, with levels and formal tests reserved for theory benchmarks. [TODO: Parametric bootstrap over the calibration parameters — perturbing p_{eq} , p_{over} , and λ within the sampling uncertainty implied by each source’s reported cell sizes, then propagating each draw through the full reconstruction — to replace the Monte-Carlo percentile bands with bands that also reflect calibration uncertainty; these bootstrap bands supply the human-anchor whiskers in Figure 8 (plan graft).]

B Simulation Procedures

We give the prompt-and-parse pipeline used to simulate the auction mechanisms with LLM agents. Each simulated bidder is queried independently. In every auction instance, the agent (i) receives the scenario and mechanism description, (ii) is asked to articulate a short bidding plan, and (iii) outputs a bid/decision in a structured format that can be parsed deterministically. This appendix describes the process for each mechanism class; the complete rule-description prompts for every mechanism and treatment are collected in Appendix F.

B.1 Simulation process in sealed-bid auctions

Prompt structure. We attach the canonical prompt template in the following. Each bidder is assigned a role (e.g., “Bidder Andy”) and informed of the other bidders’ identities. The prompt then provides (a) a mechanism-specific *rule explanation* (e.g., first-price vs. second-price vs. third-price), and (b) universal *instructions* that define the objective and response format. We require the model to respond using two exact tags: <PLAN> (a brief natural-language rationale) and <ACTION> (the bid as a number), in the chain-of-thought style of Wei et al. (2022):

```

You are Bidder Andy.
You are bidding with {OTHER_BIDDERS}.

{RULE_EXPLANATION} + {INSTRUCTIONS}

Your response must use these EXACT tags below:
```
<PLAN>[Reason out a bidding strategy and plan for this round. {% if current_round
> 1 %}Learn from your previous rounds. Remember: the values will be
different in future rounds - don't assume patterns will repeat. {% else %}
Consider the history above if available. {% endif %}Be detailed and precise.
LIMIT your plan to 100 words.] </PLAN>
<ACTION> [Your bid amount as a number, e.g., 1, 44, 50] </ACTION>
```

```

Universal objective instruction. Across all sealed-bid simulations, we append a universal instruction block that specifies the optimization objective (maximize profit) and discourages spurious cross-round pattern extrapolation (Fish et al., 2024). We report the exact text in the following. Note that the template’s references to round history are inert in the single-shot baseline (no history block is present in the prompt); they bind only in the repeated-play ablation of Appendix C.

```

Your TOP PRIORITY is to place bids which maximize the user’s profit in the long
run.
Learn from the history of previous rounds in order to maximize your total profit.
Don’t forget the values are redrawn independently each round.

```

History and feedback (multi-round ablation only). Our main experiments restrict attention to single-round auctions ($T = 1$). When we run the multi-round ablation in Appendix C, we include a standardized history transcript in the prompt. The history contains the agent’s past realized values, past bids, whether it won, the transaction price, and realized profit, as well as the list of all bids in each prior round. This transcript is appended above the response-tag requirements, so that the agent can condition on past outcomes without ambiguity.

B.2 Simulation process in ascending clock auctions

In ascending clock auctions, each auction consists of multiple clock cycles, during which every bidder is asked whether they want to stay or drop out at the current clock price. At the start of the auction, bidders are reminded of the auction rules, similar to those outlined in Appendix B.1. The auction starts with an initial price of 0, which increases in fixed increments of \$0.5 per clock tick until only one bidder remains or two bidders drop out simultaneously, in which case the winner is chosen randomly.¹⁹ The detailed prompt is listed as follows, with variables enclosed in brackets (the value shown, 26, is a sample instantiation):

¹⁹The \$0.5 tick is taken from the baseline experiment configuration files (`configs/clock/`); an earlier draft stated \$1 increments, which does not match any configuration. Some model-ablation configurations used a \$0.1 tick instead of \$0.5. [TODO: Harmonize all clock cells to the canonical \$0.5 tick of Table 2 in the re-runs (plan E2).]

```

You are Bidder Andy. You are bidding with {OTHER_BIDDERS}.

{RULE_EXPLANATION} + {INSTRUCTIONS}

Your value towards to the prize is 26 in this round.
The current price in this clock cycle is {current_price}.
The price for next clock cycle is {current_price + increment}.

The previous bidding history is: {transcript}.

Do you want to stay in the bidding?
If you choose yes, you can keep bidding for next clock. If you choose No, you will
exit and have no chance to re-enter the bidding. Your response must use
these EXACT tags below. You must output the ACTION.
'''
<PLAN>
[Write your plans for bidding strategies. Be detailed and precise but keep things
succinct and don't repeat yourself. LIMIT your plan to 50 words. ] </PLAN>
<ACTION> Yes or No </ACTION>
'''

```

In AC (the open clock), bidders who have not dropped out are shown the previous bidding history at each subsequent clock cycle. The history includes the price and how many bidders had dropped out, and the transcript variable takes the following format: *The previous biddings are: [‘In clock round 1, the price was 1, no players dropped out’].*

In AC-B (the closed/blind clock), we do not show the previous biddings; in each clock cycle, we directly query the LLM agents’ decision, so the transcript variable is None.

A clock auction ends when only one bidder remains, who is declared the winner, or when two bidders drop out simultaneously, in which case the winner is chosen randomly. Each clock auction is a *single-shot* instance with affiliated values (`rounds: 1` in the configuration files): agents receive no feedback across instances, and we run K independent repetitions per configuration with freshly drawn values ($K = 40$ in the baseline clock cells; config-verified). [TODO: Top up the clock cells to the pinned $K \geq 50$ of the harmonized design in the re-runs (plan §4, E2).]

B.3 Simulation process in eBay auctions

For eBay auctions, we discretize the continuous bidding period into 10 periods. Each period, LLM agents decide whether to increase their bid or hold their current bid. The bidding process follows a structured format, where bidders act in a predefined sequence each period (e.g., Charles, then Alice, then Betty). Agents are informed of past price changes through a transcript, such as “On day 1, the price changed to 1. On day 2, the price changed to 3.”

The prompt provides key details, including the bidder’s private value, the total number of bidding days, the current day, the bidding order, previous bids, previous proxy bids, and the current price. The decision-making process is again guided by a structured asking prompt that enforces bidding rules. The treatment-level analysis of this environment appears in Appendix I.

```

You are Bidder Andy.
You are bidding with {OTHER_BIDDERS}.

{RULE_EXPLANATION} + {INSTRUCTIONS}

Your private value for this item is ${private_value}. This is the maximum amount
you are willing to pay. Keep this value private. There are in total {
total_periods} days of bidding and this is day {current_period}. {ordering}.

Your previous bids are {previous_bid}. If you have already placed bids, you can
only increase your bid or hold your current bid. The previous proxy bids are:
{transcript}.
The current price is {current_price}.
Your response must use these EXACT tags below. Don't miss the amount.
```
<PLAN>[Write your plans for what bidding strategies to test next. Be detailed and
precise but keep things succinct and don't repeat yourself. LIMIT your plan
to 50 words.]</PLAN>
<CHECK>your last bid is {{last_bid_amount}}, you cannot bid lower than this value
</CHECK>
<ACTION> BID or HOLD </ACTION>
<AMOUNT> if BID, enter a number here, e.g. 1, 44. If HOLD, enter 0 </AMOUNT>
```

```

C Learning Effects in Repeated Play

To assess whether LLM agents exhibit learning or adaptation across rounds, we conduct a set of statistical tests comparing early-round behavior to late-round behavior and examining continuous trends over time. For the first-price sealed-bid (FPSB) auction, the mean SMAD in the first five rounds (0.2523) is essentially identical to that in the last five rounds (0.2524), with no detectable difference in either a t-test ($p = 0.9968$) or a Mann–Whitney U test ($p = 0.9131$). Trend analyses likewise show no evidence of systematic change: a linear regression yields a slope indistinguishable from zero ($p = 0.9886$) and a Spearman test finds no monotonic movement ($\rho = 0.0929$, $p = 0.7420$). Results are similar for the second-price sealed-bid (SPSB) auction. The comparison between the first and last five rounds reveals no significant difference (means 0.1282 vs. 0.1447; t-test $p = 0.3335$; Mann–Whitney $p = 0.1948$), and neither a linear trend test ($p = 0.3664$) nor a Spearman correlation test ($\rho = 0.3964$, $p = 0.1435$) indicates meaningful temporal structure. Taken together, across all specifications, we find no statistical evidence of learning effects, behavioral drift, or systematic changes in bidding deviations over the 15 rounds. LLM agents' deviation magnitudes remain remarkably stable throughout both FPSB and SPSB environments (Figure 9).

This stability is what licenses the one-shot design used throughout the paper: both the fidelity battery of Section 5 and the lever experiments underlying the ranking of Section 6 run each auction (and each DA instance) as an independent single-shot repetition with freshly drawn values and no cross-instance feedback (Appendix B). Because deviations neither shrink nor grow with repetition, the treatment effects we report cannot be confounded by within-session adaptation, and none of the cross-lever comparisons depend on assumptions about learning dynamics.

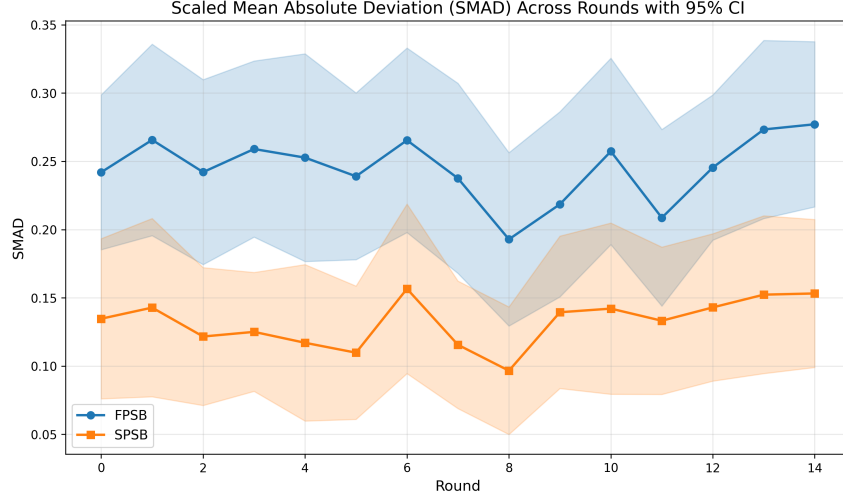


Figure 9: Repeated-play ablation: 15 consecutive rounds of first-price and second-price sealed-bid auctions with GPT-4o ($T = 0.5$), where bidders observe the full outcome history of prior rounds. The y-axis is the scaled mean absolute deviation (SMAD) between actual and benchmark bids. Deviations show no trend in either format.

D Model and Temperature Ablations

This appendix reports the stability ablations referenced in Section 5—varying the sampling temperature of the GPT-4o anchor, and varying the base model on the seven-format fidelity battery—together with a corrected accounting of the frontier tier (four reasoning models: gpt-5-mini, gpt-5, claude-sonnet-5, gemini-2.5-flash). Throughout, the human column is the moment-matched reconstruction of Section 4.5; consistent with the epistemic policy stated there, we report *no* significance tests against reconstructed human benchmarks, because such tests would treat a model-based reconstruction as if it were sampled micro-data. Comparisons to the human column should be read qualitatively—rankings, directions, and orders of magnitude. [TODO: Attach Monte-Carlo percentile bands (and, once available, the parametric-bootstrap bands of Appendix A) to the human column in both tables (plan §5.8).]

D.1 Varying temperature for the GPT-4o agent

We vary GPT-4o’s sampling temperature $T \in \{0.1, 0.5, 1.0\}$ across the seven fidelity-battery formats (Table 10; Figure 10). The temperature parameter controls the randomness of the model’s outputs, with lower values producing more deterministic responses and higher values introducing more variability.

Two facts stand out. First, temperature effects are modest and, crucially, *non-monotonic*: no temperature setting dominates. $T = 0.1$ is strictly best in three of the seven formats (both ascending-clock formats and FPSB-CV) and strictly worst in three others (SPSB-IPV, SPSB-APV, and FPSB-IPV), with the remaining format (SPSB-CV) ordered $T = 1.0 < T = 0.1 < T = 0.5$. Second, the pattern is consistent with temperature scaling the dispersion around the model’s modal action rather than changing the underlying strategy: at $T = 0.1$ the clock formats become exactly truthful (SMAD = 0.00) because near-deterministic decoding removes stochastic exits from an already-truthful modal policy, while in the sealed-bid formats

the same determinism locks in the modal bias. We therefore retain $T = 0.5$ throughout the paper as a *convention*—it is the most common setting in LLM-based simulation (Horton, 2023) and preserves comparability across models and with prior work—not because it is performance-optimal in any particular format. Temperature reappears in Section 7 as a candidate knob for *cardinal* calibration to human levels; its effects there are likewise non-monotonic (on the rebuilt grid’s normalization, SPSB-IPV SMAD is 15.1/8.9/13.2 at $T = 0.1/0.5/1.0$).

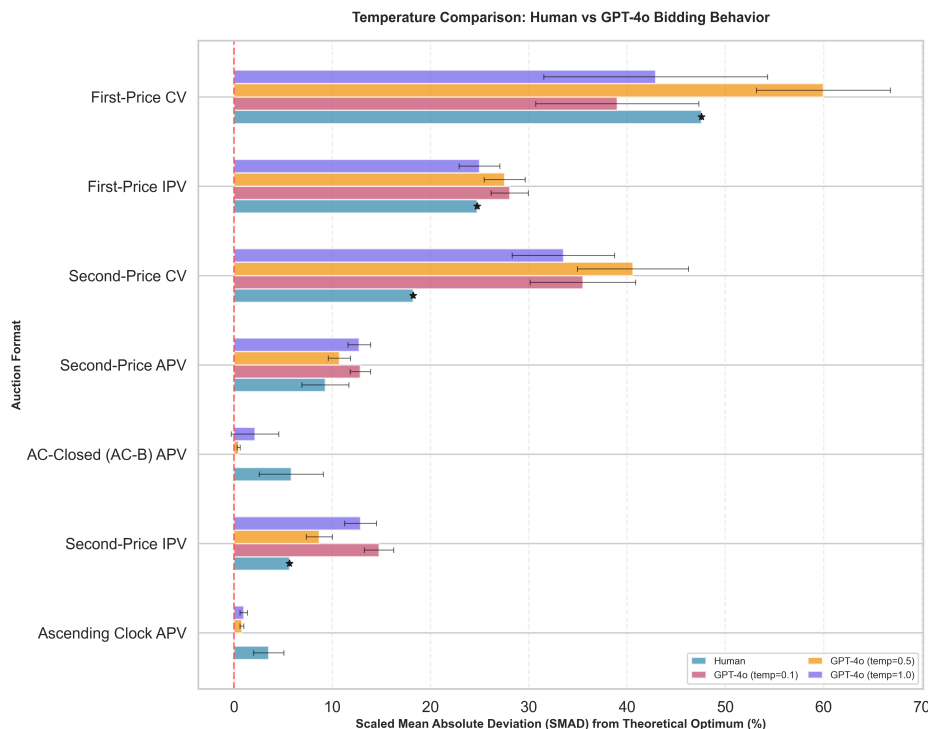


Figure 10: Comparison of the bidding behavior of reconstructed human participants and GPT-4o across three temperature settings ($T = 0.1, 0.5, 1.0$) in seven auction formats. The horizontal bars represent the scaled mean absolute deviation (SMAD) from the theoretical benchmark, with error bars indicating 95% confidence intervals (for the LLM cells only; see the text of this appendix for the human column’s epistemic status).

D.2 Varying the base model

We repeat the seven-format fidelity battery, holding prompts fixed and $T = 0.5$, for three models beyond the GPT-4o anchor: Claude 3.5 Haiku, Gemini 2.0 Flash, and Llama-3-8B-Instruct (an open-weight model).²⁰ Claude 3.5 Haiku and Gemini 2.0 Flash belong to the canonical model set of Section 4; Llama-3-8B-Instruct

²⁰Model identifiers: `claude-3-5-haiku-20241022`, `gemini-2.0-flash`, `Meta-Llama-3-8B-Instruct`, and, in the frontier tier below, `gpt-5-mini`, `gpt-5`, `claude-sonnet-5`, and `gemini-2.5-flash`. An earlier draft referred to the Anthropic model as “Claude Sonnet 3.5”; the experiment configurations confirm that the runs stored under directories named `claude_sonnet` (e.g.,

Auction Format	$T = 0.1$	$T = 0.5$	$T = 1.0$
AC-B (closed clock), APV	0.00	0.47	2.15
AC (open clock), APV	0.00	0.81	1.00
SPSB, IPV	14.77	8.69	12.89
SPSB, APV	12.87	10.73	12.75
FPSB, CV	39.00	59.99	42.92
SPSB, CV	35.51	40.60	33.53
FPSB, IPV	28.07	27.55	24.99

Table 10: Temperature effects on GPT-4o bidding behavior (SMAD, %). SMAD = scaled mean absolute deviation from the theoretical benchmark. No temperature dominates: $T = 0.1$ is strictly best in three formats and strictly worst in three others. Values use the legacy figure normalization ($\mathbb{E}[b^*] = 25$); the rebuilt cell-level grid normalizes by 24.5, making values uniformly $\approx 2\%$ larger (e.g., SPSB-IPV 15.1/8.9/13.2)—orderings and the non-monotonicity are unaffected.

and the frontier reasoning models of Section D.3 (treated separately below, because their data coverage differs) bracket the canonical set from below and above. Results appear in Table 12 and Figure 11. The fourth canonical model, Gemma 27B, ran a separate six-format sealed-bid fidelity battery (Table 11); we report it apart from Table 12 because its cells are recomputed from the cell-level dataset (`results/merged_ranking/auction_cells.csv`, normalizer $\mathbb{E}[b^*] = 24.5$), a normalization that differs from the legacy figure scaling of the Claude/Gemini/Llama columns here (the same $\approx 2\%$ offset flagged for Table 10), so the columns are not directly comparable cell-for-cell.

Llama-3-8B-Instruct serves as the *capability floor*. It exhibits substantially degraded performance across all auction formats (mean SMAD = 57.58%), exceeding the reconstructed human benchmark in six of seven formats—evidence that open-weight availability at small scale is not sufficient for competent strategic play, and that the canonical models sit well inside the band where design levers have room to operate. Claude 3.5 Haiku and Gemini 2.0 Flash behave broadly like the GPT-4o anchor: near-truthful in the clock formats, moderate deviations in the sealed-bid formats, and large deviations under common values. [TODO: The SPSB-IPV cells will be recomputed on the harmonized canonical dataset alongside signed means (plan E8, §5.3); the Claude SMAD/signed-mean contrast (worse absolute, best signed) is expected to persist and will be reported as such.] [TODO: Regenerate the figure asset `appendix2_model_comparison.png`; the legend of the current PNG carries the superseded “Claude Sonnet 3.5” label (see footnote in this appendix), and its gpt-5-mini series rests on the misattributed logs documented in Section D.3 of this appendix and must be dropped or rebuilt from the genuine cells.]

Gemma 27B and the difficulty ordering. The open-weight Gemma 27B model ran the sealed-bid fidelity battery in six formats ($K = 30$ per cell; Table 11). Its mean SMAD is 19.09%, between the canonical models and the Llama floor. The more instructive fact is not the level but the *shape*: Gemma does *not* reproduce the human difficulty ordering cleanly. In the human data the first-price format is much harder than the second-price dominant-strategy format—FPSB-IPV SMAD 24.76% versus SPSB-IPV 5.65%, a ratio of 4.4—and GPT-4o preserves that ordering (27.55%/8.69% = 3.2). Gemma nearly flattens it: 24.80% in FPSB-IPV against 19.88% in SPSB-IPV, a ratio of only 1.25. It deviates almost as much in the format where truthful

`robustness_logs/V10/*claude_sonnet*`) were in fact executed with `claude-3-5-haiku-20241022` (config-verified: `configs/clock/01_06_descending_clock_apv_claude_sonnet.yaml`). The gpt-5-mini API ignores the temperature parameter.

bidding is a dominant strategy as in the format where equilibrium requires shading, so the fidelity ordering that the auction ranking exploits is, in part, model-specific rather than a universal property of LLM bidders. The two ascending-clock cells are not yet run for Gemma (the per-tick clock harness is prohibitively slow, and reconciling the survey-style elicitation with the tick protocol is deferred to the native-key runs), so the clock rows of Table 12 have no Gemma entry. [TODO: Gemma ascending-clock (AC and AC-B) cells pending native-key runs; survey-vs-tick harmonization deferred.]

Auction Format	SMAD vs. value (%)	Signed mean dev. (\$)
FPSB, IPV	24.80	−5.89
TPSB, IPV	23.31	−5.45
SPSB, IPV	19.88	−4.87
SPSB, APV	18.67	−4.57
FPSB, CV	16.08	−2.57
SPSB, CV	11.80	−1.44

Table 11: Gemma 27B sealed-bid fidelity battery (SMAD, % of $\mathbb{E}[b^*] = 24.5$; signed mean deviation from value in dollars). Recomputed from `results/merged_ranking/auction_cells.csv` (group `gemma27b`, $K = 30$ per cell). Mean SMAD across the six formats is 19.09%. Gemma’s FPSB-IPV/SPSB-IPV ratio is 1.25, against 4.4 for reconstructed humans and 3.2 for GPT-4o: the model does not reproduce the human difficulty ordering, so the fidelity ordering is partly model-specific. The two ascending-clock cells (AC, AC-B) were not run for Gemma (per-tick harness too slow; survey-vs-tick harmonization pending native-key runs). CV deviations are relative to the private signal and carry no dominant-strategy benchmark.

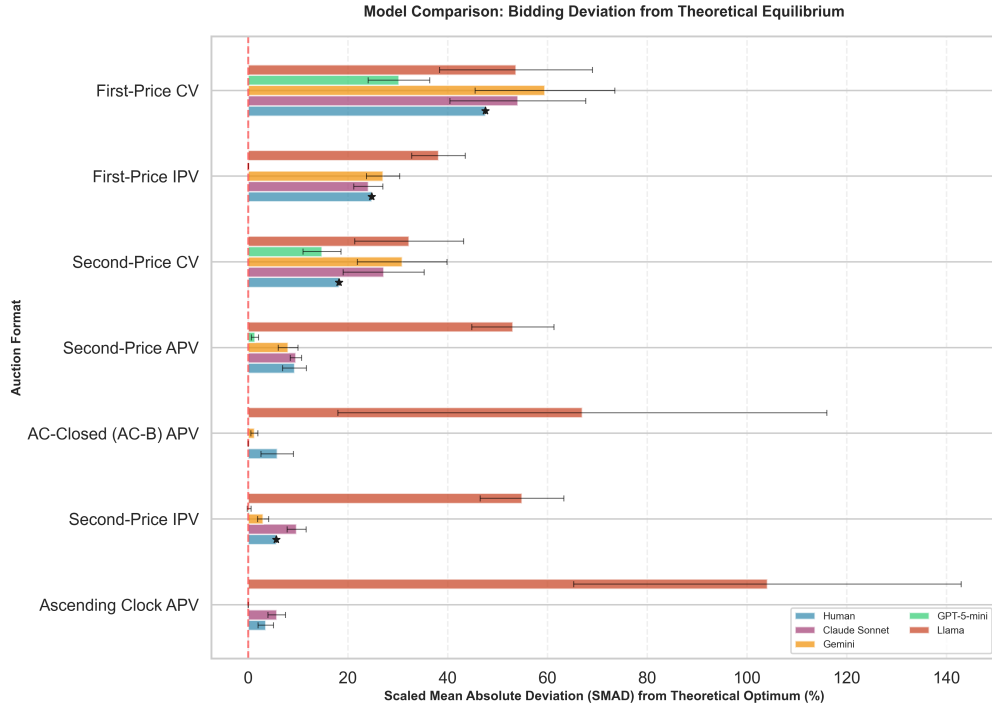


Figure 11: Comparison of the bidding behavior of reconstructed human participants and LLM agents across further foundation models in seven auction formats. The horizontal bars represent the scaled mean absolute deviation (SMAD) from the theoretical benchmark, with error bars indicating 95% confidence intervals (LLM cells only). The gpt-5-mini series shown in the current asset is superseded (see the frontier discussion in the text; regeneration pending).

Auction Format	Human (reconstr.)	Claude 3.5 Haiku	Gemini 2.0 Flash	Llama-3-8B
FPSB, IPV	24.76	24.09	27.05	38.16
SPSB, IPV	5.65	9.72	3.01	54.89
AC-B (closed clock), APV	5.83	0.00	1.26	67.00
SPSB, APV	9.31	9.58	8.02	53.08
AC (open clock), APV	3.54	5.74	0.00	104.10
FPSB, CV	47.59	54.07	59.50	53.68
SPSB, CV	18.23	27.19	30.89	32.27

Table 12: Fidelity battery across base models (SMAD, %). SMAD = scaled mean absolute deviation from the theoretical benchmark. All models run at $T = 0.5$. The human column is the moment-matched reconstruction of Section 4.5; no significance tests are reported against it because the reconstruction is model-based (see text and Appendix A). An earlier draft’s gpt-5-mini column is removed: its source logs are GPT-4o (see the frontier subsection); the genuine gpt-5-mini cells appear in Table 13.

D.3 The frontier tier: reasoning models do not uniformly dissolve the constraints

An earlier draft treated gpt-5-mini as a frontier *ceiling* that plays near-optimally across a battery of sealed cells (mean SMAD 6.68%). That figure did not survive a provenance audit: the log tree from which it was computed (`recovered_logs/experiment_logs_with_explanation/V10/`) is GPT-4o in every run configuration—not gpt-5-mini—and shares most of its run identifiers with the anchor logs, so it is discarded. Two data sources now replace it. First, a genuine gpt-5-mini sealed-bid mechanism battery (Table 13), whose across-format mean SMAD *against truthful value* is 19.09% over the six robustness formats (or 16.6% if the genuine clock cell below is included)—the 6.68% figure is retired, not merely re-sourced. Second, and new to this draft, a full intervention grid run on four frontier models—gpt-5-mini, gpt-5, claude-sonnet-5, and gemini-2.5-flash (canonical identifiers; family `es_v12` in `results/merged_ranking/auction_cells.csv`)—which replaces the earlier claim that “no gpt-5-mini intervention cells exist anywhere.” The empty gpt-5-mini clock cells that motivated that claim are now explained: a configuration typo (`configs/clock/01_07_ascending_clock_apv_closed_gpt5mini.yaml` carried the invalid model id `gpt-4-mini`) suppressed the closed-clock robustness runs; with the typo fixed and the frontier battery’s own `ascending_clock_closed` cell in hand ($K = 50$, SMAD 1.25%), a genuine gpt-5-mini clock number exists and supersedes the empty cells.

The frontier intervention grid: truthful at baseline, but the backfire rungs retain teeth. Table 14 reports the ten-cell lever grid across the four frontier models, entries recomputed from `auction_cells.csv` as mean deviation from value (dollars) with SMAD in parentheses. Three facts stand out. (i) *All four frontier models are essentially exactly truthful at every sealed baseline*: mean deviations of 0.00–0.30 and SMAD of 0–2.3% across the sealed baseline cells, and the safety, payoff-tree, second-order-belief, and clock-framing description cells are inert (they land on the same ≈ 0 play, because there is no baseline error left to preserve). (ii) *The menu restatement still breaks the frontier*, exactly the backfire the ranking predicts for its restatement rung: it moves claude-sonnet-5 catastrophically away from truthful play (mean deviation $-\$8.71$, SMAD 35.5%, $d = -3.8$, $p < 10^{-58}$) and degrades gemini-2.5-flash (SMAD 2.3% $\rightarrow$ 9.1%, $p < 10^{-8}$), while leaving gpt-5 and gpt-5-mini exactly truthful—the same sign-mixed, model-specific pattern the canonical menu cell shows, now with one model driven to a 35% deviation. (iii) *What looks like a clock inversion is a description-triggered retrieval failure*. Relative to its near-perfect sealed play, the closed clock raises claude-sonnet-5 from 0.27% to 8.6% SMAD, with 75% of exits more than a dollar early; but the legacy clock cells carry an affiliated-values *description*, and claude-sonnet-5’s exit rationales explicitly invoke the winner’s curse (“others’ persistence signals possibly similar high values, risking overpayment”)—a script that is correct in common-value clocks and inapplicable here, where the bidder knows its own value. A clean IPV-described rerun of the same cell (`ascending_clock_ipv_closed`, $K = 50$) restores near-perfect play: SMAD 2.0%, early exits 3%, marginal-reasoning traces (“price equals my value; drop”). One sentence of affiliation language costs an otherwise exactly-truthful model eight SMAD points by activating the wrong playbook—the same failure mode as the menu cell, where its traces announce the *first-price* equilibrium formula $(n-1)/n \cdot v$ and execute it flawlessly ($-\$8.71 \approx -\frac{1}{3}\mathbb{E}[v]$). gpt-5 (+\\$0.61) and gpt-5-mini (+\\$0.28) shift only by roughly the \\$0.5 tick granularity, with clean marginal reasoning throughout.

The gpt-5 cells carry reduced and uneven K : the model frequently omits the required response tags, and unparseable repetitions are dropped, so per-cell n (in bids) ranges from 15 to 150 (102 for the SPSB baseline after its final top-up) (reported in Table 14); the n column should be read as a sample-size caveat; the run set is now frozen. gpt-5’s DA cells were deliberately not run.

[TODO: gpt-5 intervention cells have reduced/uneven K (15–150 parseable bids) from response-tag omissions; top up and re-freeze n before the final draft. The frontier-battery runbook is `plan/FRONTIER_`

RUNBOOK.md.]

The genuine cells sharpen, rather than remove, the paper’s scope condition: *frontier reasoning dissolves the bounded-rationality constraints where optimal play is textbook, and only there*. In the dominant-strategy format, gpt-5-mini is essentially perfect—mean deviation $-\$0.05$ (SMAD 0.2%) in SPSB-IPV and $-\$0.34$ (1.4%) in SPSB-APV, truthful play at levels the canonical models reach only under the strongest levers. In FPSB-IPV, its bids sit $\$8.53$ below value on average, i.e., almost exactly on the risk-neutral equilibrium shading rule $\frac{n-1}{n}v$ (mean deviation from the RNE benchmark $+\$0.002$; SMAD against RNE 0.1%)—it applies the textbook formula.²¹ But in the *third-price* format, where the equilibrium bid $\frac{n-1}{n-2}v$ exceeds value and no textbook formula is rehearsed in training corpora, the frontier model fails: with $n = 3$ it overbids value by $+\$14.76$ on average yet remains $\$10.87$ below the equilibrium bid $2v$ (SMAD 60.2% against truthful play, 22.5% against the equilibrium benchmark); with $n = 5$ the same pattern appears at smaller scale ($+\$4.39$ above value, $\$4.11$ below the $\frac{4}{3}v$ equilibrium). It recognizes that third-price rules reward aggression but cannot compute how much. The common-value cells show similarly dispersed behavior (deviations relative to the private signal $-\$2.91$ and $-\$2.60$, with wide spreads; descriptive only, as no dominant-strategy benchmark exists).

The implication for the main text is a two-sided scope condition. On one side, at the frontier the gaps that the design levers close have vanished *in the strategy-proof formats the ranking studies*—confirmed directly by the intervention grid of Table 14, in which every safety, tree, and belief cell is inert because there is no baseline error left to preserve—which is why reasoning models are excluded from the main treatment grid, and why the ranking’s practical relevance turns on deployed agent fleets occupying the sub-frontier regime for cost reasons (Section 8). On the other side, the third-price cells show that frontier reasoning does *not* uniformly dissolve bounded rationality: where the required strategic computation is non-obvious, even the frontier model deviates by more than any canonical model deviates in second price. And the frontier grid shows that the backfire rungs keep their teeth—through a distinctive frontier failure mode: surface language selects which memorized playbook runs. The menu wording re-classifies the game as first-price for claude-sonnet-5 (a 35% deviation from flawlessly executing the wrong formula), and one sentence of affiliation language in the legacy clock description triggers a misapplied winner’s-curse script that a clean IPV description removes entirely (Table 14). Whether the levers of Section 6 would repair a frontier model’s third-price play—a format the grid does not yet cover—remains the natural next question.

Format	n bids	Mean dev. vs. value (\$)	SMAD vs. v (%)	Mean dev. vs. b^* (\$)	SMAD vs. b^* (%)
SPSB, IPV	87	-0.05 $[-0.15, 0.00]$	0.2	-0.05	0.2
SPSB, APV	87	-0.34 $[-0.53, -0.18]$	1.4	-0.34	1.4
FPSB, IPV	87	-8.53 $[-9.33, -7.74]$	34.8	$+0.00$	0.1
TPSB, IPV ($n=3$)	87	$+14.76$ $[12.32, 17.28]$	60.2	-10.87	22.5
TPSB, IPV ($n=5$)	130	$+4.39$ $[2.83, 6.19]$	26.5	-4.11	18.2
FPSB, CV	90	-2.91 $[-5.19, -0.74]$	36.6	—	—
SPSB, CV	90	-2.60 $[-4.93, -0.37]$	36.9	—	—

Table 13: Genuine gpt-5-mini sealed-bid mechanism cells (robustness_logs/*_gpt5mini; brackets are bootstrap 95% CIs on the mean deviation vs. value). b^* is each format’s own benchmark: truthful bidding in SPSB, $\frac{n-1}{n}v$ in FPSB, $\frac{n-1}{n-2}v$ in TPSB. CV deviations are relative to the private signal and carry no dominant-strategy benchmark. The gpt-5-mini ascending-clock and intervention cells appear separately in Table 14 (frontier grid).

²¹The cell-level data (results/merged_ranking/auction_cells.csv) report deviations against both truthful bidding and each format’s own equilibrium benchmark; against truthful bidding the FPSB cell reads SMAD 34.8%, which measures correct equilibrium shading, not error.

Lever grid cell	Family	Mean dev. vs. value, \$ (SMAD vs. v , %)			
		gpt-5-mini	gpt-5 [†]	claude-sonnet-5	gemini-2.5-flash
<i>Sealed baselines</i>					
SPSB, IPV baseline	—	0.00 (0.0)	0.00 (0.0; $n=102$)	−0.07 (0.3)	−0.30 (2.3)
Contingent baseline (axis 1)	—	0.00 (0.0)	0.00 (0.0; $n=69$)	0.00 (0.0)	−0.16 (0.9)
Beliefs baseline (axis 3)	—	0.00 (0.0)	0.00 (0.0; $n=45$)	0.00 (0.0)	0.00 (0.2)
<i>Description / scaffold levers</i>					
Payoff Safety	B2	0.00 (0.0)	0.00 (0.0; $n=132$)	0.00 (0.0)	0.00 (0.0)
Payoff Tree	C1	0.00 (0.0)	0.00 (0.0; $n=72$)	0.00 (0.0)	−0.01 (0.5)
Second-order beliefs	C3	0.00 (0.0)	0.00 (0.0; $n=54$)	0.00 (0.0)	−0.07 (0.5)
Clock framing (text)	B3	0.00 (0.0; $n=36$)	0.00 (0.0; $n=27$)	0.00 (0.0)	−0.29 (1.2)
Two-stage exit description	B3	0.00 (0.0)	0.00 (0.0; $n=87$)	0.00 (0.0)	−0.04 (0.2)
Menu restatement	B1	0.00 (0.0)	0.00 (0.0; $n=15$)	−8.71 (35.5)	−2.19 (9.1)
<i>Extensive form</i>					
Ascending clock (AC-B)	A	+0.28 (1.3)	+0.61 (2.7)	−2.08 (8.6)	−0.16 (0.8)

Table 14: Frontier intervention grid: the ten-cell lever grid on four frontier models, recomputed from `results/merged_ranking/auction_cells.csv` (family `es_v12`; canonical model ids). Entries are mean deviation from value in dollars, with SMAD (% of $\mathbb{E}[b^*] = 24.5$) in parentheses; each cell is $K = 50$ ($n \approx 150$ bids) except as noted. All four models are near-exactly truthful at every sealed baseline, and the safety, tree, belief, and clock-framing description cells are inert (no baseline error remains to preserve). Two backfire rungs retain teeth: the *menu restatement* breaks claude-sonnet-5 (mean dev $-\$8.71$, SMAD 35.5%, Cohen’s $d = -3.8$, $p < 10^{-58}$) and degrades gemini-2.5-flash (SMAD 2.3% $\rightarrow$ 9.1%, $p < 10^{-8}$) while leaving gpt-5/gpt-5-mini exactly truthful; and the apparent clock *inversion* (claude-sonnet-5 SMAD 0.3% $\rightarrow$ 8.6%) is a description-triggered winner’s-curse misfire from the legacy cell’s affiliation language, removed entirely by a clean IPV-described rerun (SMAD 2.0%; see text); gpt-5 +\\$0.61 and gpt-5-mini +\\$0.28 are tick-granularity shifts. [†]gpt-5 cells have reduced and uneven parseable K (the model often omits response tags; unparseable repetitions are dropped): per-cell n in bids is noted in the column—read n as a caveat; the run set is frozen. gpt-5’s DA cells were deliberately not run.

E Preference-Framing Interventions

In addition to the extensive-form, description, and reasoning-scaffold levers reported in the main text (Families A–C of Section 3), we test a separate family of interventions—Family D—inspired by prospect theory (Kahneman and Tversky, 1979) and behavioral departures from expected utility theory. These interventions do not scaffold strategic reasoning; instead, they reframe the payoff structure to test whether LLMs exhibit sensitivity to framing effects, loss aversion, the endowment effect, and risk attitudes. Because truthful play is a dominant strategy in both of our environments regardless of risk preferences, none of these framings should move a rational agent; expectations-based loss aversion has nonetheless been proposed as an explanation for seemingly dominated choices by humans in strategy-proof matching mechanisms (Dreyfuss et al., 2022), which makes the family a natural placebo-adjacent battery for our testbed. We test these framings across both domains (SPSB auctions and direct DA) with the same four model families as the main grid (GPT-4o, Claude 3.5 Haiku, Gemini 2.0 Flash, Gemma 27B). In the assembled ranking of Section 6.4, Family D populates the bottom rungs: the risk-averse persona is the worst lever we measure, with provisional preservation index $\rho \approx -0.76$ in auctions and $\rho \approx -3.5$ in DA (Table 5).

[TODO: All Family-D cells below are point estimates from the ES runs (auction cells: IPV sealed-bid grid, seed 1299, $K = 50$ auctions per cell; DA cells: 50 market instances per cell). E6 statistics (SEs, MW/Welch tests, Cohen’s d , bootstrap CIs on ρ) and the harmonized re-runs (plan E3/E8) are pending; report SMAD alongside signed means when recomputed.]

E.1 Loss aversion and framing effects

We test five variants that reframe the payoff description while holding the mechanism and underlying payoffs constant:

- *Gain Frame*: Presents all outcomes as gains from zero (“you cannot lose money—you can only gain”).
- *Loss Frame*: Endows the agent with an initial amount equal to their maximum possible value, then presents outcomes as losses from this reference point (“Think carefully about what you might lose”).
- *Mixed Frame*: Explicitly decomposes each outcome into a GAIN component and a LOSE component, asking the agent to weigh both.
- *Endowment*: Gives the agent an initial endowment and frames payment as a deduction “from your endowment,” targeting the endowment effect.
- *WTA/WTP*: Asks the agent to compare their willingness-to-accept (selling price) with their willingness-to-pay (buying price), invoking the classic WTA–WTP gap.

Loss Aversion Interventions

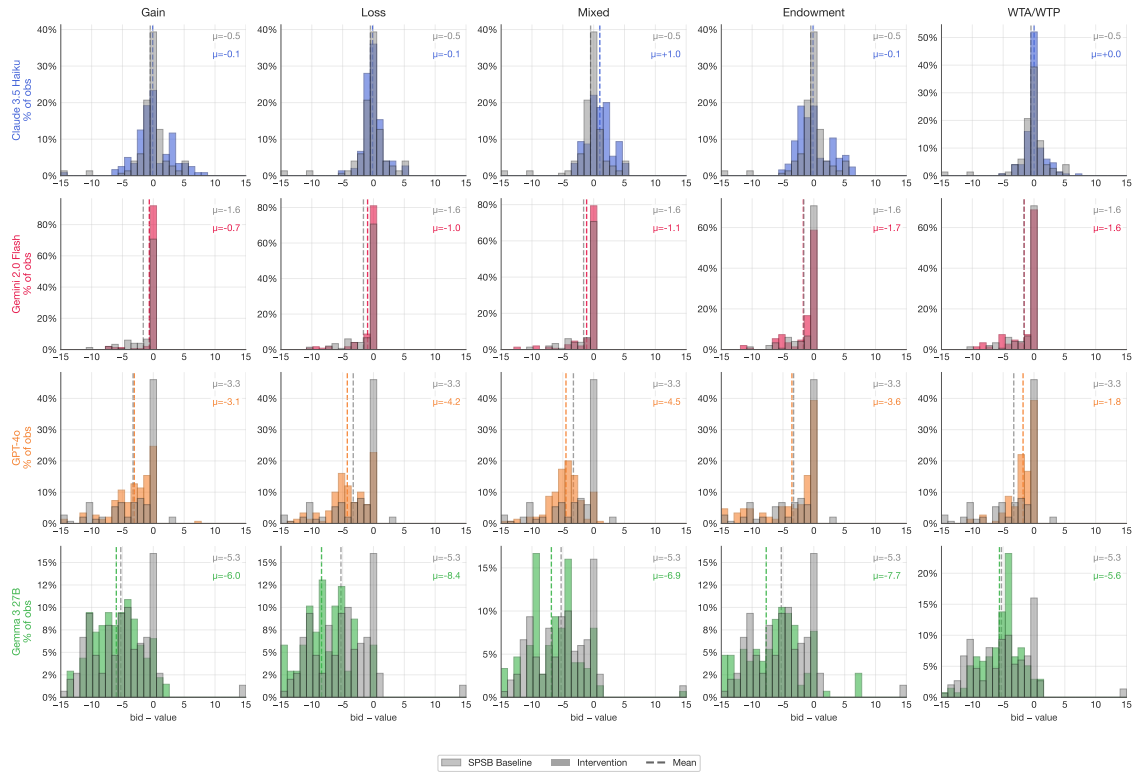


Figure 12: Loss-aversion and framing interventions in SPSB auctions (mean bid deviation from value, \$). Baseline SPSB in gray, intervention in color; dashed lines indicate means. None of the framing interventions consistently improve play relative to the SPSB baseline ($\mu = -2.67$). The loss frame ($\mu = -3.44$) and endowment ($\mu = -3.40$) worsen underbidding, while the gain frame ($\mu = -2.51$) and WTA/WTP ($\mu = -2.17$) show modest, inconsistent effects.

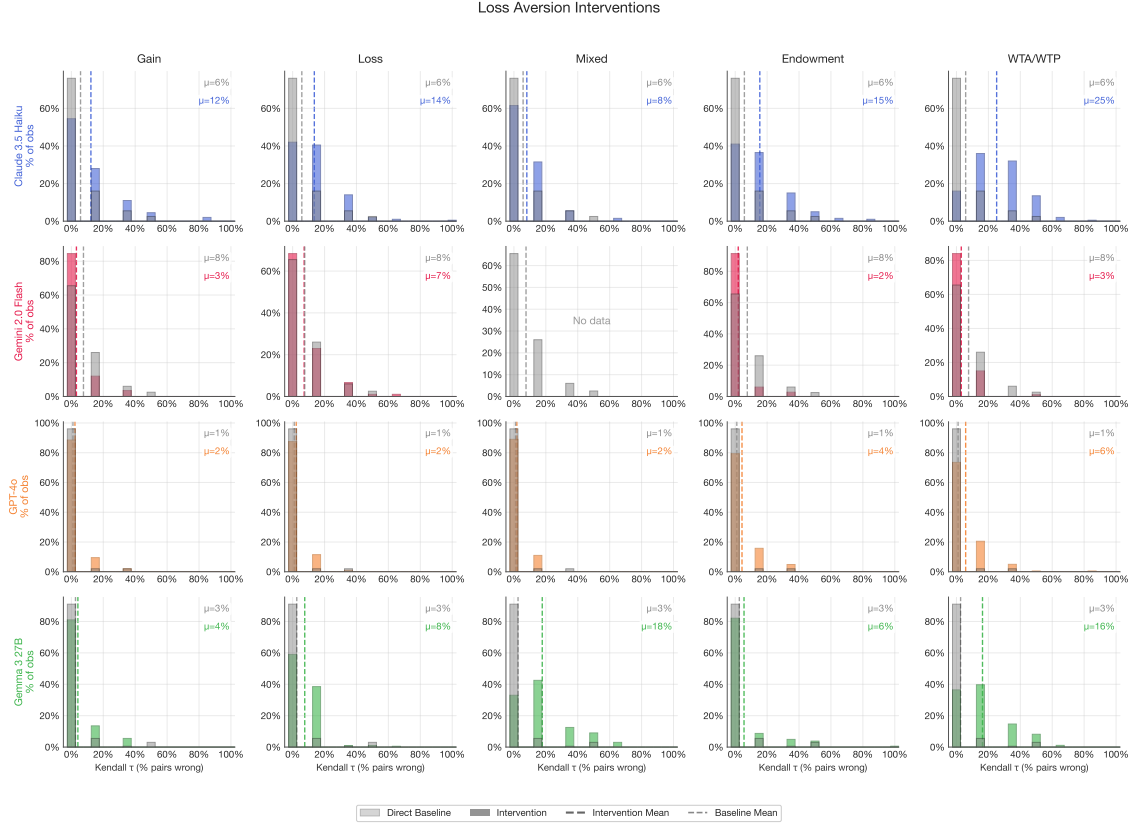


Figure 13: Loss-aversion and framing interventions in deferred acceptance (mean normalized Kendall- τ error, %). Baseline direct DA in gray, intervention in color. The results are noisy and largely negative: most framing interventions worsen play relative to the baseline ($\mu = 4.2\%$), with WTA/WTP producing the worst outcomes ($\mu \approx 12.6\%$ across models with data). Claude 3.5 Haiku is particularly sensitive to these framings.

In auctions, the framing interventions do not produce consistent improvements. The loss frame and endowment interventions worsen underbidding (overall means of $\mu = -3.44$ and -3.40 , compared to the SPSB baseline of -2.67), opposite to what a naive application of loss aversion would predict. The gain frame and WTA/WTP interventions produce small, model-dependent effects. In DA, the interventions are uniformly harmful, with WTA/WTP and endowment producing the largest increases in error.

The overall picture is that prospect-theoretic framings do not improve play—and often worsen it. This contrasts with the levers in the strong tiers of the ranking (Section 6): contingent-reasoning scaffolds and safety-exposing mechanism descriptions produce large, consistent improvements. The difference is informative: the cognitive barrier in these mechanisms is not about how payoffs are framed, but about understanding the mechanism’s strategic structure.

E.2 Risk-preference personas

We also test whether assigning explicit risk attitudes to LLMs changes behavior. Each intervention instructs the model to adopt a calibrated risk preference via a concrete coin-toss example (full persona text in Appendix F):

- *Risk Averse*: “You would only pay \$4 for a coin toss worth \$0 or \$10 (expected value \$5).”
- *Risk Neutral*: “You would pay \$5 for a coin toss worth \$0 or \$10.”
- *Risk Seeking*: “You would pay \$6 for a coin toss worth \$0 or \$10 (expected value \$5).”

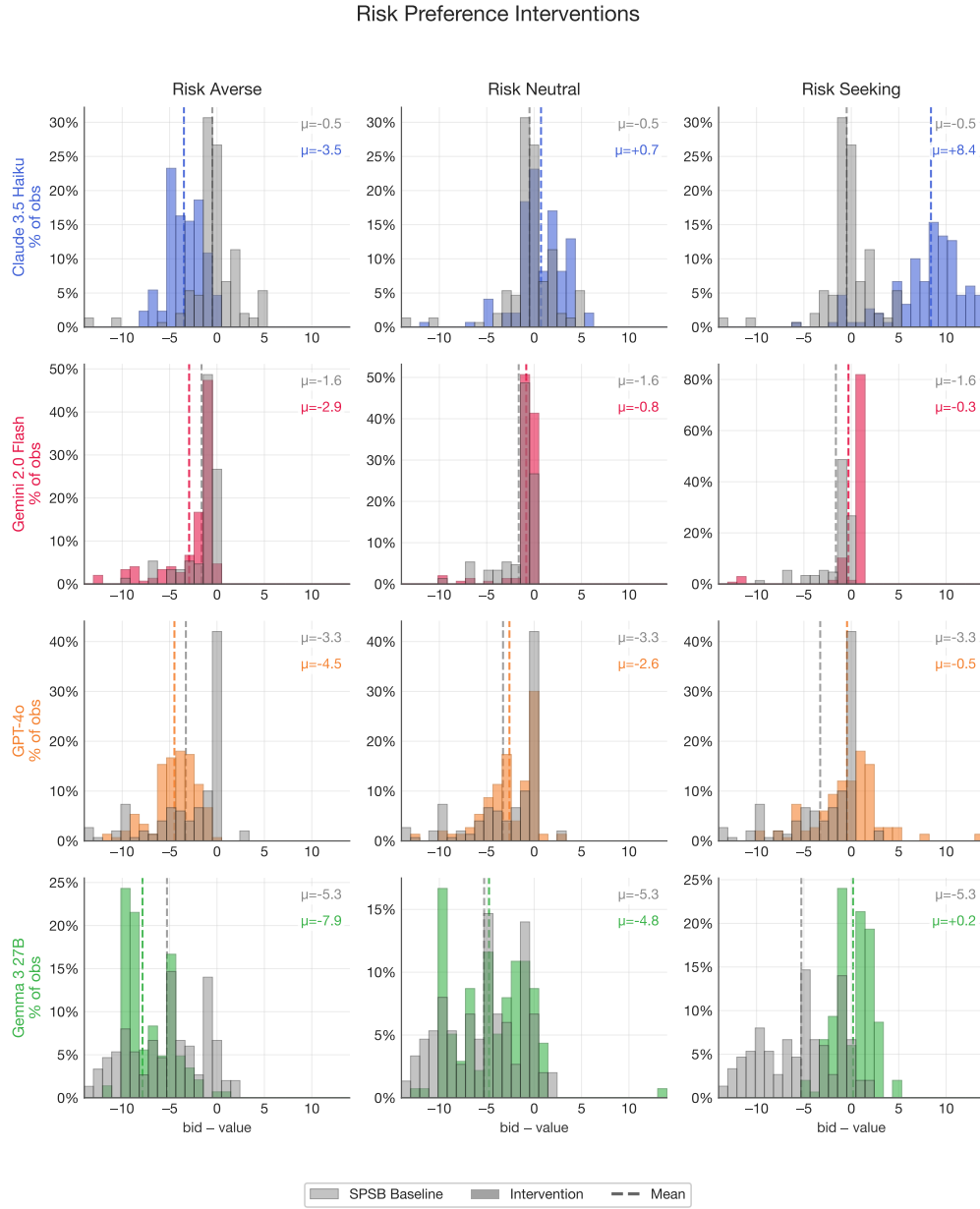


Figure 14: Risk-preference personas in SPSB auctions (mean bid deviation from value, \$). Risk aversion increases underbidding ($\mu = -4.71$), while risk-seeking dramatically shifts Claude 3.5 Haiku to overbidding ($\mu = +8.40$). Risk neutrality ($\mu = -1.82$) modestly improves play relative to the SPSB baseline ($\mu = -2.67$).

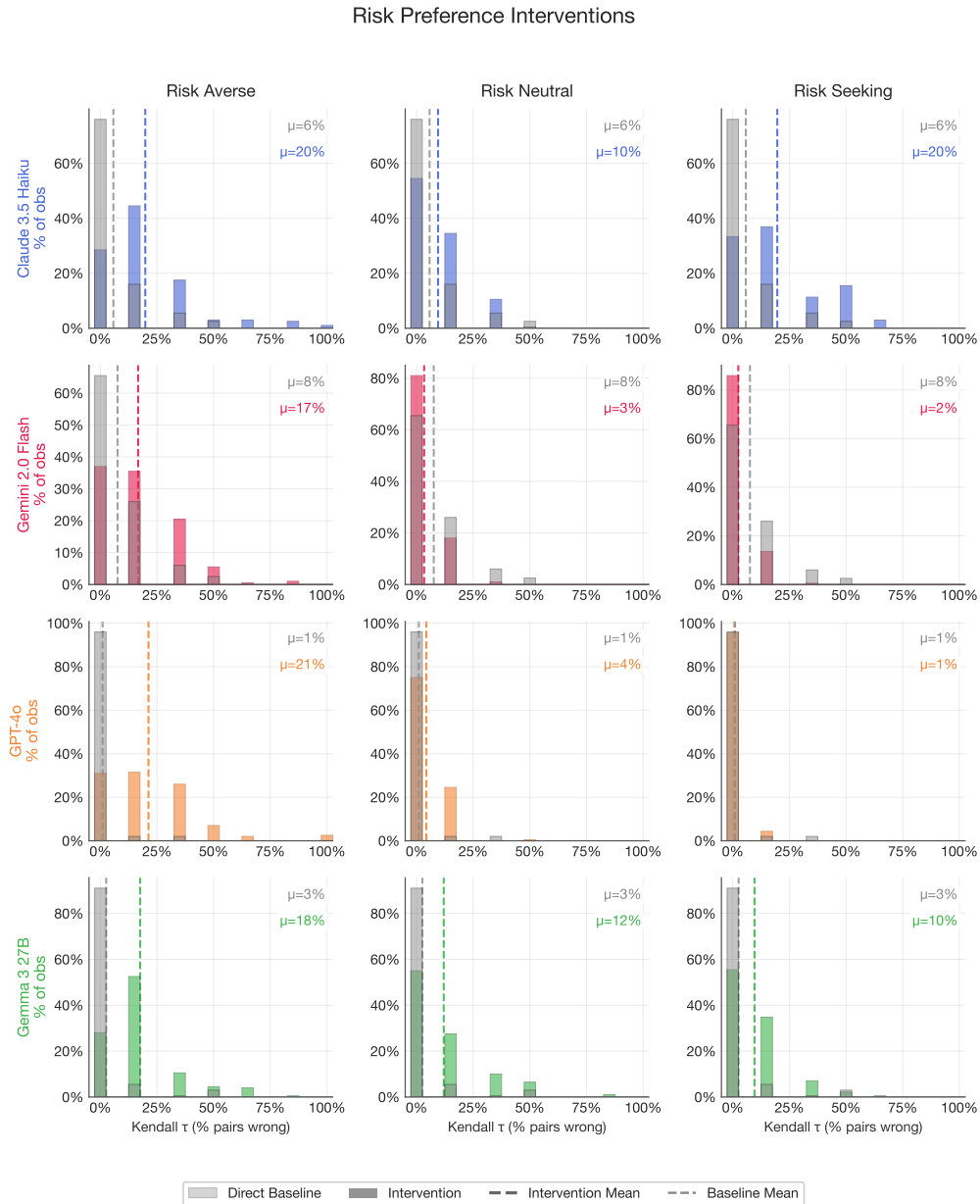


Figure 15: Risk-preference personas in deferred acceptance (mean normalized Kendall- τ error, %). Risk aversion dramatically worsens play ($\mu \approx 18.8\%$, compared to baseline 4.2%). Risk neutrality and risk-seeking also worsen play, though less severely.

The risk interventions reveal that LLMs are highly responsive to persona-based instructions about risk attitudes, but that this responsiveness does not improve play. The risk-averse persona produces the most

dramatic deterioration: in auctions, it increases underbidding from $\mu = -2.67$ to -4.71 ;²² in DA, it roughly quadruples the error rate ($\mu \approx 18.8\%$ against the 4.2% baseline)—these two cells are the source of the bottom-rung ρ values quoted above. The risk-seeking persona shifts Claude 3.5 Haiku to massive overbidding ($\mu = +8.40$), while the other models are less affected. Risk neutrality modestly improves auction play ($\mu = -1.82$) but worsens DA play. In aggregate, the models—while strategizing—are extremely susceptible to risk-preference priming.

These results underscore that the LLMs’ failures in strategy-proof mechanisms are not driven by implicit risk attitudes. Truthful play is dominant regardless of risk preferences in both environments, so the fact that risk-persona instructions change behavior at all confirms that the models do not understand the mechanism’s incentive structure. The risk interventions change *how* models deviate from truth-telling, but do not address *why* they deviate in the first place. For the ranking, the practical reading is a warning: persona-style “preference” instructions—a common ingredient in deployed agent stacks—are not neutral wrappers, and the risk-averse persona in particular is the most reliably harmful lever in our entire battery.

F Prompt Library and Intervention Inventory

This appendix documents the prompt templates behind every design lever used in the paper, organized by the taxonomy of Section 3: Family A (extensive-form implementations), Family B (mechanism descriptions: B1 menu restatement, B2 safety-exposing, B3 clock-framing), Family C (reasoning scaffolds: C1 contingent, C2 forward planning, C3 beliefs), and Family D (preference framings, Appendix E). We present the full base-rule prompts for both domains; for each intervention we show only the text that is *added to or modified from* the corresponding base prompt, since the surrounding context (bidder identity and response format in auctions; preference ordering, priorities, popularity signal, and output format in DA) remains identical. Template variables (e.g., `{{student_id}}`, `{{preference_order}}`, `{{increment}}`) are populated at runtime. The simulation pipeline that embeds these prompts is described in Appendix B; the DA protocol details are in Appendix H. Section F.6 closes with the complete inventory (Table 15), which maps every repo-level template name to its final taxonomy family and records whether its results appear in this paper.

F.1 Base rules

SPSB auction (IPV baseline). The canonical baseline for the auction ranking cells (repo template `private_second_price`):

²²The per-model risk-averse cells reproduce from the imported pipeline (`writeup/auction-v2-numbers.txt`): Claude -3.49 , Gemini -2.94 , GPT-4o -4.49 , Gemma -7.86 , pooling to -4.69 , which matches the -4.71 shown in Figure 14 up to rounding and cell-completion differences. [TODO: Finalize on the harmonized re-run (plan E3/E8); note the per-axis baseline-pooling caveat flagged in Section 6.]

In this game, you will participate in an auction for a prize against `{{num_bidders}}` other bidders. You will play this game for `{{n}}` rounds.

At the start of each round, bidders will see their value for the prize, randomly drawn between \$0 and `{{private}}`, with all values equally likely.

After learning your value, you will submit a bid privately at the same time as the other bidders. Bids must be in `{{increment}}` increments.

The highest bidder wins the prize and pays the second-highest bid. If you win, your earnings will increase by your value for the prize, and decrease by the second-highest bid. If you don't win, your earnings will remain unchanged.

After each auction, we will display all bids. Ties for the highest bid will be resolved randomly.

Figure 16: SPSB auction, base rule (`private_second_price`). Standard second-price sealed-bid auction with independent private values. The FPSB, TPSB, APV, CV, and eBay rule prompts used in the validation exhibits (Section 5) differ only in the allocation/payment paragraph and the value-model paragraphs; the eBay treatment prompts are summarized in Appendix I.

Direct DA (traditional description). The canonical DA baseline (`da_direct_traditional`) describes the full five-step student-proposing algorithm:

You are Student `{{student_id}}`. There are 4 schools: w, x, y, z.
You will be matched to at most one school.

HOW THE MATCHING WORKS (Deferred Acceptance):

1. All students simultaneously submit a ranking of schools
2. Each student "proposes" to their top-ranked school
3. Each school tentatively accepts the proposer it ranks highest (by priority), rejecting others
4. Rejected students propose to their next choice
5. This repeats until all students are matched

Your preference ordering (most preferred to least preferred):
`{{preference_order}}`

Your priority rank at each school (1 = highest priority):
w: `{{pw}}`, x: `{{px}}`, y: `{{py}}`, z: `{{pz}}`

Moreover, most applicants seem to favor `{{global_ranking}}`.

Submit a complete ranking of schools from most preferred to least preferred.

Format: Ranking: <1st> > <2nd> > <3rd> > <4th>

Figure 17: Direct DA, base rule (`da_direct_traditional`). A *null* variant (`da_direct_null`) is identical except that the "HOW THE MATCHING WORKS" block is removed entirely, providing no mechanism explanation; its results appear in Appendix H (Table 16).

F.2 Family A: extensive-form implementations

Ascending clock (AC-B), per-tick query. In the true iterative clock (final ID `osp_clock_iterative`), each auction consists of multiple clock cycles; at each posted price every remaining bidder is queried, and no dropout information is displayed, as in Breitmoser and Schweighofer-Kodritsch (2022):

```
Your value towards to the prize is {value} in this round.  
The current price in this clock cycle is {current_price}.  
The price for next clock cycle is {current_price + increment}.  
  
Do you want to stay in the bidding?  
If you choose yes, you can keep bidding for next clock. If you  
choose No, you will exit and have no chance to re-enter the  
bidding. Your response must use these EXACT tags below. You  
must output the ACTION.  
    <PLAN> [...limit 50 words...] </PLAN>  
    <ACTION> Yes or No </ACTION>
```

Figure 18: Ascending clock (AC-B), per-tick stay/exit query (final ID `osp_clock_iterative`). The bidder is first shown the auction rules (value model plus clock payment rule); the query above is repeated at every clock tick. See Appendix B for the full clock simulation loop and the AC (open) variant. On the `intervention_proxy_breitmoser` name collision affecting this lever, see Section F.6.

Iterative DA, binary-offer query. The primary prompt at the binary-offer nodes of the OSP decision tree of Ashlagi and Gonczarowski (2018); it explicitly states the consequence of each answer, providing the “obviousness witness” of that construction:

```

You are Student {{student_id}}. We are running a step-by-step
matching procedure.

Current remaining schools: {{remaining_set}}

Your preference ordering (most preferred to least preferred):
  {{preference_order}}

Your priority rank at each school (1 = highest priority):
  w: {{pw}}, x: {{px}}, y: {{py}}, z: {{pz}}

Moreover, most applicants seem to favor {{global_ranking}}.

Question: Among the remaining schools ({{remaining_set}}), is
  {{candidate}} your most preferred?
- If you answer YES: you are immediately matched to
  {{candidate}} and leave the process.
- If you answer NO: {{candidate}} remains available to you.
  You continue in the process with all current remaining
  schools ({{fallback_set}}).

<REASON> ... </REASON>
<DECISION> Answer: YES / NO </DECISION>

```

Figure 19: Iterative DA, yes/no query with consequence explanation (da_osp_yesno). At serial-dictatorship nodes, a companion prompt (da_osp_choice) instead asks the student to name their single most preferred school among the currently available set (“Do not list a full ranking. Do not condition on hypothetical future steps.”).

F.3 Family B: mechanism descriptions

B1 — Menu restatement (auction). Based on Gonczarowski et al. (2023), this replaces the payment paragraph of the SPSB base rule with a menu description (intervention_menu):

```

Your "price to win" the item will be set to the highest bid
placed by any other player. If your bid is higher than this
"price to win," then you will win the item and pay this price.
If you don't win, your earnings will remain unchanged.

```

Figure 20: Menu restatement for SPSB (intervention_menu, Family B1). Reported in Section 6.2 for all four models: small, sign-mixed effects across models and null on average.

B1 — Menu restatement (DA). Two DA analogues exist as templates, da_direct_menu_property and da_direct_menu_mechanics; both restate DA through the obtainable-schools menu, following Gonczarowski et al. (2023). Both are reported in Appendix H, where they split sharply by content. The *property* variant reads:

KEY PROPERTY OF THIS MECHANISM:

Given the other students' rankings and school priorities, there is a fixed set of schools you could possibly be matched with (your "Obtainable Schools").

IMPORTANT: Your ranking CANNOT change what is in your Obtainable Schools. Your ranking ONLY determines which school you receive WITHIN that set - you get your highest-ranked school among the obtainable ones.

Figure 21: Menu restatement for DA, property variant (da_direct_menu_property, Family B1). The *mechanics* variant states the same construction procedurally ("Step 1: ...the mechanism computes your Obtainable Schools. Step 2: You receive the school you ranked highest among your Obtainable Schools.") *without* the invariance statement. Both are reported in Appendix H: the property variant significantly improves truthful reporting (pooled 4.2% $\rightarrow$ 1.8%), the mechanics variant significantly worsens it (4.2% $\rightarrow$ 6.9%).

B2 — Payoff Safety (auction). Surfaces the pivotality property behind Vickrey (1961)'s strategy-proofness argument (old labels axis2_forward_onestep / axis1_contingent_onestep; final ID *Payoff Safety*):

****How the auction works:****

Think of your bid as setting your "maximum price." The auction will find the highest price below your maximum that still wins. Specifically:

- If your bid is the highest, you win and pay the second-highest bid
- If your bid is not the highest, you don't win and pay nothing

This means: Your bid only matters for determining IF you win, not HOW MUCH you pay. You can think of it simply as: "At what price would I no longer want the item?"

Figure 22: Payoff Safety description for SPSB (Family B2). Despite the "axis-2 / forward-planning" repo filename, this is a mechanism description, not a reasoning scaffold: it states *what is true* of the payment rule rather than instructing a mode of reasoning (see the mapping notes in Section F.6).

B2 — Rejection Safety (DA). Surfaces the property that distinguishes DA from the Boston mechanism (Abdulkadiroğlu and Sönmez, 2003): rejections are safe (old label axis2_da_monotonic_safety; final ID *Rejection Safety*):

KEY INSIGHT - REJECTIONS ONLY REDIRECT, NEVER ELIMINATE:
 In this matching process:
 - Being rejected from a school you ranked highly does NOT hurt your chances at schools ranked lower
 - You cannot be "knocked out" of a school that would have accepted you
 - The algorithm protects you: rejections only happen BEFORE you would have been matched

Your ranking determines the ORDER in which you're considered, not WHETHER you're considered.

Figure 23: Rejection Safety description for DA (Family B2). Two sibling “monotonicity” templates (axis2_da_monotonic_options: “your options never shrink”; axis2_da_monotonic_outcome: “the obtainable set determines the outcome”) were run but are not reported (Table 15).

B3 — Clock-framing (auction). Based on Breitmoser and Schweighofer-Kodritsch (2022), this *describes* the sealed bid as an automatic exit threshold in a simulated clock—a description change only; the game remains a one-shot sealed-bid auction (repo template `intervention_proxy_breitmoser`; final ID `desc_clock_framing`):

```

**FIRST STAGE: Sealed Bid**
You will submit a sealed bid privately at the same time as the
other bidders. This bid will serve as your automatic exit price
in the next stage.
**SECOND STAGE: Ascending Clock (Simulation)**
After the sealed bid stage, we will simulate an ascending clock
auction. The clock will start at $0 and increase in increments
of ${{increment}}. The clock will display the current price.
You will also see that there are a total of {{num_bidders}}
bidders participating, although you do not know other bidder's
values. If the current price on the clock reaches or exceeds
your sealed bid from the first stage, you will automatically
exit the auction. The auction ends when only one bidder is left
remaining in the second stage based on their bid from the first
stage.
**END OF AUCTION**
If you win, your earnings will increase by your value for the
prize and decrease by the clock price at the end of the auction.
If you don't win, your earnings will remain unchanged.

```

Figure 24: Clock-framing description for SPSB (Family B3; final ID `desc_clock_framing`). The elicited action is still a single sealed bid; no per-tick queries occur. Never to be conflated with the true iterative clock of Figure 18; see the name-collision note in Section F.6.

F.4 Family C: reasoning scaffolds

C1 — Payoff Tree (auction). Presents the explicit decision tree of outcomes (old labels `axis2_forward_tree` and `axis1_contingent_tree`; final ID *Payoff Tree*; classified C1 contingent reasoning—see

Section F.6):

****Decision Tree:****

Consider your decision as follows:

You choose bid B . Then one of two things happens:

PATH A: Your bid $B >$ all other bids
-> You WIN
-> You pay: second-highest bid (call it P)
-> Your profit: (Your Value) - P
-> Note: $P < B$, so if Value $\geq B$, then profit ≥ 0

PATH B: Your bid $B \leq$ some other bid
-> You LOSE
-> You pay: \$0
-> Your profit: \$0

Key insight: In PATH A, you pay P (not B). So bidding higher than your value only matters if it changes whether you're in PATH A vs PATH B.

Figure 25: Payoff Tree scaffold for SPSB (Family C1). Non-ASCII arrows in the repo template are transliterated as $\rightarrow$.

C1 — Payoff Tree (DA). The DA analogue (`axis1_da_tree`) replaces the algorithm description with a tree view:

HOW THE MATCHING WORKS (Decision Tree View):
Think of DA as a tree of possibilities:
- ROOT: You propose to your 1st choice
- If ACCEPTED: You're matched to that school
- If REJECTED: You propose to your 2nd choice
- If ACCEPTED: You're matched to that school
- If REJECTED: You propose to your 3rd choice
- And so on...

At each branch, acceptance/rejection depends on the school's priorities and who else proposed there.

Figure 26: Payoff Tree scaffold for DA (Family C1). Additional C1 variants in both domains (*Enumerate*, *Dominated*, *Worst-Case*, *Backward-Induction* framings) were run but are not reported; see Table 15. The *Enumerate* variant, for example, asks: "What are the possible bids the other players might submit? For each possible bid they might make, what would be your best response?" (auctions), or the analogous question over rankings (DA).

C2 — Forward-planning lookahead (DA only). Scaffolds k -step simulation of the DA algorithm's internal rounds (Pycia and Troyan, 2023); SPSB has no analogous internal dynamics, so C2 is DA-only (Section 3). The $k=1$ variant (`axis2_da_1step`):

HOW THE MATCHING WORKS:

You submit a ranking. The algorithm processes your ranking sequentially:

- First, you "propose" to your top-ranked school
- If accepted, you're matched there
- If rejected, you propose to your second choice

THINK ONE STEP AHEAD:

Before finalizing your ranking, consider: If you are rejected from your first choice, which school would you propose to next? Does this affect how you should rank schools?

Figure 27: One-step lookahead scaffold for DA (Family C2, axis2_da_1step). The $k=2$ variant (axis2_da_2step) extends the framing to the first two potential rejections ("Plan your ranking with these two steps in mind"). The $k=0$ and $k=\infty$ (full-simulation) variants were run but are not reported.

C3 — Belief scaffolds (both domains). First- and second-order belief prompts appended to the base rules. In auctions (axis3_beliefs_firstorder, axis3_beliefs_secondorder):

[First-order]

The other bidders are rational agents who are also trying to maximize their earnings.

Before bidding, ask yourself: What do I expect the other bidders to bid? Given their values are also drawn randomly between \$0 and $\$ \{ \text{private} \}$, and they want to maximize their earnings, how do I think they will behave? Does this affect what I should bid?

[Second-order]

Before bidding, consider the following: What do the other bidders think YOU will bid? They know you are rational and trying to maximize your earnings. They might try to anticipate your strategy. Does their belief about your behavior affect what they will do? And does that, in turn, affect what you should do?

Figure 28: Belief scaffolds for SPSB (Family C3). The DA analogues (axis3_da_firstorder, axis3_da_secondorder) ask "What rankings do you think the other students will submit?" and "What do you think the other students believe YOU will rank?" respectively. A *Common Knowledge* variant in each domain (emphasizing common knowledge of rationality) was run but is not reported.

F.5 Family D: preference framings

The Family-D battery (results in Appendix E) replaces or augments the objective framing rather than the mechanism explanation. As one representative, the risk-averse persona (intervention_risk_averse, identical wording in both domains modulo context):

You are risk-averse. This means that you are the type of person that would only pay \$4 for a coin toss where you got \$0.00 on tails and \$10.00 on heads (even though the expected value is \$5.00). You prefer to avoid uncertainty and would rather have a guaranteed outcome than take a risky gamble, even if the gamble has the same or slightly higher expected value.

Figure 29: Risk-averse persona (Family D). Risk-neutral and risk-seeking personas change only the coin-toss price (\$5, \$6); the loss-aversion battery (gain/loss/mixed frames, endowment, WTA/WTP) is described in Appendix E.1.

F.6 Intervention inventory and old-to-new mapping

The prompt templates accumulated across two experimental programs and several repo versions, and three of the resulting labels are actively misleading. We fix the mapping here once, and Table 15 applies it to every template in the repo.

Taxonomy corrections (plan reconciliation R7). (i) *PayoffTree* carries the repo filename `axis2_forward_tree.txt` (V12) / `axis1_contingent_tree.txt` (imported pipeline), but SPSB is a one-shot game with no own future decision points, so there is no forward-planning axis to scaffold; the intervention lays out payoff-relevant *contingencies* and is classified C1. (ii) *Payoff Safety* (`axis2_forward_onestep / axis1_contingent_onestep`) and *Rejection Safety* (`axis2_da_monotonic_safety`) state true properties of the payment/rejection rule without instructing any mode of reasoning; both are descriptions (B2), not scaffolds. (iii) Clock-framing is a description (B3), not an extensive-form change.

The intervention_proxy_breitmoser name collision (plan reconciliation R6). The single template name `intervention_proxy_breitmoser` has been used for two different objects: (a) in the ES appendix, it labels the *true iterative clock* prompt (per-tick stay/exit queries, Figure 18); (b) in the repo template files (`rule_template/V10/`, `V12/`, and the imported pipeline’s `rule_template/auctions/`), it is the *two-stage sealed-bid clock-framing description* (Figure 24). We split the ID: `osp_clock_iterative` (Family A) vs. `desc_clock_framing` (Family B3). Our audit of the imported pipeline supports the split as follows: the logs stored under `experiment_logs/<model>/intervention_proxy_breitmoser/` record exactly one sealed bid per bidder per auction (no tick-by-tick decisions), i.e., they are `desc_clock_framing` runs; whereas the clock column of the OSP comparison (Figure 3) reproduces numerically from the pipeline’s `ascending_clock_closed` cells, which are true per-tick clock runs (in the APV environment—see the provenance flag in Section 6.1). The audit is settled: the pipeline template behind `intervention_proxy_breitmoser` is the *description* variant (`desc_clock_framing`), and the OSP clock column is computed from the `ascending_clock_closed` cells. With both IDs now reported for all four models (Sections 6.1 and 6.2), ranking runs A (true clock) and B3 (clock-framing) are empirically distinguished model by model.

Run-status audit. Table 15 records, for every declared template, whether its results are reported in this paper (with the section), were *run but not reported* (logs exist in `engineer_simplicity-main/experiment_logs/` or in the git-recovered writeup logs, but the cells do not appear in any exhibit), or were *not run* on the canonical grid. This implements the no-phantom-results rule (plan §7.8): no template is silently declared.

Comparison-baseline declaration. Two families of sealed-bid SPSB baselines coexist in the pipeline, and the paper uses them for different purposes: OSP and extensive-form comparisons quote the *dedicated* `spsb` cells (mean deviations $-0.49/-1.63/-3.28/-5.27$ for Claude/Gemini/GPT-4o/Gemma, matching the previously published $-0.5/-1.6/-3.3/-5.3$), while intervention contrasts use the *axis-specific* baselines (each `axisK_*` treatment against its same-axis baseline; the pooled axis baseline for the `intervention_*` cells). Both baseline sets, and every treatment cell, are recorded with bootstrap confidence intervals in `results/merged_ranking/auction_cells.csv`.

G Reasoning-Trace Analysis: Details

Every bid in our experiments is accompanied by a short free-text plan (median 68 words) in which the agent explains its choice. This appendix documents the measurement apparatus behind the trace results reported in the main text (Section 6.5): the corpus, the keyword dictionary, full prevalence tables, the pooled regression with its honestly modest explanatory power, the mediation detail behind the dissociation, and robustness to duplicate traces. Throughout, the plans are stated strategies, not full chains of thought; models may compute more (or less) than they verbalize, so every claim here concerns what models *say*, validated against what they *do*.

G.1 Corpus and Feature Dictionary

The corpus contains 21,990 traces from three sources: (i) 13,416 traces from the harmonized four-model grid (92 run directories covering the sealed-bid SPSB baseline and intervention cells plus the closed clock); (ii) 1,167 traces from the menu-restatement (B1) and clock-framing (B3) cells, all four models; and (iii) 7,407 recovered GPT-4o traces from the FPSB (3,711) and TPSB (3,696) variants of the same grid, which provide the cross-mechanism contrast. Duplicate second-price cells present in both (i) and (iii) were excluded from (iii) to prevent double counting. Each trace row carries its run directory as a cluster identifier; all standard errors below are cluster-robust at the run level. Ascending-clock traces (510 rows) are per-decision exit rationales, structurally different from sealed-bid plans; they are retained in the corpus but excluded from the pooled regressions.

We score each trace with eighteen binary keyword features (Table 17), applied as transparent regular expressions (full patterns in `analysis/build_trace_features.py`). The features were tuned on the language models actually produce: an initial dictionary keyed to the theory’s vocabulary (“no risk,” “can’t lose”) turned out to be essentially absent from real traces—the `safety_recognition` feature, which detects an echo of the Payoff Safety invariant, fires on 3 of 21,990 traces—so the safety/risk group was re-anchored on the words models use (“overpay,” “profit margin,” “buffer,” “zero profit”). The dictionary does no negation handling by design: `overpay_concern` counts “avoid overpaying” as overpayment rhetoric either way, which is the intended reading.

G.2 Prevalence: Model Fingerprints and Mechanism Insensitivity

Table 18 reports feature prevalence in the baseline SPSB cells by model; Table 19 reports prevalence by sealed-bid mechanism for GPT-4o. Pooling all 14,073 sealed second-price traces, the modal story a model tells itself is textbook *first-price* logic applied to a second-price auction: 53% state an intent to shade below value, 55% worry about overpaying, and 31% invoke a profit margin, while dominance language appears in 2.1% of traces, correct payment-rule rehearsal in 3.8%, and stated truthful intent in 6.8%.

Old label (repo template)	Final taxonomy family	Domain	Reported in paper?
<i>Panel A: Auctions</i>			
private_second_price	Baseline (SPSB, IPV)	Auction	Yes (§5, §6)
affiliated_spsb;	Baseline (APV); A (AC); A	Auction	Yes (§5, §6.1)
affiliated_ac;	(AC-B)		
affiliated_ac_closed			
intervention_proxy_breitmoser (ES-appendix usage) →	A (extensive form)	Auction	Yes (§6.1); clock column = ascending_clock_closed cells (provenance settled; see below)
osp_clock_iterative			
intervention_proxy_breitmoser (repo template) →	B3 (clock-framing)	Auction	Yes, all four models (§6.2); large improvement in every model ($p < 0.001$)
desc_clock_framing			
intervention_menu	B1 (menu restatement)	Auction	Yes, all four models (§6.2); sign-mixed, null on average
axis2_forward_onestep =	B2 (safety-exposing)	Auction	Yes (§6.2)
axis1_contingent_onestep →			
Payoff Safety			
axis2_forward_tree =	C1 (contingent)	Auction	Yes (§6.3)
axis1_contingent_tree → Payoff Tree			
axis1_contingent_{enumerate, dominated, worstcase}	C1 (contingent)	Auction	Run; the <i>Worst-Case</i> cell enters the trace analysis as a manipulation check (App. G); <i>Enumerate/Dominated</i> not reported
axis2_forward_backward_induct	C1 (two-stage framing)	Auction	Run, not reported
=axis1_contingent_backward_induct			
axis3_beliefs_firstorder;	C3 (beliefs)	Auction	Yes (§6.3)
axis3_beliefs_secondorder			
axis3_beliefs_common_knowledge	C3 (beliefs)	Auction	Run, not reported
axis1_contingent_baseline;	Per-axis baselines	Auction	Yes, as comparison baselines for axis-matched treatment cells (see the comparison-baseline declaration above)
axis2_forward_baseline;			
axis3_beliefs_baseline;			
loss_aversion_baseline			
intervention_risk_{averse, neutral, seeking}	D (risk personas)	Auction	Yes (App. E)
loss_aversion_{gain_frame, loss_frame, mixed_frame, endowment, WTA_WTP}	D (loss/endowment frames)	Auction	Yes (App. E)
intervention_NE_strat_reveal;	Excluded: advice (reveals or misstates the strategy; not a simplicity-preserving lever)	Auction	Run, not reported
intervention_wrong_strat_reveal;			
intervention_nash_deviation			
first_price_interventions/*;	Out of ranking scope (BNE benchmark, not dominant-strategy)	Auction	Partially run, not reported; parked per plan §6 scope decision
third_price_interventions/*			
robust_* (currency/language);	Legacy V10 robustness	Auction	Run in earlier drafts, not reported (cut, plan §7.9)
private_all_pay*			

Table 15: Intervention inventory (Panel A: auctions): every declared prompt template, its final taxonomy family (Section 3), and its run/reporting status. “Run, not reported” means raw logs exist in the imported pipeline (Engineering_simplicity/engineer_simplicity-main/experiment_logs/) or the git-recovered writeup logs, but the cell appears in no exhibit of this paper. Old labels are repo file-names; where the V12 writeup name and the imported-pipeline name differ, both are given with “=”. Panel B (deferred acceptance) continues in Table 16.

Old label (repo template)	Final taxonomy family	Domain	Reported in paper?
<i>Panel B: Deferred acceptance</i>			
da_direct_traditional	Baseline (direct DA)	DA	Yes (App. H)
da_direct_null	Baseline variant (no explanation)	DA	Yes (App. H); heterogeneous across models
da_direct_textbook_sp	Direct SP statement (reveals the dominant strategy)	DA	Yes, as a diagnostic ceiling (App. H): 0.0% error for all four models; excluded from the lever ranking
da_direct_menu_property; da_direct_menu_mechanics	B1 (menu restatement)	DA	Yes (App. H); property variant improves play, mechanics variant backfires
da_osp_yesno (+ A (iterative DA) da_osp_choice node prompt; da_osp_yesno_guaranteed)		DA	Yes (App. H); the pick-based protocol is the canonical iterative-DA cell, the yes/no tree appears in the decision-level accounting
axis1_da_tree → Payoff Tree (DA)	C1 (contingent)	DA	Yes (App. H)
axis1_da_{enumerate, dominated, worstcase, onestep, backward_induct}	C1 (contingent)	DA	Run, not reported
axis2_da_1step; axis2_da_2step	C2 (forward planning)	DA	Yes (App. H)
axis2_da_0step;	C2 (forward planning)	DA	Run, not reported
axis2_da_fullsim			
axis2_da_monotonic_safety → Rejection Safety	B2 (safety-exposing)	DA	Yes (App. H)
axis2_da_monotonic_options;	B2 variants	DA	Run, not reported
axis2_da_monotonic_outcome			
axis3_da_firstorder;	C3 (beliefs)	DA	Yes (App. H)
axis3_da_secondorder			
axis3_da_common_knowledge	C3 (beliefs)	DA	Run, not reported
intervention_risk_{averse, neutral, seeking} (DA)	D (risk personas)	DA	Yes (App. E)
loss_aversion_* (DA)	D (loss/endowment frames)	DA	Yes (App. E)
da_cardinal_variants (pipeline)	Environment variant (cardinal payoffs)	DA	Run, not reported

Table 16: Intervention inventory, continued (Panel B: deferred acceptance). Conventions as in Table 15. The eBay treatment prompts (T1–T4) belong to the eBay environment rather than the lever taxonomy and are catalogued in Appendix I.

Group	Feature	Fires when the plan...
Normative recognition	<code>dominance_language</code>	invokes dominance (“dominant strategy,” “regardless of others”)
	<code>truthful_intent</code>	states an intent to bid exactly one’s value
	<code>payment_rule_correct</code>	correctly rehearses that the winner pays the second-highest bid
	<code>second_price_mention</code>	name-drops “second-highest”/“second price” (weak rule mention)
	<code>first_price_mention</code>	calls the auction “first-price” (mechanism confusion)
Opponents & beliefs	<code>opponent_modeling</code>	speculates about other bidders’ values or bids
	<code>probability_reasoning</code>	uses chance/likelihood language
	<code>expected_value_reasoning</code>	computes or invokes expected value
Stated intent	<code>shading_intent</code>	states an intent to bid <i>below</i> value (value-anchored)
	<code>overbid_intent</code>	states an intent to bid <i>above</i> value (value-anchored)
Safety / risk rhetoric	<code>worst_case</code>	reasons about the worst case
	<code>safety_recognition</code>	echoes the B2 invariant (“bid determines if you win, not what you pay”)
	<code>overpay_concern</code>	worries about overpaying
	<code>zero_profit_fallacy</code>	argues that bidding one’s value yields zero profit
	<code>margin_language</code>	invokes a profit margin or buffer
Style	<code>conservative_language</code>	self-describes as cautious/conservative
	<code>aggressive_language</code>	self-describes as aggressive/competitive
	<code>risk_language</code>	uses risk vocabulary

Table 17: The trace feature dictionary: eighteen binary keyword features in five groups. Exact regular expressions and per-cell prevalence are in the replication package (`analysis/build_trace_features.py`; `results/traces/trace_features.csv`).

Each model has a recognizable fingerprint (pooled sealed second-price cells). Gemini is a near-pure shader (`shading_intent` 86%, `overbid_intent` 4.5%). Claude is the only model that explicitly mislabels the mechanism “first-price” (10.0% of its traces versus at most 0.1% elsewhere) and has both the most stated overbidding (30%) and the most stated truthful intent (12.6%)—the noisiest stated strategy. Gemma produces the least value-anchored language (`shading_intent` 10.6%) but the most opponent modeling (49%), conservative self-description (75%), and overpayment concern (83%); its vague, unanchored plans coincide with the largest bid errors. GPT-4o shades in 67.5% of traces and has the highest rate of the zero-profit fallacy (2.6%: “bidding my value risks zero profit”).

Mechanism-insensitive scripts. Restricting to GPT-4o’s *baseline* cells in each format, shading language appears in 70%/79%/72% of second-/first-/third-price traces, with mean bid-to-value ratios of 0.88/0.87/0.84 and mean deviations of $-2.78/-3.15/-3.40$: one script for all payment rules, under-shading where shading is optimal (FPSB) and over-shading where truthfulness is optimal (SPSB, TPSB). A useful side effect is a validity check on the dictionary itself: `shading_intent` prevalence is nearly identical where shading is correct (FPSB, 66.2%) and where it is an error (SPSB, 67.5%), so the feature measures the stated strategy, not its appropriateness. The cross-mechanism contrast is currently GPT-4o only (the FPSB/TPSB grids were recovered for that model alone); see Section G.6.

G.3 Stated Intent, Realized Bids, and the Pooled Regression

The traces are not cheap talk. Among the 7,469 sealed second-price traces stating an intent to shade, 97.1% of realized bids fall below value (mean deviation $-\$1.84$); among the 1,797 stating an intent to bid above value, 87.6% fall above (mean $+\$1.88$); among the 950 stating truthful intent, 77.6% land within $\pm\$0.50$ of

Feature (% of traces)	Claude	Gemini	GPT-4o	Gemma
dominance_language	0.0	0.7	0.0	0.0
truthful_intent	8.0	2.7	2.0	0.0
payment_rule_correct	11.3	0.0	9.3	0.0
second_price_mention	50.7	10.0	39.3	46.0
first_price_mention	18.7	0.0	0.0	0.0
opponent_modeling	8.0	4.7	20.0	73.3
probability_reasoning	49.3	22.7	38.0	18.0
expected_value_reasoning	3.3	2.7	2.0	0.0
shading_intent	54.0	88.0	61.3	4.0
overbid_intent	25.3	0.0	2.7	2.0
worst_case	6.0	8.0	2.7	1.3
safety_recognition	0.0	0.0	0.0	0.0
overpay_concern	59.3	22.7	45.3	89.3
zero_profit_fallacy	0.0	0.0	1.3	0.0
margin_language	54.7	29.3	30.7	13.3
conservative_language	6.0	14.7	12.7	86.0
aggressive_language	2.7	5.3	2.0	17.3
risk_language	52.0	6.7	32.7	22.7
Traces	150	150	150	150

Table 18: Feature prevalence in the baseline SPSB cells, by model (percent of traces; Claude 3.5 Haiku, Gemini 2.0 Flash, GPT-4o, Gemma 27B). Prevalence for every model $\times$ lever-family cell is in `results/traces/feature_prevalence_model_family.csv`; the model fingerprints quoted in the text pool all sealed second-price cells, not only the baselines shown here.

value (mean $|b - v|$ of \$1.03). The largest errors come from the 4,541 traces that state no value-anchored intent at all: mean deviation $-\$4.84$, mean absolute deviation \$5.51, with plans anchored on uninformative quantities (“bid slightly above half my value”). In a signed-deviation regression with model and lever-family fixed effects, stated overbid intent shifts the realized deviation by $+\$4.72$ (SE 0.36) relative to unanchored traces, with shading and truthful intents at $+\$0.99$ and $+\$0.91$; direction of stated intent, not eloquence, is what the bids track.

Table 20 reports the pooled feature regression of $|b - v|$ on the full dictionary with model and lever-family fixed effects, over the 14,073 sealed second-price traces. Two facts summarize it. First, articulating *any* value-anchored plan—shading, overbidding, or truthful—is associated with \$1.5–\$2.2 smaller absolute deviations than an unanchored plan, and correct payment-rule rehearsal with a further $-\$0.77$; conservative self-description is the one style feature with a robust positive (worse) association ($+\$0.76$). Second, the overall explanatory power is honest but modest: $R^2 = 0.251$ with features against 0.176 with fixed effects only, an incremental R^2 of ≈ 0.075 . Stated reasoning is informative about *how* an agent errs; it is far from a sufficient statistic for the error itself. We present this table as measurement discipline behind the prevalence and dissociation results, not as a headline finding.

G.4 Mediation Detail for the Dissociation

The main text reports a complete dissociation: across every lever tested, stated reasoning and bids never move together—Payoff Safety (B2) and the Payoff Tree (C1) move bids without moving stated reasoning, the worst-case scaffold moves stated reasoning without moving bids, and the menu restatement moves neither. All quantities below are computed against the *corrected* pooled axis baseline (axis-1 and axis-3 cells, $n =$

Feature (% of traces)	SPSB	FPSB	TPSB
dominance_language	0.1	0.2	0.3
truthful_intent	2.4	2.9	1.9
payment_rule_correct	6.1	0.0	0.0
second_price_mention	23.8	0.0	0.5
first_price_mention	0.1	0.2	0.0
opponent_modeling	22.0	24.5	22.2
probability_reasoning	37.7	43.4	39.5
expected_value_reasoning	3.7	8.3	5.2
shading_intent	67.5	66.2	60.0
overbid_intent	3.5	1.4	3.3
worst_case	4.2	4.3	4.9
safety_recognition	0.0	0.0	0.0
overpay_concern	49.5	32.3	45.6
zero_profit_fallacy	2.6	4.7	2.7
margin_language	31.1	30.2	27.5
conservative_language	12.6	13.0	14.6
aggressive_language	2.4	4.0	4.0
risk_language	49.2	44.7	46.7
Traces	3,600	3,711	3,696

Table 19: Feature prevalence by sealed-bid mechanism, GPT-4o, all cells pooled within mechanism (percent of traces; results/traces/feature_prevalence_mechanism_gpt4o.csv). The reasoning script is nearly invariant to the payment rule, although the optimum differs sharply across formats (truthful in SPSB; $\approx \frac{2}{3}v$ in the three-bidder FPSB; above value in TPSB). The rule-rehearsal rows (payment_rule_correct, second_price_mention) are mechanically zero outside SPSB, since the phrase they detect describes the second-price rule.

Feature	Coefficient (\$)	Cluster SE	<i>p</i>
shading_intent	−2.06	0.24	< 0.001
overbid_intent	−2.18	0.34	< 0.001
truthful_intent	−1.50	0.31	< 0.001
payment_rule_correct	−0.77	0.23	0.001
dominance_language	−0.84	0.53	0.118
zero_profit_fallacy	−0.82	0.37	0.026
overpay_concern	−0.29	0.13	0.026
worst_case	−0.28	0.21	0.179
second_price_mention	−0.21	0.21	0.307
first_price_mention	−0.14	0.34	0.688
opponent_modeling	+0.10	0.20	0.619
probability_reasoning	+0.10	0.10	0.347
expected_value_reasoning	+0.11	0.19	0.569
margin_language	+0.17	0.16	0.277
conservative_language	+0.76	0.14	< 0.001
aggressive_language	+0.01	0.59	0.980
risk_language	+0.06	0.14	0.640
safety_recognition	+8.26	8.24	0.316

Table 20: Pooled OLS of absolute bid deviation $|b - v|$ (in dollars) on the trace features, sealed second-price cells ($n = 14,073$), with model and lever-family fixed effects; standard errors cluster-robust by run directory. Intercept \$4.05 (SE 0.50). $R^2 = 0.251$; fixed effects only, 0.176; incremental R^2 of the text features ≈ 0.075 . The `safety_recognition` coefficient is meaningless noise (the feature fires on 3 of 21,990 traces) and is retained only for completeness. Full statsmodels output: `results/traces/ols_results.txt`.

1,188 traces): the legacy axis-2 “baseline” cell was itself a two-stage clock-exit description (a treated cell; see Section 6.4 and `results/merged_ranking/_axis2_baseline_provenance.md`) and is excluded from every baseline pool and analyzed as a treatment. Inference uses wild-cluster bootstrap p -values clustered at the run level with model-cluster permutation cross-checks and duplicate-trace robustness; the full pipeline is `analysis/build_trace_mediation.py` (seed 1299) with outputs in `results/traces/mediation/`.

Payoff Safety: behavior without verbalization. Treated traces ($n = 600$) improve from a mean deviation of -2.67 to -1.53 against the dedicated SPSB baseline cell (the published comparison), and from mean $|b - v|$ of 3.20 to 1.95 against the corrected pooled axis baseline (wild-cluster $p = 0.015$; model-cluster permutation $p = 0.36$, an honest four-cluster power limit). Yet the first stage of any classical mediation through verbalized understanding is zero: the invariant the description asserts is echoed in 0 of 600 treated traces (`safety_recognition` prevalence 0.000 treated vs. 0.000 control), correct payment-rule rehearsal is flat (2.8% treated vs. 3.9% control, $p = 0.60$), and rule name-dropping is flat (28.2% vs. 33.0%, $p = 0.61$). The product-of-coefficients decomposition accordingly bounds the comprehension-mediated share of the effect: point estimate ≈ 0 , 95% upper bound 35.4% on its absolute value (`results/traces/mediation/mediation_b2.csv`). The plans keep the same shading rhetoric and simply shrink the shade toward zero. Within the treated group, the minority of traces that do rehearse the payment rule err less (within-treatment gap in $|b - v|$ of $-\$1.40$; cluster-bootstrap 95% CI $[-2.15, -0.56]$)—a cross-sectional association, not evidence of mediation. One sign flag for the record: in the trace-aligned logs, Claude’s Payoff-Safety cell has a mean deviation of $+0.30$ (mild *overbidding*), whereas the ES manuscript

reports -0.30 ; the magnitude, and every absolute-deviation quantity used here, is unaffected, but signed summaries of that cell should quote $|\Delta| = 0.30$ with the sign discrepancy noted.

Worst-case scaffold: verbalization without behavior. The `axis1_contingent_worstcase` scaffold passes its manipulation check—worst-case language rises from 2.0% in the axis baseline to 43.0% under treatment (wild-cluster $p = 0.007$)—while behavior does not move: mean $|b - v|$ goes from 3.24 to 3.07 ($p = 0.81$; signed mean -2.42 to -2.81).

Payoff Tree and menu. The Payoff Tree is a second bids-only lever: mean $|b - v|$ falls from 3.20 to 1.29 (wild-cluster $p = 0.017$) while payment-rule name-dropping stays exactly at its baseline rate (33.0% vs. 32.7%, $p = 0.90$). An earlier draft reported the tree as moving language too (7.5% $\rightarrow$ 26.7%); that contrast was an artifact of the contaminated axis-2 cell, whose clock-exit narrative crowded out rule vocabulary and depressed the apparent baseline rate. The reclassified two-stage clock-exit cell itself passes a manipulation check as a treatment (clock-and-exit language 0% $\rightarrow$ 14.3%, $p = 0.002$) with a pooled behavioral null that conceals offsetting model effects (Gemma strongly helped, Gemini strongly harmed; Section 6.2). The menu restatement moves neither bids ($p = 0.66$) nor language, serving as the placebo row—with one measurement caveat: the menu prompt *removes* the “second-highest” vocabulary from the rules text, so its 0% rule-rehearsal prevalence is prompt echo by construction, and cross-lever language comparisons involving B1 are not meaningful. Within-lever comparisons (Payoff Safety versus the axis baselines, which share the standard rule text) are unaffected by this confound.

G.5 Robustness: Duplicates, Clusters, and Measurement Caveats

Byte-identical duplicates. Roughly 40% of sealed-bid traces are byte-identical duplicates within model $\times$ cell $\times$ value draw: at temperature 0.5 the models are near-deterministic conditional on the value, and every duplicate group shares a single value and a single bid. Full-sample standard errors are therefore optimistic, and the effective sample per cell is closer to the number of distinct value draws than to the row count. Re-running the pooled regression on deduplicated data ($n = 8,839$) leaves every headline coefficient stable: `shading_intent` $-2.06 \rightarrow -1.90$, `overbid_intent` $-2.18 \rightarrow -2.17$, `truthful_intent` $-1.50 \rightarrow -1.46$, `payment_rule_correct` $-0.77 \rightarrow -0.77$, `conservative_language` $+0.76 \rightarrow +0.72$ (all $p \leq 0.001$); `overpay_concern` attenuates to -0.20 ($p = 0.10$). Incremental R^2 is 0.069 deduplicated versus 0.075 in the full sample. The stated-intent consistency shares (97.1%/87.6%) are computed over traces and are unaffected in sign or approximate magnitude by deduplication.

Cluster counts. The combined grid provides 92 run-directory clusters, but the menu and clock-framing cells are stored as one flat directory per model $\times$ cell (8 clusters total), so standard errors for the B1/B3 family fixed effects lean on few clusters and should be read accordingly.

Scope. Keyword features are surface-level measurements of stated plans; the cross-mechanism invariance result is GPT-4o only; and clock traces are excluded from the regressions. None of the trace results are used as causal mediation estimates—the first-stage zero for Payoff Safety is precisely a *failure* of the mediation pathway through verbalized understanding, which is the point.

G.6 Further Directions

Three extensions would upgrade this apparatus from regexes to a measurement instrument, listed in descending referee-value per unit of effort. First, an *LLM-judge strategy taxonomy*: classify every trace into a small closed set of strategies (truthful-dominance, shading-for-margin, opponent-anchored, mechanism-confused, ...), validated against a ~ 100 -trace human-labeled set; this fixes the dictionary’s blind spots (negation, paraphrase, and especially the 4,541 unanchored traces where the largest errors live) and would sharpen the dissociation into a statement about strategy shares. Second, *cross-mechanism script invariance for all models*: FPSB/TPSB cells for Claude, Gemini, and Gemma with the frozen dictionary (~ 50 auctions $\times$ 2 formats $\times$ 3 models), turning the GPT-4o-only invariance result into a general claim about model “house styles.” (The formal mediation analysis proposed in earlier drafts is now implemented; this subsection reports it.) [TODO: Co-author decision: which of the two remaining trace extensions (LLM-judge taxonomy; cross-mechanism runs for the remaining models) to execute for the next draft; inputs and cost estimates in results/traces/BRAINSTORM.md.]

H Descriptions in a Second Domain: The Menu-Content Split in Deferred Acceptance

Auctions are this paper’s domain. This appendix asks whether the description finding of Section 6.2—that what moves behavior is not additional text about the mechanism but text that states the *invariance* making truthful play safe—is specific to auctions, by replicating it in a second, non-auction strategy-proof environment: school choice under student-proposing deferred acceptance (DA). The setup is deliberately minimal. In each instance, four LLM students submit rank-order lists over four schools with unit capacity, under Ergin-acyclic priorities (Ergin, 2002; Ashlagi and Gonczarowski, 2018); student-proposing DA is strategy-proof for the proposing side (Roth, 1982), so a truthful list is dominant. We measure misreporting as the normalized Kendall- τ distance between the submitted and the true preference ranking (the fraction of the six school pairs ranked in opposite order; the DA analogue of the auction deviation metric of Section 4.4), and we run the same four model families as in the main text (Claude 3.5 Haiku, Gemini 2.0 Flash, GPT-4o, Gemma 27B). The pooled direct-DA baseline error is 4.2%. All numbers below are recomputed from the raw per-instance logs by `analysis/build_da_cells.py` (market-cluster percentile bootstrap, $B = 2000$, seed 1299); cross-model figures are unweighted means of per-model means.

H.1 The Invariance Content Is the Active Ingredient

The description cells (Family B; prompts in Appendix F) replicate, and sharpen, the auction-domain finding of Section 6.2: what moves behavior is not additional text about the mechanism but text that states the *invariance* making truthful play safe.

Safety-exposing description (B2). The *Rejection Safety* description states the property that distinguishes DA from the Boston mechanism (Abdulkadiroğlu and Sönmez, 2003): rejections only redirect, never eliminate—a rejection from a school does not hurt the student’s chances at her next-ranked school. It nearly eliminates misreporting, from 4.2% to 0.2% pooled (Claude 5.8% $\rightarrow$ 0.8% [0.3, 1.7]; Gemini, GPT-4o, and Gemma all to exactly 0.0%).

Menu descriptions (B1) split by content. Gonczarowski et al. (2023) propose describing a strategy-proof mechanism through the *menu* it induces: given others’ reports, a student faces a fixed set of obtainable schools, and her report selects from that set. We implement two variants that bracket the informational content of this idea. The *invariance-property* variant states the menu’s incentive content explicitly (“your ranking *cannot change* your obtainable set; it only determines which school you receive within it”). The *mechanics* variant restates the same construction procedurally (“Step 1: the mechanism computes your obtainable schools. Step 2: you receive the school you ranked highest among them”) without asserting the invariance. The two variants move behavior in opposite directions:

- *Menu with the invariance property improves play*, behaving like a safety-exposing description: pooled error falls from 4.2% to 1.8% (Mann–Whitney $p < 10^{-4}$), with large significant gains for the two weak-baseline models (Claude 5.8% $\rightarrow$ 2.1%, MW $p = 10^{-4}$; Gemini 7.6% $\rightarrow$ 0.3%, $p < 10^{-4}$) and null effects for the models already near truthful (GPT-4o $p = 0.23$; Gemma $p = 0.43$).
- *Menu mechanics alone worsens play*: pooled error rises from 4.2% to 6.9% ($p < 10^{-4}$), driven by GPT-4o (1.0% $\rightarrow$ 4.2%) and especially Gemma (2.6% $\rightarrow$ 13.7%). Restating the outcome computation gives the models material to strategize over without the property that would make strategizing pointless.

In auctions the menu restatement is null on average and sign-mixed across models (Section 6.2); the DA data explain why: the standard menu text bundles a computation with an invariance, and only the invariance helps. Where the auction cell delivers the bundle and finds nothing, the DA cells unbundle it and find two opposing effects.

Description	Family	Mean Kendall- τ error, % [95% CI]				Mean
		Claude	Gemini	GPT-4o	Gemma	
Direct DA (baseline)	—	5.8 [4.3, 7.5]	7.6 [6.1, 9.2]	1.0 [0.4, 1.7]	2.6 [1.4, 4.0]	4.2
Menu, mechanics only	B1	6.7 [5.3, 8.2]	3.1 [2.0, 4.2]	4.2 [2.9, 5.7]	13.7 [11.0, 16.7]	6.9
Menu, invariance property	B1	2.1 [1.3, 3.0]	0.3 [0.1, 0.7]	0.3 [0.1, 0.7]	4.2 [2.6, 6.1]	1.8
Rejection Safety	B2	0.8 [0.3, 1.7]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.2
Textbook strategy-proofness	(advice)	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0

Table 21: The menu-content split in DA: mean normalized Kendall- τ error in percent, with market-cluster bootstrap 95% confidence intervals ($B = 2000$, seed 1299); “Mean” is the unweighted cross-model mean. Models: Claude 3.5 Haiku, Gemini 2.0 Flash, GPT-4o, Gemma 27B. Cells are 50 markets $\times$ 4 students (a few cells have 42–49 markets after parse-failure drops), recorded in `results/merged_ranking/da_cells.csv`. A menu description that carries the invariance property improves play like a safety-exposing description; a menu description that restates only the mechanics worsens it. The textbook strategy-proofness cell states the dominant strategy outright and is included only as a diagnostic ceiling (see text).

Diagnostic ceiling: the textbook statement. A *textbook strategy-proofness* description—a plain statement that submitting one’s true ranking is optimal regardless of what others do—produces 0.0% error for *all four* models, marginally below even Rejection Safety (0.2%). This cell reveals the dominant strategy outright and is not a simplicity-preserving lever, but it serves as a useful ceiling: when told the answer, every model complies perfectly, confirming that baseline misreporting reflects a failure to *derive* strategy-proofness rather than an inability to execute a truthful report.

Removing the explanation. Conversely, a *null* description that states the rules with no explanation of the algorithm shifts play heterogeneously (pooled 4.2% $\rightarrow$ 5.2%, MW $p = 0.03$): it worsens Claude (8.7%), GPT-4o (3.3%), and Gemma (7.0%) but *improves* Gemini (2.1%)—description content interacts with model-specific priors rather than acting uniformly.

Robustness: cardinal presentation. An earlier variant of the grid presented raw dollar values (“ $w = \$72$, $x = \$61$, ...”) instead of a sorted preference ordering, forcing the model to derive its own ranking. Baseline errors rise by an order of magnitude (12–24% across models), and the description split replicates: the invariance-property menu again produces the large improvement (Gemini 18.8% $\rightarrow$ 1.2%; GPT-4o 11.9% $\rightarrow$ 1.9%) while the mechanics variant barely moves the needle. The cardinal variant is incomplete (several cells were not run) and is documented, cell by cell, in `results/merged_ranking/da_cells.csv` (variant=da_cardinal).

I eBay-Style Closing Rules: Field Microstructure Robustness

As a bonus field-anchored robustness check, we confirm that the LLM bidders reproduce the classic eBay-style sniping pattern in a stylized proxy-bidding environment. Each of three GPT-4o bidders holds an independent private value and may raise or hold a maximum bid over a fixed sequence of discrete bidding periods, with the platform bidding on each bidder’s behalf at the second-highest maximum plus an increment. Under a standard hard close, decisive bids concentrate at the deadline: 63.3% of winners place their final maximum-bid change on the scheduled last day (GPT-4o baseline treatment). A soft-close (auction-extension) rule collapses this last-day share to 7.6% and shifts the distribution of final-winning-bid times substantially earlier (Mann–Whitney $p = 3.5 \times 10^{-5}$), spreading competition across the auction rather than compressing it into the final moments. This directional response echoes the field evidence that hard-close eBay auctions elicit far more sniping than soft-close (auction-extension) formats (Roth and Ockenfels, 2002; Ockenfels and Roth, 2006). Companion hidden-reserve treatments show only limited behavioral and revenue sensitivity, consistent with the large-sample field evidence of Einav et al. (2011). We report this exhibit only as a field-microstructure robustness check; the auction laboratory of Sections 5–6 carries the paper’s main results.

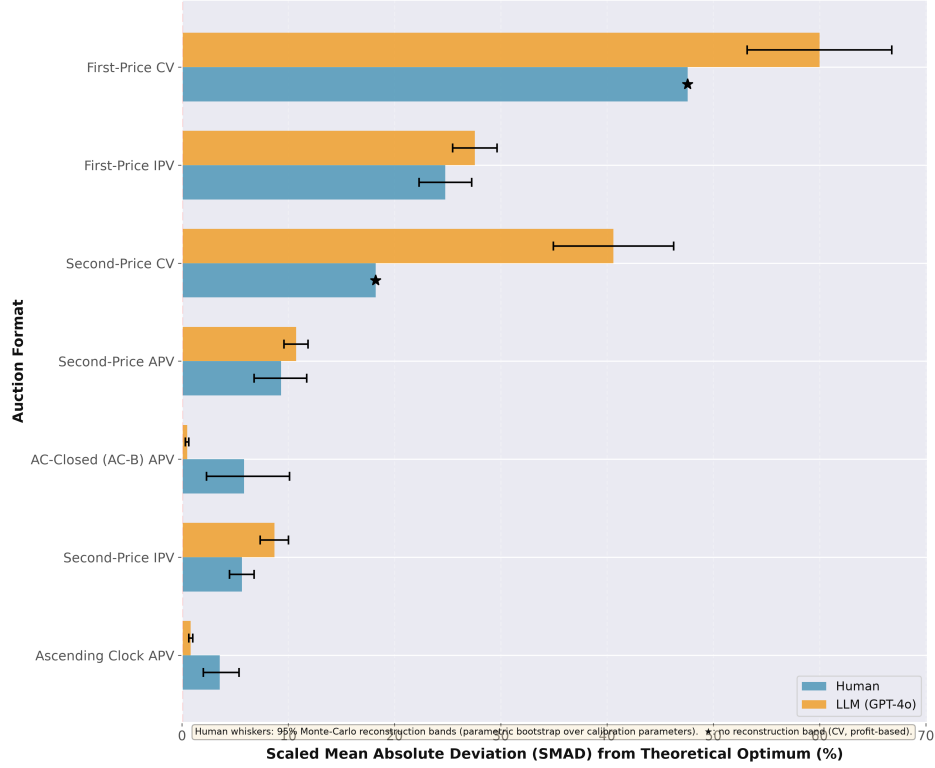


Figure 1: **Human vs. LLM bidding behavior across seven auction formats, with reconstruction-uncertainty bands on the human anchors.** Blue bars: scaled mean absolute deviation (SMAD) between human bidding and the theoretical benchmark, computed from the moment-matched reconstructions of Section 4.5. Yellow bars: SMAD between GPT-4o agents and the same benchmark. Formats are ordered by human SMAD; LLM simulations use $N = 3$ bidders. Black whiskers on the human bars are 95% Monte-Carlo reconstruction-uncertainty bands: for each source-paper calibration we re-draw $B = 2000$ vectors of the mixture parameters $(p_{eq}, p_{over}, p_{under}, \lambda, \sigma, \delta)$ within the ranges the reported source statistics admit and propagate each draw through the reconstruction (Appendix A). The bands are [22.30, 27.25] (FPSB IPV, point 24.76), [4.46, 6.79] (SPSB IPV, 5.65), [6.78, 11.74] (SP-APV, 9.31), [2.00, 5.36] (AC, 3.54), and [2.30, 10.11] (AC-B, 5.83). The FPSB $\gg$ SPSB difficulty ordering (ratio 95% band [3.54, 5.65]) and the AC clock improvement over sealed SP-APV hold in 100% of the re-draws; the AC-B closed-clock improvement holds in 92.6% of re-draws, its wider band inherited from the large reported standard error on the Breitmoser–Schweighofer–Kodritsch mean absolute deviation. Common-value formats (marked $\star$) use a profit-based deviation metric and carry no reconstruction band. These are calibration bands, not sampling bands on human behavior; all comparisons remain at the level of orderings and comparative statics (Section 4.5).

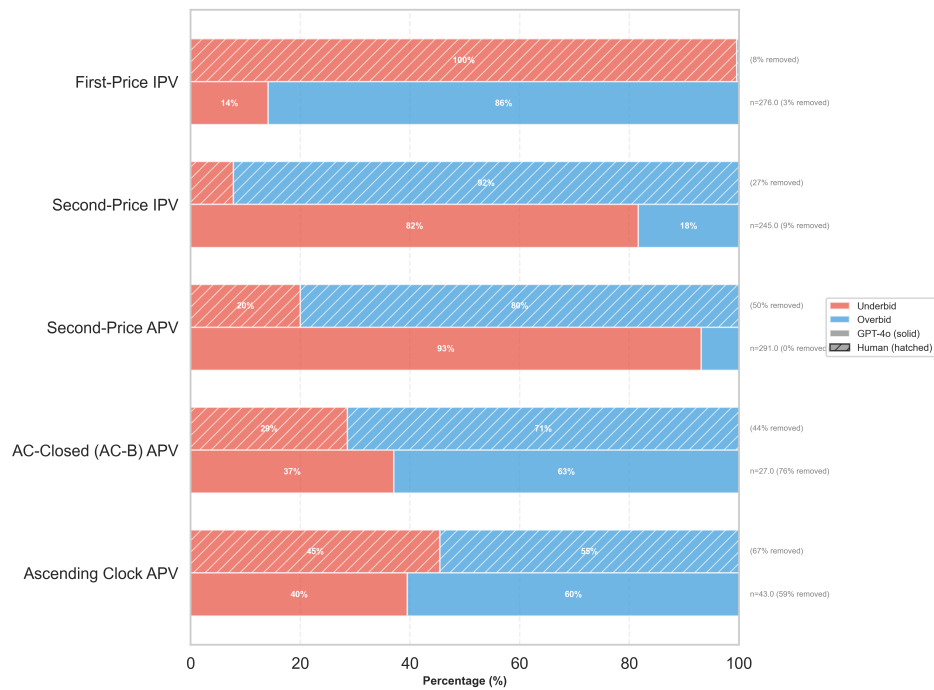


Figure 2: **Directional decomposition of deviations, humans vs. GPT-4o.** Share of underbids vs. overbids across auction formats after removing bids within a $\pm 2\%$ band of the theoretical benchmark. Humans overbid where GPT-4o underbids in the strategy-proof sealed-bid formats—the direction divergence carried into Table 4 and Section 7.

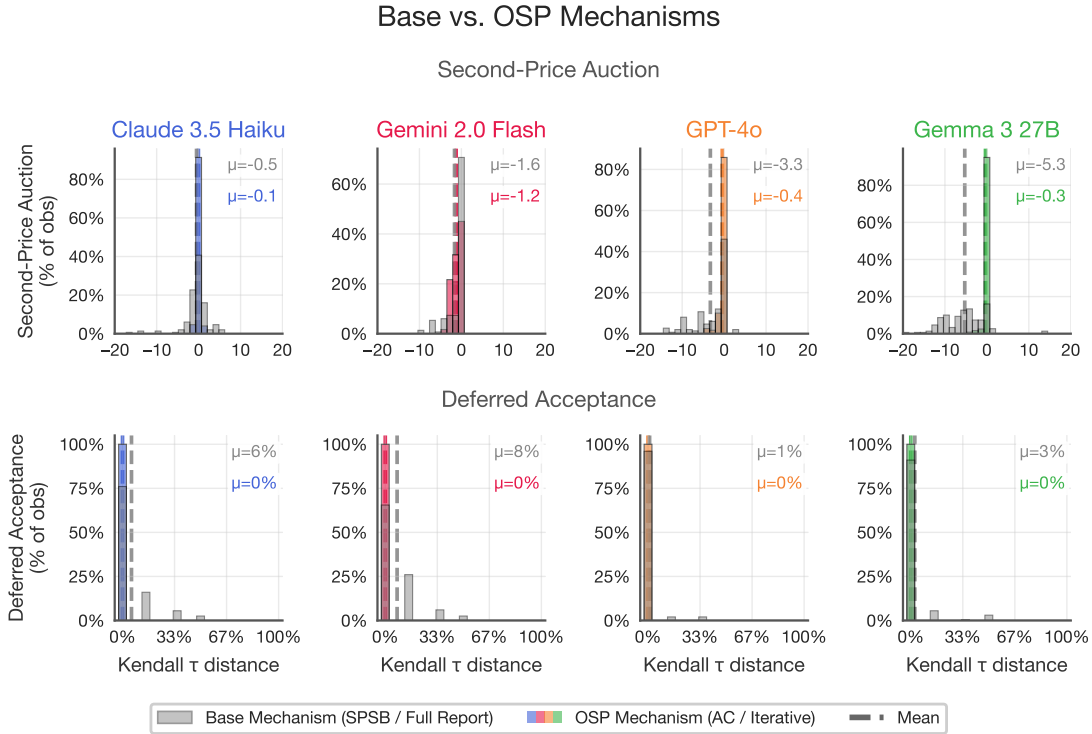


Figure 3: Comparing base (non-OSP) and OSP mechanisms across four model families. Distribution of bid deviations from value ($\text{bid} - \text{value}$) in second-price sealed-bid (gray) and ascending clock (colored) auctions. Dashed lines indicate means. All models deviate from truthful play in the non-OSP sealed-bid format; the ascending clock corrects toward truth-telling.

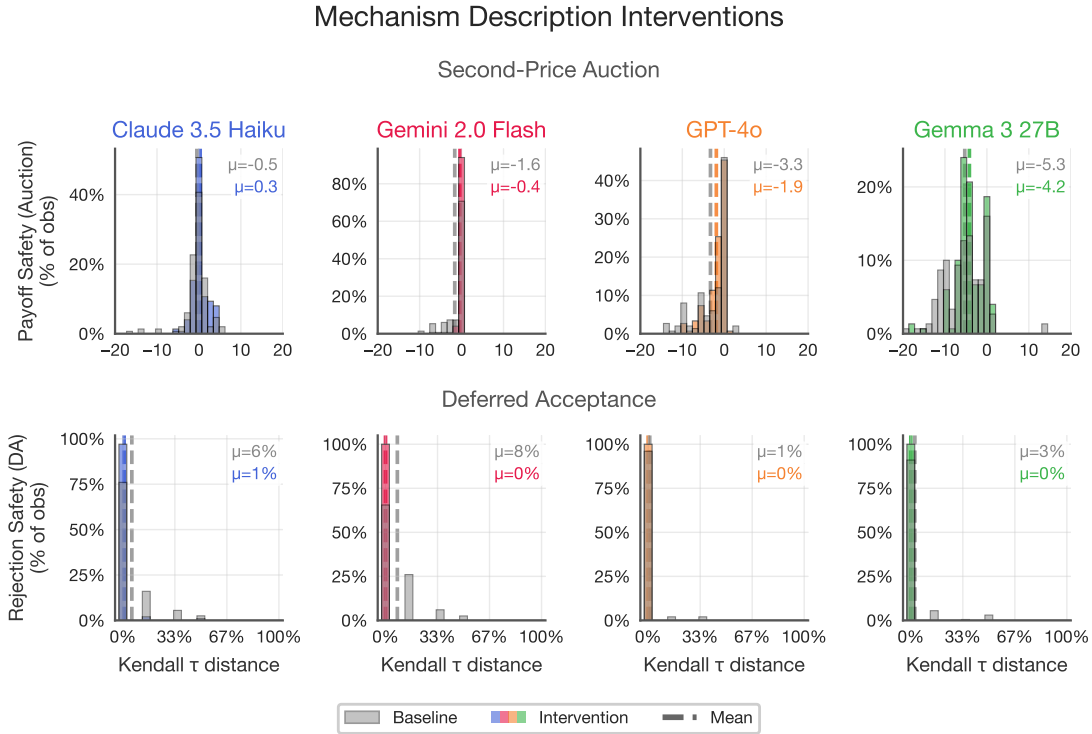


Figure 4: Safety-exposing description (B2): *Payoff Safety* in second-price auctions (“your bid determines IF you win, not WHAT you pay”), across the four model families. Baseline in gray, intervention in color; dashed lines indicate means. The intervention improves play in every model, closing roughly half the gap to the full OSP result of Figure 3.

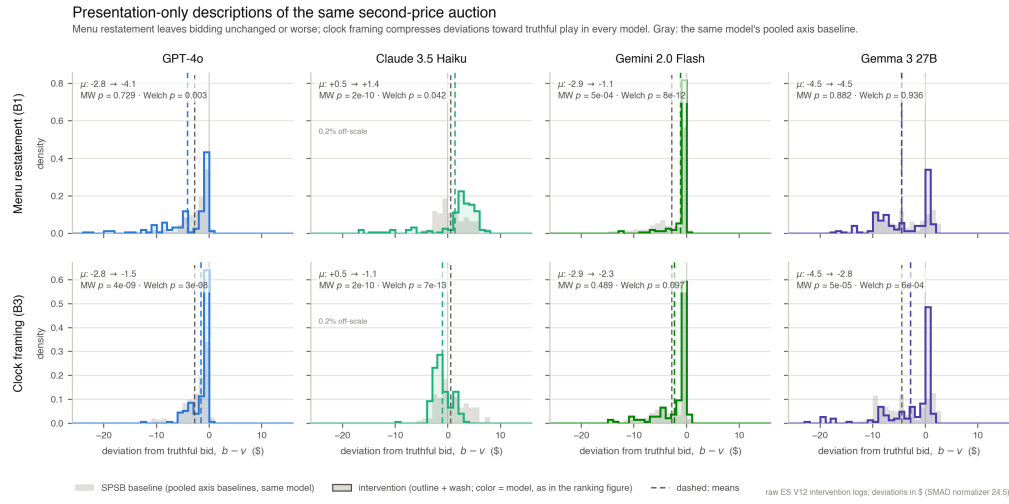


Figure 5: Presentation-only descriptions of the same SPSB mechanism, across all four model families (2×4 panels; deviations in dollars). *Top row*: menu restatement (B1) vs. the pooled axis baseline—small, sign-mixed shifts (Gemma null, Claude and GPT-4o slightly away from truthful play, Gemini toward it). *Bottom row*: clock-framing (B3)—a same-signed improvement in every model, significant in three of four. Only LLM simulation data are shown.

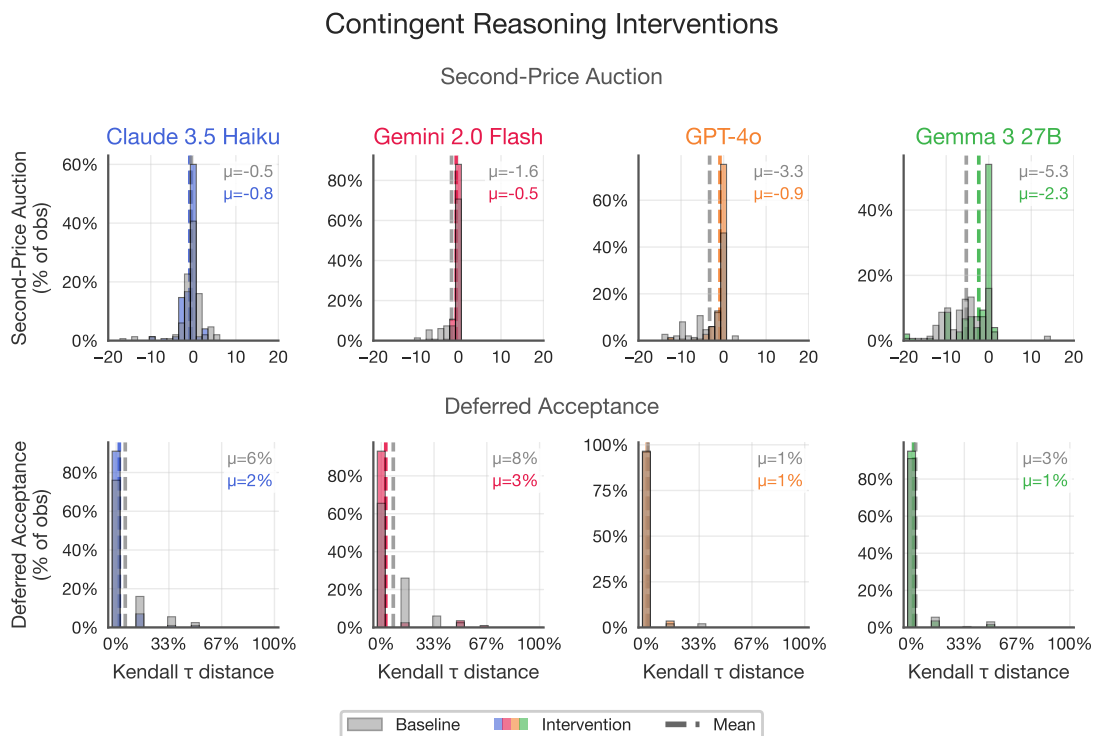


Figure 6: Contingent reasoning: *Payoff Tree* intervention (C1). Second-price auction bid deviations (bid – value) across the four model families; baseline SPSB in gray, intervention in color. The payoff-tree scaffold consistently improves play across models, roughly halving the distance from truthful.

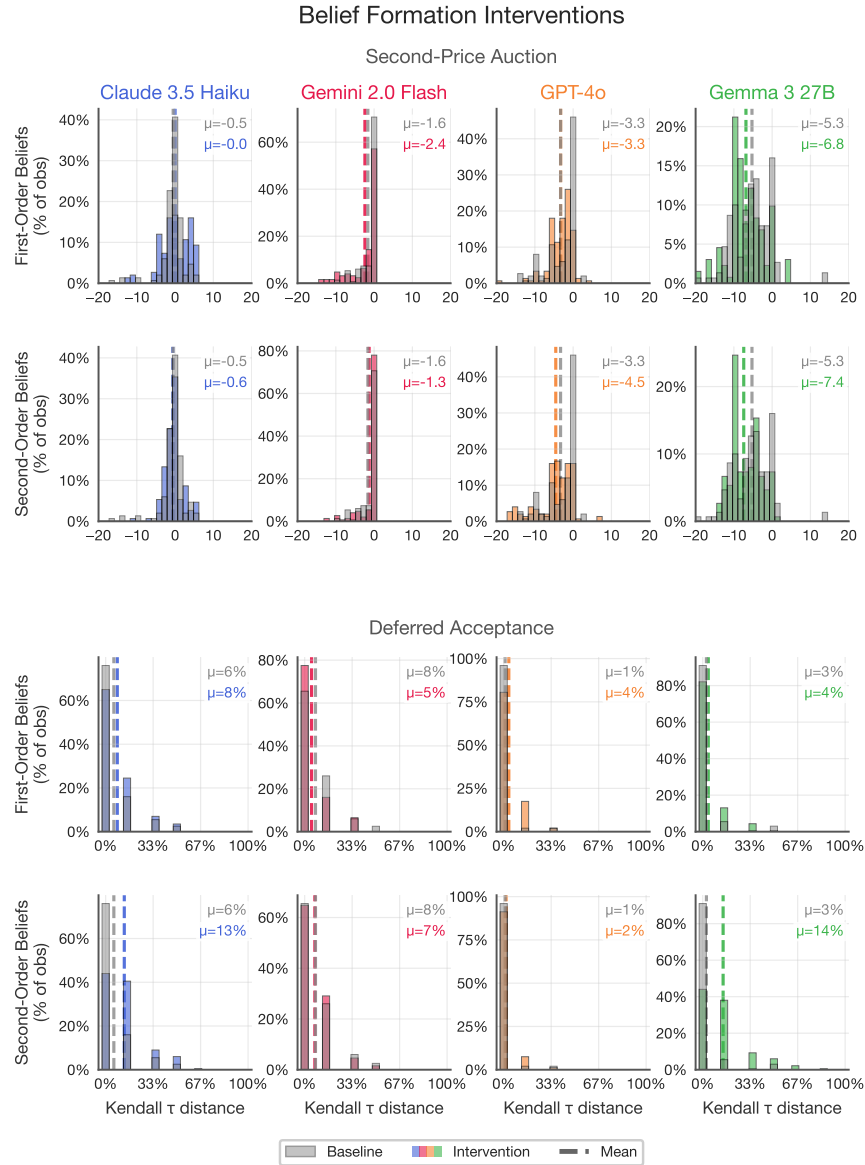


Figure 7: Belief-formation interventions (C3): second-price auction bid deviations under first-order and second-order belief scaffolds, across the four model families. Baseline distributions in gray, interventions in color. Belief scaffolds almost uniformly worsen play, with second-order beliefs causing the most damage. Gemini is the notable exception, showing near-immunity to belief scaffolding.

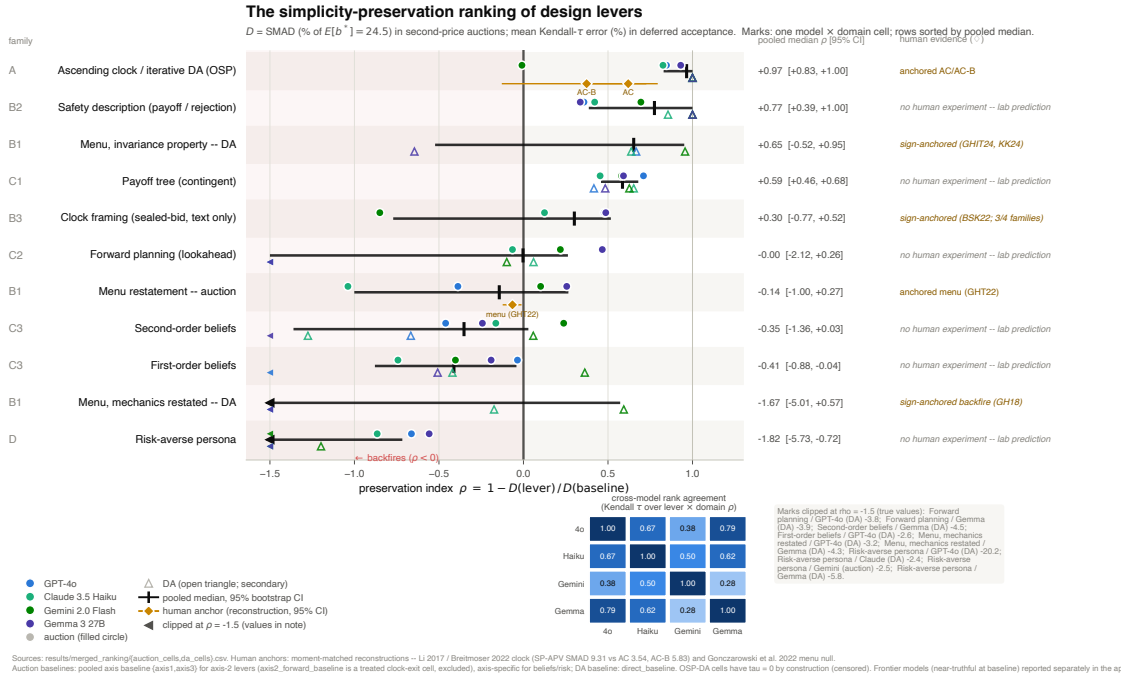


Figure 8: The simplicity-preservation ranking of design levers. *Panel A*: rows are levers sorted by pooled median $\rho = 1 - D(\text{lever})/D(\text{baseline})$; individual marks show each model $\times$ domain cell (color = model, shape = domain), with the pooled median and double-bootstrap 95% CI per row; the shaded region $\rho < 0$ marks levers that backfire. Gold diamonds are human anchors computed from the moment-matched reconstructions (Section 4.5): AC $\rho = +0.62$, AC-B $+0.37$, menu -0.07 ; rungs without a diamond have no human experiment and are issued as lab predictions (Section 7). Extreme negative matching-domain cells are clipped for display (see text). *Panel B (inset)*: cross-model rank-agreement (Kendall τ) over the lever ordering. See Table 5 for the underlying cells.